Teaching Portfolio

Edison Hauptman

November 10, 2026

Introduction to the Reader

I have been an instructor of record for seven courses across six terms. This is more experience than any current or recent graduate student at the University of Pittsburgh. While graduate Teaching Fellows at Pitt are often instructors of record during the summer terms, it is rare to be an instructor of record during the fall or spring.

This Teaching Portfolio contains my teaching philosophy statement; teaching resume; statement on diversity, equity, inclusion, and accessibility; and two sample syllabi, including for a proposed course in Network Science. I also include three previous assignments; three previous assessments; and a summary of my student evaluation scores. A full record of student evaluations from my courses taught as an instructor of record appears in the appendix.

Thank you,

Edison Hauptman

November 10, 2026

Table of Contents

1 Teaching Philosophy Statement **2**

2 Teaching Experience **4**

3 Diversity, Equity, Inclusion, and Accessibility **5**

4 Sample Syllabi **8**
4.1 Proposed Course: Network Science 8
4.2 Fall 2025: Calculus 1 15

5 Sample Assignments **27**
5.1 Group Work: The Yellow Light Dilemma 27
5.2 Project: "When Will I Ever Use This?" 32
5.3 Take-Home Assignment: Extra Practice Problem Sets 34

6 Sample Assessments **44**
6.1 Sample Progress Check: Business Calculus 44
6.2 Sample Progress Checks: College Algebra 1 46

A Student Evaluations of Teaching **50**
A.1 Full Student Evaluation Reports 51

1 Teaching Philosophy Statement

I am refining a recipe for my course design and teaching practices. I have two main objectives as an instructor. The first is to build up and evaluate the core learning outcomes of the course. The second is to get students to engage with math in a way that is interesting to them. In my intro-level courses, I use a standards-based grading structure with 8-10 "Big Ideas" alongside a broad "Engagement Points" category for every other assignment in the course. My students report that this structure supports a growth mindset in mathematics and encourages their independent exploration and thought. For more information on my course design, recent syllabi are available on my website under each course.

Equipment: Like a kitchen, math has tools and appliances that a competent chef must learn to use properly. The core of all my courses is therefore a list of clearly-defined standards—one set for each Big Idea—that students are required to achieve by the end of the course. There is no partial credit, and students can reattempt Big Ideas throughout the course. My students therefore interpret my grading as a temporary diagnosis, not a punitive judgment, as evidenced by students' end-of-semester comments: "I think that having this structure allows you to not become defeated easily if you do bad on one exam. That keeps people who are not traditionally from a math background motivated to keep going in this course."

Flavor: Once students know the basic techniques, we can add flavor! My lesson on bipartite graphs is enriched by building towards the Stable (Queer) Marriage Problem. The theorem that a random selection of vectors in high-dimensional space are mutually almost orthogonal gives insight into how language models work in machine learning. In my intro algebra class, I use mechanics from a video game to help explain slope. My research on bootstrap percolation was initially inspired by seeing water percolate in a video game. I discuss my research in small-world networks by asking students to evaluate whether their own social networks have the small-world property. I do all of this to show students how people's diverse perspectives make math useful and beautiful.

My students then add their own flavors—whether from a skate park, chemistry, architecture, or fMRI data—to make math their own. In my Calculus 2 course, one of my students made an interactive Bézier curve tutorial in Unity, inspired by the curve tool he'd seen in Photoshop. Every assignment students complete earns them Engagement Points, which shows them that I want them to take charge of their own learning. I wrote about this on the Grading for Growth blog, a hub for alternative grading discussion.

Furthermore, math is powerful because it is widely applicable. When students bring their own flavors into the course, they are better-equipped to do mathematical research; their research question is already interesting to them. My own research experience largely comes from encountering puzzles in the wild and using the mathematical skills developed in my coursework to solve them.

Simmering: I use clicker questions in class to get students to pause, ponder, and then convince their neighbor. I make students answer their own questions, so they have space to think for themselves. "Even when I gave a wrong answer, [Edison] respected my train of thought and built off of that answer rather than just telling me I'm wrong and moving on." By adapting my lectures in this way, I make the material more memorable and foster a culture of belonging.

Cleanup: After solving any question, I give a recap, zeroing in on the essential skills, reminding students what we noticed to come up with the key step of a solution, and explaining what was

usual vs. unusual in the question. This cleanup step keeps my class grounded and facilitates their practice.

Adjusting: I adapt my teaching to meet my class's needs. For example, in my College Algebra course this semester, I am spending more time presenting full examples in class. My students told me they wanted to see more correctly-worked examples; I realized that students are more able to solve new examples through their own practice at the Calculus level.

My teaching is also informed by attending my school's STEM education seminar, dB-SERC; they awarded me a grant to implement alternative grading in my Calculus courses, and I have used ideas from instructors across disciplines to refine my course structure. My structure has in turn served as a model for faculty in the SEISMIC Collaboration, a multi-institutional, multi-disciplinary consortium that promotes assessment equity in large-enrollment STEM courses. I am also researching whether completing an extra-credit project on a topic of interest improves students' self-concepts in large-enrollment Calculus 2 courses, again supported by dB-SERC with a Course Transformation Award.

Iterate, Iterate, Iterate: Some recipes take a few attempts to get right, but by engaging my students in feedback loops, they make adjustments and improve. I remind my students throughout the semester that math is hard, but practice makes the Big Ideas easier to understand. By facilitating a growth mindset via my course design and classroom practices, I make sure that all my students can succeed. Instead of seeing math as a boring, required, eat-your-vegetables course, my students become mathematical chefs with the tools and recipes to make their own masterpieces!

2 Teaching Experience

(Key: F = Fall, S = Spring, Su = Summer)

University of Pittsburgh: Teaching Fellow 2021 - present

Awards and Honors
2025 Teaching Assistance Excellence Award: Department of Mathematics
2024 Elizabeth Baranger Teaching Award: Dietrich School of Arts & Sciences
2023 Teaching Assistance Excellence Award: Department of Mathematics

Math Education Research Grants
Hauptman, Edison; Heck, Lizzie; Sampaio, Rafaella; Wheeler, Jeffrey: University of Pittsburgh dB-SERC Course Transformation Award, 2025 ($9250)
Hauptman, Edison; Wheeler, Jeffrey: University of Pittsburgh dB-SERC Course Transformation Award, 2024 ($2563)

Instructor of Record (6 terms, 7 total sections)

College Algebra 1	F 26
Business Calculus	S 26
Analytic Geometry & Calculus 1 (2 sections)	F 25
Analytic Geometry & Calculus 2 (3 sections)	Su 24, Su 23, Su 22 6-week-1

Teaching Assistant

Prep for Scientific Calculus (2 sections)	S 25, F 21
Business Calculus (3 sections)	S 23, F 22
Analytic Geometry & Calculus 1 (10 sections)	F 24, S 24, F 23, S 23, F 21
Analytic Geometry & Calculus 2 (2 sections)	S 25, F 24
Analytic Geometry & Calculus 3 (2 sections)	F 24, S 24
Intro Theory of 1-Variable Calculus	Su 22 6-week-2

Mentorship

Mentor: CMU-Pitt Math Directed Reading Program (Mathematical Biology)	F 26
TA Mentor (graduate TAs): Department of Mathematics	Su 26
Math Assistance Center Assistant (graduate TAs): Department of Mathematics	S 26
Mentor: CMU-Pitt Math Directed Reading Program (Combinatorial Optimization)	F 25
TA Mentor (graduate TAs): Department of Mathematics	Su 25

Grader

Combinatorial Mathematics	F 26
An Introduction to Optimization (2 sections)	S 25, F 22

3 Diversity, Equity, Inclusion, and Accessibility

I am a representative of the math community when I teach mathematics. When students take intro or elective courses in math, I am one of the only mathematical authorities they'll interact with in their college experience, so I prioritize creating a welcoming and encouraging environment in my classroom.

Identity and Safety

I believe that signaling my support for marginalized groups is important because it helps members of those groups feel safe around me. Folks from marginalized communities often have a harder time feeling like they belong. As a queer person, I greatly appreciated that many of my undergraduate math faculty had rainbow "ally" stickers outside of their offices. I didn't come out to them as a student, but I've talked openly with them on return trips, and I know I have a sizable professional network that won't discount me because of who I am. I am invested in math partly because I know there's a place for me as myself. I aim to foster that sense of belonging for all my students.

Most of the way through the term, one of my students once told me that I'd saved his ability to get a degree in his intended major, and that he looked forward to coming to a class he'd once feared. Very early on in the term, he'd asked about the decorative stickers on my laptop, and among my personal interests and alma mater, I was proud to explain that I had a heart with the trans pride flag because of my support for trans rights. Just as I felt a bit more comfortable every time I walked past my old faculty's offices with their stickers, I noticed my student's increased comfort being open in class and in my presence. The increasing hostility towards trans people in this country sadly means this isn't something he can take for granted.

Different Skills and Perspectives

The core of my commitment to fostering a welcoming environment in mathematics is my flexible course structure. I tell my students that math is more than just drilling problems and completing exams, and then I back that up with a course structure that rewards different skills. Depending on the context, the assignments have changed slightly, but my mindset has been consistent. "In this course, you're going to get the opportunity to flex different mathematical muscles, such as communicating your results, resilience, creativity, and teamwork. These are all skills I use *every day* as an aspiring mathematician!"

When I was assigned an instructor position and the corresponding TA position for a summer course, my students could complete a "Teach the Recitation" assignment by presenting a practice problem. In my small-enrollment courses, I have assigned challenge problems which emphasize mathematical writing skills. Most importantly, if my students want to pursue a topic beyond the scope of the course, there is always a way for me to give them credit for it.

For example, a first-semester transfer student in my summer course wrote a short paper on Integrated Rate Laws for his old Chemistry professor and classmates. In my large-enrollment calculus courses this fall, I am chatting with my students about the related rates of change involved in dance or in window design via my shorter "A New Perspective" assignment. It is critical to me that my students have space and encouragement to find connections from the math they are learning to some aspect of their own lives. I believe this because this is what fuels my own interest in math, including my current research. This priority also informs my research in math education; my current Course

Transformation Award grant involves coordinating with my department and others to create and implement applied projects in the University of Pittsburgh's large-enrollment Calculus courses.

Another one of my students emailed me to say that he'd unexpectedly failed his Calculus 2 course in the spring and now had to start my 12-week course a week late due to a family vacation. This is not a great way to start the summer, so I insisted on meeting him in my Drop-In Hours. There, we discussed his previous experience in the course and figured out that his lack of algebra skills had caused problems last time. He took it upon himself to work through the algebra review packet I always include on my Canvas page, building a much stronger foundation. I still remember his enthusiasm and charisma when presenting difficult problems to the class for my recurring Teach the Recitation assignment, and his joy exploring how insect populations change from year to year using logistic difference equations. He cleaned up his weaknesses, played to his strengths, and earned an A in my course.

Another student of mine picked up on the material quickly, but she rarely participated in an otherwise lively class. There were two pivotal times when she spoke up. The first was while solving the rough integral $\int \sqrt{1+x^2}dx$, where she came up with the key insight to get the class unstuck. The second was halfway into the course in my recitation, where I was preparing to solve a problem that no one signed up to present. She'd mentioned having solved it and offered to present instead, so I happily gave her the stage; she later went on to teach the recitation three more times. Towards the end of the semester, she told me that the structure of the course and the environment I'd created eventually encouraged her to come out of her shell, and that she was glad for this.

My course structure has enabled all the students I've mentioned and many others to improve their math ability, because I explicitly recognize and nurture skills other than taking exams. It is self-evident to me that giving students the freedom to explore math in ways that interest them will broaden participation in the mathematical sciences. I will note that although most of my practices are not directly targeted towards helping minoritized students, fostering a sense of belonging for everyone closes achievement gaps. It is no coincidence that the students who benefit most from my non-traditional course structure are students with minoritized identities.

Mentorship

As the TA Mentor for graduate Teaching Assistants over the past two summers, I put specific emphasis on sharing institutional knowledge—that is, the policies, procedures, and resources that can take several years to learn. I proactively shared resources and opportunities that I knew of, whether I'd used them myself or not. I discussed our department's Teaching Seminar and dB-SERC's lunch seminar, and encouraged my peers to attend these as well. Before the start of the Fall 2025 term, I created and led an orientation session on teaching expectations and important resources for the incoming graduate students—this inspired my department's teaching professors to create a more formal workshop for this year. I also led the committee to redo our department's Graduate Student Guide, so that future graduate students will be better-prepared than I was during graduate school.

I see all of this work as a crucial way to broaden participation in mathematics—specifically when it comes to retaining folks from minoritized backgrounds—because there is an unequal understanding of these norms at the start of any program. My own unawareness of some norms caused some issues at the start of my grad school experience, and I actively insist that future cohorts have a better experience. As a professor, I intend to actively share institutional knowledge early and often with all of my students, so that I can close achievement gaps.

Future Work

As I begin the next stage of my career, I will have to adapt to new contexts and refine my course structure accordingly. There is always more to learn, but I believe the broad strokes of my work will be the same—namely, to contribute to an open, welcoming environment for everyone that I interact with!

4 Sample Syllabi

4.1 Proposed Course: Network Science

Syllabi from recent courses I have taught are available on my website under each course.

MATH XXX-XXX	**Network Science**	**Term XXX**
Instructor: Edison Hauptman (he/him/his)		**Contact Me:** XXX
My Drop-In Hours: TBD		
Textbook: Network Science, by Albert-László Barabási (freely available online)		
Supplemental Resources: To be shared during the course		

Meeting Times

- **Lecture:** Days XXX at times XXX in location XXX

Drop-In Hours

Are you having trouble with the problem sets? Would you like a place to study with your peers, where you can ask questions if you get stuck? Do you want to discuss your ideas for a project? Have you discovered something cool that you want to show me? Drop-In Hours are a great way to chat with me outside of class! They are also a great place to find study partners! Working with peers on this material will help you learn it faster and better. Math is not an individual pursuit; sharing ideas with one another makes all of us stronger.

I will announce my Drop-In Hours once I establish my schedule during the first week of classes.

What is Network Science?

How do scientists model the spread of diseases like COVID-19? How can a new union of workers build its membership? Why do we often discover unexpected connections with random strangers? **Network Science** is used by scholars from a wide range of disciplines, and they study questions like these using networks as a model.

To become confident with the tools we'll learn in this class, some background with computer programming (especially Python), linear algebra, probability, and differential equations would be helpful. The first part of this course will focus on another background topic, graph theory. If you find yourself getting stuck on these or other background topics, I am happy to help you fill in the gaps. I will **never** demean you for not knowing something that others consider "basic" or "easy".

Note: Math is hard, and this course will be a lot of work. This is a friendly reminder that **YOU CAN DO IT!** If you are in this classroom, then you belong here, even if you don't feel fully prepared. Every single mathematician I know has struggled with math and *currently* struggles with math, because math is very hard! **It will take consistent hard work to learn and apply the material in this course, but I promise you that you can!** Let's find out how on the next page:

How to Succeed in this Course

My main objective as your instructor is that you learn this material well enough to use it later in your life, wherever you happen to end up! If you can build a solid foundation with these tools, then you can develop a new lens through which you view and navigate the world around you. *(That might sound a bit pretentious, but I'm not kidding!)*

Since this is an interdisciplinary subject, you will be able to flex different mathematical muscles throughout this course, such as communicating your results, teamwork, resilience, and creativity. These are all skills I use *every day* as a mathematician!

Personally, I don't put much stock in grades. However, I still have to report a grade at the end of the term. I'll do that based on how well you do three things:

1. Demonstrate understanding of the **Big Ideas** in Network Science,
2. Complete an **Applied Project** in Network Science, and
3. Earn **Engagement Points** by doing extra assignments that will help *you* learn this material well.

I will be measuring your *progress* throughout the term. This means that for the Big Ideas and Applied Project, you will have many chances to show me that you have *improved* your understanding of the course material, and whenever you do, that will be reflected in your grade! Let's see what that means:

1. Achieving the Big Ideas

In most weeks, we will have a Progress Check on a **Big Idea** in this course. Progress Checks will take place at the beginning of a lecture, and each Progress Check will be 20 minutes long. The detailed schedule for all Progress Checks will be posted on Canvas.

To achieve (pass) a Big Idea, you must answer at least 3 of the 4 questions on the Progress Check correctly. You will have three attempts to achieve each Big Idea:

1st: Soon after I cover the material in lecture,

2nd: 1-2 weeks later, after you have feedback from your first attempt,

3rd: On our final exam, which will solely consist of reattempts for all Big Ideas.

Note: **Whenever I talk about a "correct answer," I really mean "essentially correct."** In other words, if you can demonstrate full understanding of the relevant concepts in the question, you will be marked "essentially correct" even if there is a minor error unrelated to the Big Idea. Whenever it's important enough to give a perfect answer in this course, you'll be able to double-check your notes.

To facilitate your studying, each Big Idea is described using several Learning Outcomes, which are *highly suggestive* of the questions you should expect on each Progress Check. The full list is available on page 6.

2. Completing an Applied Project

During the term, you will complete an applied project. There is a detailed schedule for the final project available on Canvas; it includes three check-in deadlines. In between each check-in deadline, I will give you feedback to improve your project along with a (tentative) Applied Project grade (A, B, C, D, or F). The standards for each of these grade levels are described in Canvas.

You may submit your project to as many of the check-in deadlines as you wish, but **your final version of your project must be submitted by the last day of the Final Exam period**.

Note: If you make no changes to your project before submitting it as a final version, the last tentative grade I assigned to it becomes your grade for the Applied Project. **I strongly recommend submitting to the first or second check-in so that you have plenty of time to revise your work**, even if you're not finished.

3. Earning Engagement Points

There will be many other assignments in this course which will earn you **Engagement Points**. **I do not expect you to complete *all* of these assignments.** You should choose the assignments that will help you with your own learning, and focus on those!

For more information on these assignments, see Canvas.

Math Problem Sets (75 pts): To facilitate your understanding of the mathematics of network science, several problem sets will be assigned. These assignments will be marked per problem, using the marks, "successful" (full points), "revise" (chat with me to revise to "successful"), or "no credit" (0 points).

Programming Assignments (75 pts): Real-world applications of Network Science often involve massive data sets, so implementing them in code is very useful for ease of computation. Lab assignments will be marked per problem, using the marks, "successful" (full points), "revise" (chat with me to revise to "successful"), or "no credit" (0 points).

Practice Problem Sets (50 pts): To help you prepare for each Big Idea, I will include a list of practice problems for each one. These problems are graded for completion, but you should still give your best effort. If you complete 80% of the practice problems in a set, you will earn the full Engagement Points for that set. All practice problem sets are due before the final exam.

Teach the Example (25 pts each): In some lectures, I will invite a student to solve an example on the white board. These examples will be posted to Canvas with at least one week's notice and require advance sign-up via email. For more information on this assignment, see Canvas.

Note: The Engagement Point totals required for each final grade are minimum standards. Once again, **you should not complete *all* these assignments**. Instead, choose the ones that will help your learning, whether that's for the Big Ideas or the Applied Project.

Final Course Grade

To earn a given letter grade, you must satisfy **all** of the requirements in the corresponding row.

Final Grade	Big Ideas Achieved	Applied Project	Engagement Points Earned
A	5	A-level work	100
B	4	B-level work	75
C	3	C-level work	50
D	2	D-level work	0
F	Did not meet all of the requirements for a D		

Note: If your final marks are short of an A, B, or C in any **one** category, by at most **one** grade level, you will earn the corresponding "minus" grade instead. If your final marks are above a given letter grade in the **Big Idea** or **Applied Project** category (not Engagement Points), you will earn the corresponding "plus" grade instead. Following all of these rules, you will earn the highest grade that applies.

Other Course Policies

Canvas: Most of this course, including announcements, resources, due dates, and assignments, will be on our course's Canvas page. You are responsible for checking Canvas often. You will also be able to see scores for all assignments on Canvas. However, Canvas is not compatible with this course's grading system, so **your overall grade will *not* appear on Canvas!**

Keeping Track of Your Grade: I will hand out a colored sheet of paper on the first day of class (see page 7). Copies of this sheet will be on our Canvas page. If you have never taken a course like this one before, then this grading system may feel a bit intimidating or complicated. That's okay! **Just like Network Science, if this grading system is new to you, then it will take some time to learn, and I don't expect you to understand it immediately!** Throughout the term, I will dedicate class time to discussing the grading system. Please don't hesitate to ask me any questions!

Note: If you are ever worried about your grade in this course, please reach out earlier than you think you need to. I am happy to help you make a plan!

Absent from Class: If you miss a class meeting, whether excused or not, you are responsible for using the resources available to you (your peers, Drop-In Hours, the textbook, etc.) to ensure you do not fall behind on the material. If you are absent for an extended period of time, or if you miss the final exam day, please keep me in the loop. I am willing to work with you!

Make-up/Late Assignments: You will not be allowed to submit assignments late or make up missed Progress Checks. I have already built a lot of flexibility built in this course because I know things will go awry at some point. Please take care of yourself first, and use the flexibility in this course to your advantage!

Academic Integrity and External Resources: For our purposes, you may consider "cheating" to be any action that could give you an unfair advantage on a Progress Check or exam, and you may consider "plagiarism" to be any presentation of work as produced by you when it was not. **Neither of these are acceptable in a college course.**

You are likely familiar with Large Language Models such as ChatGPT. When calculators were invented, they did not make math obsolete, and the same will be true of LLMs. [In fact, it is my own knowledge of math that informs me that LLMs do not actually *know* anything, and therefore they should *never* be trusted to make decisions or answer questions!]

You should never copy solutions from an LLM or anywhere else for *any* assignment in this course—that's plagiarizing! I have no issue with you using external resources to study or practice. Just remember that math is not a spectator sport, and understanding someone (or something) else's explanation is only half the battle. **The only way to learn the material in this course is by building your own understanding!**

Also, there is flexibility and leniency built into this grading structure specifically so you can succeed! If you find yourself in a place where you're considering cheating or plagiarism, please ask me for help instead, and let's find a fair way to succeed in this course.

Equity, Diversity, and Inclusion: I have designed this course with the recognition that you all have different opinions about, background knowledge of, and strengths in math. Your diversity of perspectives, intersected with your disability status, race, color, religion, national origin, ancestry, genetic information, marital

status, familial status, sex, age, sexual orientation, veteran status, gender identity, etc., can make this course more enriching for all of us. I strive to make my class a space that is respectful, welcoming, and inclusive of every member of the class, and **you are expected to contribute to this!**

If you or someone else does not feel respected, welcomed, or included in this class, and you are not comfortable talking to me first, you can reach out to our Title IX Coordinator by emailing XXX. If you wish to maintain complete confidentiality, you may also contact our Counseling Services Office (XXX-XXX-XXXX).

Course Accommodations: If you would like to request accommodations for this course, you should contact the Accommodations Office XXX as early as possible in the term. You should also contact me so that we can work together to determine reasonable accommodations for this course. As with all other things in this course, proactive communication is key!

I wish you all a wonderful term!

List of Big Ideas and Learning Outcomes (Textbook Chapters)

1. **Graph Theory** (Chapter 2)
   (a) I can determine whether two networks are isomorphic and provide an isomorphism if they are.
   (b) I can translate between the adjacency matrix representation of a network and a drawing of the network.
   (c) I can use the adjacency matrix to calculate degrees, lengths of walks, and connectivity in a network.
2. **Random Networks** (Chapter 3)
   (a) I can calculate probabilities and statistics of Erdős-Rényi random networks, perhaps by using binominal and Poisson distributions.
   (b) I can find shortest paths between a given pair of nodes in a network.
   (c) I can compute the clustering coefficient of a node or network.
3. **Scale-Free Networks** (Chapters 4-5)
   (a) I can calculate statistics of networks built from the configuration model.
   (b) I can use the configuration model to construct networks or prove that no construction is possible.
   (c) I can calculate statistics of networks built from the Barabási-Albert model.
4. **Communities** (Chapter 9)
   (a) I can use the spectral bisection method to split a network into two communities.
   (b) I can use hierarchical clustering to create communities in a network.
   (c) I can use the Girvan-Newman algorithm to split a network into two communities.
5. **Spreading Phenomena** (Chapter 10)
   (a) I can calculate the characteristic time and basic reproductive number for an infection in a network with a homogeneous mixing assumption.
   (b) I can model how infections will spread in a network using a degree block approximation.
   (c) I can model how an infection will spread in a small temporal network.

Note: The Learning Outcomes for each Big Idea are *highly suggestive* of the questions you should expect on the Progress Checks. To decide whether you are prepared for each Big Idea, consider whether you can do the things the Learning Outcomes say you can.

Note: You can find a complete schedule with dates for all Progress Checks on Canvas!

Final Grade Tracker

Big Ideas
☐ BI 1: Graph Theory
☐ BI 2: Random Networks
☐ BI 3: Scale-Free Networks
☐ BI 4: Communities
☐ BI 5: Spreading Phenomena

Achieved Big Ideas:

1	2	3	4	5

Engagement Points:

Math Problem Sets (∼15 each) ____+____+____+____+____=_____

Programming Assignments (∼15 each) ____+____+____+____+____=_____

Practice Problem Sets (50) ☐☐☐☐☐x10=_____

Teach the Example (25 each) ____+____=_____

Total Engagement Points (mark every 5 earned)**:**

					C					**B**					**A**				
5	*5*	*5*	*5*	*5*	*5*	*5*	*5*	*5*	*5*	*5*	*5*	*5*	*5*	*5*	*5*	*5*	*5*	*5*	*5*

Final Course Grade

To earn a given letter grade, you must satisfy **all** requirements in the corresponding row.

Final Grade	Big Ideas Achieved	Applied Project	Engagement Points Earned
A	5	A-level work	100
B	4	B-level work	75
C	3	C-level work	50
D	2	D-level work	0
F	Did not meet all of the requirements for a D		

Note: If your final marks are short of an A, B, or C in any **one** category, by at most **one** grade level, you will earn the corresponding "minus" grade instead. If your final marks are above a given letter grade in the **Big Idea** or **Applied Project** category (not Engagement Points), you will earn the corresponding "plus" grade instead. Following all of these rules, you will earn the highest grade that applies.

4.2 Fall 2025: Calculus 1

MATH 0220-1025 Analytic Geometry & Calculus 1 Fall 2025

Instructor: Edison Hauptman (he/him/his) **Contact Me:** ERH129@pitt.edu
My Drop-In Hours: TBD, 4 per week
Prerequisite: Minimum math placement score of 76 or Math 0200 with minimum grade of C.
Textbook: James Stewart, Essential Calculus, Early Transcendentals, Second edition.

Meeting Times

- **Lecture:** Monday/Wednesday 7:30pm-8:45pm, Benedum Hall G29
- **Recitation:** Wednesday 6:00pm-6:50pm, check your TA's Canvas page
- **Lab:** Monday 6:00pm-6:50pm, Posvar Hall 1200A

Drop-In Hours

Are you having trouble understanding an example in the book? Would you like a place to study with your peers, where you can ask questions if you get stuck? Have you discovered something cool that you want to show me? Drop-In Hours are a great way to chat with me or your TA outside of class! All graduate instructors and TAs at Pitt will have Drop-In Hours in the Math Assistance Center (MAC) or their office every week, but you can also stop by the MAC whenever you are free!

Drop-In Hours are also a great place to find study partners! Working with classmates on this material will help you learn it faster and better. You won't have the exact same view on the course as your partners, so you'll be able to help one another fill in the gaps and advance everyone's knowledge together!

What is Calculus?

I think of Calculus as the math that answers two big questions. Take a minute to ponder each of these:

1. **The "Instantaneous" Rate of Change Paradox**: It is intuitive to think of velocity as the distance covered divided by the time it took to travel that distance. But it always takes some small amount of time to travel any distance, so how can we describe the *current* velocity of anything?

2. **The Area Problem**: We have formulas to calculate the area of simple shapes like rectangles and triangles. Is it possible to calculate the areas of more complicated shapes?

As we will discover, the tools developed to solve these questions are closely related to each other. They have also been applied to every branch of science, and in this course, we will lay the foundation to solve both!

To accomplish this, we will use techniques from Algebra, Geometry, Trigonometry, and more. However, your TA and I cannot spend lecture or recitation time reviewing these past tools. If you find yourself getting stuck on this background material, you should work to improve those skills *as soon as possible*! I am happy to use Drop-In Hours to help review these, and I will never demean you for not knowing something "basic" or "easy".

Friendly reminder: Math is hard, but YOU CAN DO IT! Some of you may think that you are not "math people." However, the vast majority of the world's population will never study calculus, and a little over 300 years ago, no one knew how to do any of this! Like it or not, if you are enrolled in this course, then you *are* "math people," and you belong here even if you don't feel fully prepared! Every single mathematician I know has struggled with math and *currently* struggles with math, because math is very hard! It will take consistent hard work to succeed in this course, but I promise you that you can! Speaking of which...

How can I succeed in this course?

Personally, I don't put much stock in grades. My main objective as your instructor is that you learn this material well enough to use it later in your life, wherever you happen to end up! If you can build a solid foundation with the derivative and the integral—the two tools I alluded to earlier—then you can develop a new lens through which you view the world. *(That might sound a bit pretentious, but I'm not kidding!)*

I also want to show you that there is so much more to math besides the memorization and pattern-matching that it's often known for. So in this course, you're going to get the opportunity to flex different mathematical muscles, such as communicating your results, resilience, creativity, and teamwork. These are all skills I use *every day* as an aspiring mathematician!

Sadly, there are other people in our lives who *do* care about your grades. So, to appease those folks, I will report your grade based on how well you do three things:

1. Master the **Big Ideas** of differential calculus (that's the fancy name for Calculus 1),
2. Achieve the associated **Learning Outcomes** for each Big Idea, and
3. Earn **Engagement Points** by doing assignments that will help *you* gain something from this material!

I will be measuring your *progress* throughout the semester. There won't be a big midterm that "mathematically eliminates" you from earning a B if you do poorly. Instead, you will have many chances to show me that you have *improved* your understanding of the course material, and whenever you do, that will be reflected in your grade! Let's see what that means:

1. Mastering the Big Ideas

Every week, we will have Progress Checks on 1 or 2 **Big Ideas**. Most Progress Checks will occur in your weekly lab session. The schedule for all Progress Checks will be posted on Canvas.

To master a Big Idea, you must answer at least 3 of its 4 questions correctly. You will have three attempts to master each Big Idea:

1st: Soon after I cover the material in lecture.

2nd: One week later, after you have feedback from your first attempt.

3rd: In the last week before finals week.

Once you master a Big Idea, you do not need to master it again!

2. Achieving the Learning Outcomes

Each Big Idea consists of 2-4 **Learning Outcomes** (for a complete list, see pages 7-8). When you master a Big Idea, you will immediately also achieve each Learning Outcome where *every answer is correct.* If you would like to achieve an additional Learning Outcome on a later Progress Check, you must **master the same Big Idea again**, with correct answers in the Learning Outcome you haven't achieved yet.

Note: **Whenever I say "correct answer," I really mean "essentially correct."** If you make a meaningless sign error or algebra error, but the rest of your work is sound, that's good enough for me! In other words, if you would have only lost a point or two on a traditional exam question, you will be marked "essentially correct". What I really care about in these Progress Checks is whether you understand the main ideas in each question. Whenever it's important enough to give a perfect answer, you'll be able to double-check your notes.

Example: Big Ideas and Learning Outcomes

Let's look at **Big Idea 4**, which has two Learning Outcomes (**D7** and **A1**; see the appendix):

- On Alice's first attempt, she answers the two D7 questions correctly, along with one of the two A1 questions. Since Alice has 3 questions correct, she has mastered **Big Idea 4** and achieved Learning Outcome **D7**.
- On Alice's second attempt, she answers the two A1 questions correctly, but not the two D7 questions. Since Alice has only 2 questions correct, she has not achieved another Learning Outcome. However, her previous mastery of **Big Idea 4** and achievement of Learning Outcome **D7** are not affected.
- On Bob's first attempt, he answers the two **D7** questions correctly, but not the two **A1** questions. Since Bob only has 2 questions correct, he has not mastered the Big Idea, and therefore, he has not achieved any of the Learning Outcomes yet.

3. Earning Engagement Points

Besides the weekly Progress Checks, there are many other assignments in this course which will earn you **Engagement Points**. **I do not expect you to complete *all* of these assignments.** You should choose the assignments that will help you with your own learning, and focus on those!

Calculus Readiness Bootcamp (20 pts): For extra practice with Algebra, Trigonomtry, and Precalculus, you can complete a free resource offered by our textbook publisher, Cengage. See Canvas for more information.

Extra Learning Outcomes (20 pts each): I'll count each Learning Outcome you achieve *above* a given grade threshold as 20 bonus Engagement Points, in whatever way gives you the highest grade.

Final Exam (200 pts): In Pitt's math department, all students in day sections of Calculus must take a common exam during Final Exam Week. Since this is a night section, I will write my own final exam in a similar style, suited for our lecture. The final exam is scheduled for **Wednesday, December 10th, from 6-7:50pm**.

LON-CAPA Problem Sets (150 pts): LON-CAPA is the computer system where you will complete online homework. Each of you will have different randomly-generated numbers, but the problems are the same across all Calculus 1 courses. LON-CAPA can be accessed through my.pitt.edu or Canvas. In most weeks, a problem set will be due on Thursday night at 11:59pm.

If you want to access LON-CAPA off campus, you must download Global Protect, which is a Virtual Private Network (VPN). Your TA will show you how to set up LON-CAPA and Global Protect in your first lab session. If you have any trouble using LON-CAPA, let me or your TA know!

Practice Problem Sets (5 pts each, 1/week): For each section we cover, there are recommended textbook problems. If you submit solutions to a few of these in a given week (check Canvas for how many), you will earn 5 points. There are many outside resources with practice problems too; you may use these with prior approval.

Top Hat Tally (∼60 pts): In most lectures, I will ask a few quick questions which you can respond to anonymously using the Top Hat app, for 1 point per response. Active engagement in the lecture will be more difficult than passively listening to me, but it's better for your learning!

Recitation Check In's (5 pts each, 1/week): In the first five minutes of each recitation, your TA will have you reflect on the course material with a partner. This will give you a chance to think about your progress in the course and prepare questions for your TA.

A New Perspective (20 pts each): The Calculus 1 course at Pitt typically has a wide diversity of students across majors and mathematical backgrounds. Therefore, for any topic your TA or I may cover in class, you may have a different example or explanation that helps you or a classmate understand it better. If you want to discuss one of these with me or your TA in some detail (ideally in Drop-In Hours), we can give your points for your independent exploration and thought!

Group Excursions (20 pts each, ∼4-6 total): My lectures will often give a broad overview of the course material, and your TA's recitations will often focus on problem-solving strategies. The Group Excursions will be a change of pace, where you will work in small teams and take a deeper dive into a specific part of Calculus.

In my opinion, this assignment is the soul of this course! In Pitt's math department, Calculus 1 is a highly structured and coordinated course, and we do not have the full freedom to explore this subject however we wish. **In these Group Excursions, we get to take some steps off the beaten path, and I want *you* to choose where we go!** I would be happy to explore any topic in our curriculum (or anything *just* outside of it) in greater detail, so you should let me and your TA know what you're interested in! Group Excursions may occur in lectures or recitations, and I'd like for us to complete 4-6 of them.

Typical Weekly Schedule

Here is an outline of all the regularly-occurring events and assignments in this course:

Monday	Tuesday	Wednesday	Thursday	Friday
Practice Problem Sets (*due at 6pm*)				
Lab at 6pm (*1-2 Progress Checks*)		Recitation at 6pm (*Recitation Check In*)		
Lecture at 7:30pm (*Top Hat Tally*)		Lecture at 7:30pm (*Top Hat Tally*)		
			LON-CAPA (*due at 11:59pm*)	

Final Grade

In order to earn a given letter grade, you must satisfy **all** of the requirements in the corresponding row.

Final Grade	Big Ideas Mastered (/9)	Learning Outcomes Achieved (/26)	Engagement Points Earned
A+ (4.0)	9	22	550
A (4.0)	9	22	500
A-	8	21	500
B+	8	20	450
B	8	19	450
B-	7	18	450
C+	7	18	400
C	7	17	400
C-	6	16	400
D+	6	16	350
D	6	15	350
D-	5	14	350
F	Did not meet all of the requirements for a D-		

Other Course Policies

Canvas: Most of this course, including announcements, resources, due dates, and assignments, will be on our lecture's Canvas page. You are responsible for checking Canvas often. Your TA may have additional information on your recitation's Canvas page as well. You will also be able to see scores for all assignments on Canvas. However, Canvas is not compatible with this course's grading system, so **your overall grade will *never* appear on Canvas!**

Keeping Track of Your Grade: I will hand out a colored sheet of paper on the first day of class (see page 9). You may also want to track your progress digitally—if so, you can download and fill out a Microsoft Excel spreadsheet which is based on my own grade tracker. Copies of both of these sheets will be on our Canvas page.

If you have never taken a course like this one before, then this grading system may feel intimidating or complicated. That's okay! **Just like Calculus, if this grading system new to you, then it will take some time to learn, and I don't expect you to understand it immediately!** Throughout the term, I will dedicate class time to discussing the grading system, but please don't hesitate to ask me or your TA any questions!

If you are ever worried about your grade in this course, please reach out earlier than you think you need to. I am happy to help you make a plan!

Absent from Class: If you miss a class meeting, whether excused or not, you are responsible for using the resources available to you (your peers, Drop-In Hours, the textbook, etc.) to ensure you do not fall behind on the material. If you are absent for an extended period of time, or if you miss the final exam day, please keep me in the loop. I am willing to work with you!

Make-up/Late Assignments: Since you will have three attempts at each Big Idea in this course, **you will not be able to make up missed Progress Checks**. The only assignment that can be made up or submitted late are the Practice Problem Sets (up to 24 hours). There is a lot of flexibility in this course because I know some things will go wrong. Please take care of yourself and use the flexibility in this course to your advantage!

Textbook: You were automatically opted in to receive an electronic version of the textbook through our Canvas page. If you do not want an electronic copy, you *must* opt out early in the term. You will receive an email with the subject line, "Access your MATH 0220-10771 Materials," which will have more information on this. **Note that the course textbook is *not* required to complete any assignment in this class**.

Calculators: Calculators are not permitted on exams or Progress Checks (and are not very useful in most of our assignments), but you may benefit from having a scientific calculator handy for some parts of the course. Unless I specifically tell you to do so, you will not have to simplify your answers on Progress Checks or exams!

Academic Integrity and External Resources: For our purposes, you may consider "cheating" to be any action that could give you an unfair advantage on a Progress Check or exam, and you may consider "plagiarism" to be any presentation of work as produced by you when it was not. **Neither of these are acceptable in a college course.**

You are likely familiar with Large Language Models such as ChatGPT. When calculators were invented, they did not make math obsolete, and the same will be true of LLMs. [In fact, it is my own knowledge of Calculus that informs me that LLMs do not actually *know* anything, and therefore they should never be trusted to make decisions!]

You should never copy solutions from an LLM or anywhere else for *any* assignment in this course—that's plagiarizing! It is fine with me if you are using external resources to study or practice. Just remember that math is not a spectator sport, and understanding someone else's explanation is only half the battle. The only way to achieve the Learning Outcomes in this course is if you build your own understanding.

Also, there is flexibility and leniency built into this grading structure specifically so you can succeed! If you find yourself in a place where you're considering cheating or plagiarism, please talk to me or your TA instead, and let us help you find a fair way to succeed in this course.

Equity, Diversity, and Inclusion: I have designed this course with the recognition that you all have different opinions about, background knowledge of, and strengths in math. Your diversity of perspectives, intersected with your disability status, race, color, religion, national origin, ancestry, genetic information, marital status, familial status, sex, age, sexual orientation, veteran status, gender identity, etc., can make this course more enriching for all of us. I strive to make my class a space that is respectful, welcoming, and inclusive of every member of the class, and you are expected to contribute to this!

If you or someone else does not feel respected, welcomed, or included in this class, and you are not comfortable talking to me first, you can reach out to Pitt's Title IX Coordinator by calling 412-648-7860, or emailing titleixcoordinator@pitt.edu. Reports can also be filed online. If you wish to maintain complete confidentiality, you may also contact the University Counseling Center (412-648-7930).

Disability Resource Services: If you have a disability for which you are or may be requesting an accommodation, you are encouraged to contact me *and* Disability Resources and Services (DRS), 140 William Pitt Union (412) 648-7890, drerecp@pitt.edu, (412) 228-5347 for P3 ASL users, as early as possible in the term. DRS will verify your disability and determine reasonable accommodations for this course. As with all other things in this course, proactive communication is key!

I wish you all a wonderful semester!

Appendix: List of Big Ideas and Learning Outcomes (Textbook Sections)

1. Limits and Continuity

- **L1** I can compute the limit of a function at a point using its graph. (1.3)
- **L2** I can compute the limit of a function at a point using algebraic techniques. (1.4)
- **L3** I can use the definition of continuity to determine whether functions are continuous. If a function is not continuous, I can identify the type of discontinuity. (1.5)
- **L4** I can determine the end behavior of functions by computing limits "at" infinity. (1.6)

2. Definition of the Derivative

- **L5** I can use the limit definition of the derivative to compute derivatives of simple functions. (2.1)
- **D1** I can determine where a function is differentiable using the limit definition of the derivative or the graph of the function. (2.2)
- **D2** I can find the equation of the tangent line to a graph at a given point. (2.2)

3. Derivative Rules

- **D3** I can use the power rule to compute derivatives of algebraic functions (polynomials and roots). (2.3)
- **D4** I can use derivatives of sine and cosine functions to compute derivatives of other trig functions. (2.3)
- **D5** I can compute derivatives of products or quotients of two or more functions. (2.4)
- **D6** I can compute derivatives of nested functions using the chain rule. (2.5)

4. Related Rates of Change

- **D7** I can compute the equation of a tangent line to an implicit curve. (2.6)
- **A1** I can use implicit differentiation to compute related rates of change in model problems. (2.7)

5. Exponentials, Logarithms, and Inverses

- **D8** I can compute derivatives of exponential and logarithmic functions. (3.1-3.3)
- **D9** I can use logarithm rules to compute derivatives efficiently. (3.1-3.3)
- **D10** I can compute derivatives involving inverse trig functions using implicit differentiation. (3.5)

6. Shape of Graphs

- **A2** I can use information about the first derivative of a function to determine where it is increasing or decreasing. (4.2-4.3)
- **A3** I can use information about the second derivative of a function to determine whether it is concave up or concave down. (4.2-4.3)
- **A4** I can calculate limits using L'Hospital's Rule. (3.7)
- **A5** I can approximately graph any function using information about its intercepts, asymptotes, and first and second derivatives. (4.4)

7. Optimization

A6 I can use the first and second derivatives of a function to identify its local and global maximia/minima on a given domain. (4.1)

A7 I can find optimal solutions to model problems using single-variable derivatives. (4.5)

8. The Integral as Area

I1 I can use Riemann sums to approximate the area underneath a curve. (5.1)

I2 I can compute exact values of definite integrals using properties of area. (5.2-5.3)

9. The Fundamental Theorem of Calculus

I3 I can use antiderivative formulas to compute antiderivatives of functions. (4.7)

I4 I can compute derivatives and integrals using the Fundamental Theorem of Calculus. (5.3-5.4)

Bonus: Integration Techniques*

I5* I can use the u-substitution technique to compute relevant integrals. (5.5)

I6* I can use the integration by parts technique to compute relevant integrals. (6.1)

I7* I can compute integrals of products of trig functions using trig identities and u-substitution. (6.2)

*Since we will learn integration techniques very late in the semester, there will only be two Progress Checks where you can master this Big Idea and achieve the corresponding Learning Outcomes. So, while there are technically 10 Big Ideas and 29 Learning Outcomes in total, you are only officially responsible for 9 Big Ideas and 26 Learning Outcomes.

You can find a complete schedule with dates for all Progress Checks on Canvas.

Appendix: Final Grade Tracker

Big Ideas	Learning Outcomes			
☐ BI 1	☐ L1	☐ L2	☐ L3	☐ L4
☐ BI 2	☐ L5	☐ D1	☐ D2	
☐ BI 3	☐ D3	☐ D4	☐ D5	☐ D6
☐ BI 4	☐ D7	☐ A1		
☐ BI 5	☐ D8	☐ D9	☐ D10	
☐ BI 6	☐ A2	☐ A3	☐ A4	☐ A5
☐ BI 7	☐ A6	☐ A7		
☐ BI 8	☐ I1	☐ I2		
☐ BI 9	☐ I3	☐ I4		
☐ BI B	☐ I5	☐ I6	☐ I7	

Total Big Ideas:

1	2	3	4	5	6	7	8	9	10

Total Learning Outcomes:

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	

Total Engagement Points:

Calculus Readiness Bootcamp (20) ☐ Final Exam (200) _____

LON-CAPA Problem Sets (150) _____ Extra LOs (20 each) _____

Top Hat Tally (∼60) _____ Bonuses _____

A New Perspective (20 each) ☐☐☐x20=_____ **Total** _____

Group Excursions (20 each) ____+____+____+____+____+____=_____

Practice Problem Sets (∼60) ☐☐☐☐☐☐☐☐☐☐☐☐x5=_____

Recitation Check In's (∼60) ☐☐☐☐☐☐☐☐☐☐☐☐x5=_____

Final Grade	Big Ideas Mastered (/9)	Learning Outcomes Achieved (/26)	Engagement Points Earned
A+ (4.0)	9	22	550
A (4.0)	9	22	500
A-	8	21	500
B+	8	20	450
B	8	19	450
B-	7	18	450
C+	7	18	400
C	7	17	400
C-	6	16	400
D+	6	16	350
D	6	15	350
D-	5	14	350
F	Did not meet all of the requirements for a D-		

Rationale/Reflection (Calculus 1 Syllabus)

This is the syllabus from one of my two Fall 2025 Calculus 1 courses. There are two separate aspects to the course grade: the required Learning Outcomes and the extra Engagement Points. The Learning Outcomes are the core of the course. Students can earn Engagement Points through their study habits for the Learning Outcomes and/or through exploring the subject further. In previous courses, which were offered during the summer term to at most 25 students, students had more options in the latter category, such as completing projects and presenting practice problems to their peers. This was the first term I taught courses with a large enrollment cap (75), and I had to coordinate with 3 TAs for each section. It was also my first semester teaching Calculus 1, and I wanted to make sure I did not stretch myself too thin before adding additional assignments.

The course structure is also well-coordinated with the other Calculus 1 sections at Pitt. I covered all the same material so that my students were prepared for the department-wide final exam. By focusing on the Learning Outcomes, I am supporting my department's desire to measure learner competencies.

I had a few specific objectives for this syllabus:

- **Communicate why I think this course is important, hopefully in a way that makes my students think it's important.** Without student buy-in, none of my extra effort would matter.
- **Explain the rationale for my atypical course structure and why it's worth the trouble for my students to learn a new system.** I also try to make the course feel as straightforward as possible. I've learned that including a lot of structure, such as a consistent weekly schedule, is an enormous benefit in a course with a lot of options.
- **Give my students a growth mindset and the confidence to keep trying to master this material, even when it gets hard.** Many of my students are first-years in their first semester of college, and this course can set the pace for their entire college career.
- **Take some of the coordination pressure off of my TAs.** This course is an experiment, and while I believe that an ideal education in Calculus is far more personalized than this, I want my TAs to be juggling as little as possible while I figure out how to work with many different people in a new course structure.

In previous courses (Calculus 2), I exclusively used a points system, but I learned that a standards-based grading system is more appropriate for mathematics, and it scales more effectively in a large-enrollment course. In past semesters, I'd used a similar assessment structure, but I had not explicitly tied those assessments to Learning Outcomes like I have done here. I used my knowledge of the large space of alternative grading practices to create a course structure attuned to my, my students', my TAs', and my department's needs.

I learned several lessons from this course:

- **Requiring students to achieve the Big Idea before they could earn individual learning outcomes was unfairly harsh.** At the time, my mindset was that since I've grouped my standards across similar mathematical concepts, students should demonstrate an understanding of the overall picture before I am confident that they earned the Learning Outcomes. However, even if the mathematical concepts are similar, I did not respect the difficulties students could have with the problem-solving aspects of the questions, even after

understanding the basic mathematical concept. In future terms, I simplified my model to focus on the overall Big Ideas, though I still include individual Learning Outcomes as a guideline for studying.

- **The final exam should be tied to the standards in my course, but I don't necessarily have control over what the final exam looks like.** Towards the end of the course, many students asked whether the final exam could be used to achieve a Big Idea they had missed. Making this promise resulted in an extra round of grading when the department-wide final exam wasn't arranged this way.
- **My Engagement Points system was successful because it was flexible.** I speak to this in more detail in a post I wrote for the Grading for Growth blog. In this course, group assignments were less successful at engaging the students than I'd hoped, so I decided to run fewer of these assignments than I'd planned, and this had no great impact on the course.

5 Sample Assignments

5.1 Group Work: The Yellow Light Dilemma

Math 0230 Summer 2024, Section 1015
Team Challenge 3: The Yellow Light Dilemma (25 Points)

Instructions: Form groups of 2 to 3 students, with one worksheet per group. These challenge questions are meant to be solved as a group. If you can solve a challenge problem very quickly, then you should work with your group-mates. For these challenge problems, the process is much more important than the answer, because there will be multiple promising strategies/ideas. If you need a hint, raise your hand to ask for help! **Submit this to Canvas, then fill out an Individual Reflection Form on Canvas before Thursday.**

According to the Pennsylvania Driver's Manual, "if you are driving toward an intersection and a yellow light appears, slow down and prepare to stop. If you are within the intersection or cannot stop safely before entering the intersection, continue through carefully." In their 1959 paper, "The Problem of the Amber Signal Light in Traffic Flow," researchers Gazis, Herman, and Maradudin (GHM) analyze what can happen if a yellow light changes to red too quickly, making it impossible to either

1. **Stop** before the intersection, or
2. **Continue** through the intersection before the light turns red.

In this Team Challenge, we will use differential equations to derive Equations (1) and (2) in GHM's paper. If we do the additional algebra they did, we would see that if a yellow light changes to red too quickly (we'll call this the **yellow change interval**), then there will be spots on the road where a driver cannot do either of the options above!

The US Federal Highway Administration recommends that the yellow change interval is between 3 and 6 seconds, and says that the change interval should be determined according to "engineering principles." However, when some cities have installed red light cameras (used to fine drivers who are in the intersection when the light is red), they've shortened yellow change intervals at those intersections to make more money from fines. This concern was raised in Pittsburgh when a 2013 bill was passed to trial-run red light cameras.

Variables/Parameters:

- Let $\boldsymbol{t}$ be the time. Respectively, $\boldsymbol{y(t)}$, $\boldsymbol{y'(t)}$, and $\boldsymbol{y''(t)}$ are the position, velocity, and acceleration at time t. Let $\boldsymbol{\tau}$ be the duration of the yellow change interval for the traffic light.
- Let $\boldsymbol{a}$ be the (positive) deceleration if the driver decides to stop. If the driver decides to continue through the light, it is not safe to accelerate, so we'll assume acceleration is 0 in this case.
- Let $\boldsymbol{v_0}$ be the initial velocity of the driver before seeing the yellow light.
- Let $\boldsymbol{\delta}$ be the reaction time plus decision time the driver needs to choose one of the two options.
- Let $\boldsymbol{x}$ be the distance to the intersection, let $\boldsymbol{w}$ be the width of the intersection, and let $\boldsymbol{L}$ be the length of the vehicle. **We will assume the initial position of the driver is 0 at time $\boldsymbol{t = 0}$.**

Use this page as a reference, and write your solutions to the exercises on the other piece of paper.
If this topic interests you, I'd be happy to work with a group to extend this into an independent project! I will post GHM's paper and some of the sources on the reverse side of this page to Canvas.

References:

- D. Gazis, R. Herman, A. Maradudin: "The Problem of the Amber Signal Light in Traffic Flow", November 1959.
- J. Beeber, "Report to CTCDC on Minimum Yellow Light Change Interval Timing for Signalized Intersections", submitted for the California Traffic Control Devices Committee Meeting, February 2018.
- Pennsylvania Department of Transportation, "Pennsylvania Driver's Manual" (Chapter 2, page 7), 2024.
- "Short Yellow Lights Project" (https://www.shortyellowlights.com/), accessed June 2024.
- Van Osdol, "Plan to bring red-light cameras to Pittsburgh has supporters, critics", WTAE article, July 2013.
- Federal Highway Administration, "Manual on Uniform Traffic Control Devices, 11th Edition" (Section 4F.17, page 708), December 2023.

Math 0230 Summer 2024, Section 1015
Team Challenge 3: The Yellow Light Dilemma (25 Points)

NAMES: _____

_____ **DATE:** _____

1. If a driver opts to stop before the intersection, we should make sure that when their velocity reaches 0, they haven't crossed into the intersection. That is, when $y'(t) = 0$, we want to have $y(t) \leq x$.

   (a) Set up a differential equation with $y''(t)$ to model the driver deciding to stop at the light.

   (b) What is the velocity of the driver when $t = \delta$? Use this to write down an initial condition.

   (c) What is the position of the driver when $t = \delta$? Use this to write down an initial condition.

   (d) Find the solution to the differential equation you've made, using the initial conditions.

   (e) **Bonus:** Use algebra to show that when $y'(t) = 0$, we want to have $x - v_0\delta \geq \frac{v_0^2}{2a_2}$, which is Equation (1) of GHM's paper *(come back to this at the end).*

2. If a driver opts to continue through the intersection, we should make sure that when the light changes to red, they have completely passed through the intersection.

   (a) Write down the meaning of the phrase above using equations, as I did for you in question 1.

   (b) Set up a differential equation with $y''(t)$ to model the driver continuing through the light.

   (c) Write down the initial conditions for the position and velocity of the driver at time $t = \delta$. They are same as for questions 1b and 1c—explain why this is the case.

   (d) Find the solution to the differential equation you've made, using the initial conditions.

   (e) **Bonus:** Use algebra and the equations in part (a) to show that we want $x + w + L - v_0\delta \leq v_0(\tau - \delta)$, which is equivalent to Equation (2) of GHM's paper *(come back to this at the end).*

3. For what values of t are the differential equations you've solved in questions 1 and 2 valid?

4. In words, how would you interpret the left-hand side of Equations (1) and (2)? Use this interpretation to explain why GHM might have written Equations (1) and (2) the way they did.

Rationale/Reflection:

In my Summer 2024 course, I gave five Team Challenges. Most of these could have been a student's project in another context, and I was separately encouraging my students to complete projects of their choosing. So, I chose to build the Team Challenges from my own interests. In my Calculus 1 course, which does not have projects, the Group Excursions that my students will complete in teams will be built from my students' interests.

There are some application problems in Pitt's Calculus 2 course that are covered every semester and appear on the common homework and final exam, and they are widely disliked by students (and TAs). Sometimes, when instructors answer the age-old question, "When will I ever use this?", with practical model problems, our students become even *less* interested.

In contrast to introducing a model problem for the sake of the math, so that it can appear on an exam later, I motivate the model problem on its own merits, and I do not bring this model problem back on an exam. I deliberately include my references and discuss the problem in a Pittsburgh context so that students can see that, even if they couldn't care less about this application, some people care a lot about it, and they care for sensible reasons!

Those sensible reasons that people care about this problem reinforce some of my philosophy of math, which I talk about in class: math can be political! In this case, we can use math to advocate for a policy choice. The introduction to this worksheet also suggests a follow-up question: if the current policy does not follow sound mathematical principles, what *is* driving the policy? I think of this as the most important part of the assignment, because it unambiguously breaks the artificial barrier between our classroom and the world outside of it.

5.2 Project: "When Will I Ever Use This?"

Below, I include the instructions from my College Algebra 1 course's assignment from the Fall 2026 semester.

The "When Will I Ever Use This?" assignment is a 5-20 minute live discussion between you and me on a topic of your interest, which incorporates ideas from at least one Big Idea in our course.

To successfully complete this assignment, you should do all of the following:

- Discuss a mathematical application which was not specifically covered in a lecture or recitation (it should be **new** to our course).
- Discuss your personal motivation for or interest in your application.
- Demonstrate clear understanding of the outside-of-math concepts related to your application, if you introduce any.
- Do not read previous notes on your application. You are welcome to **reference** notes.
- Give a detailed mathematical argument or presentation, which is your own work, without major errors, and which is connected to your topic of interest and the material in our course.
  - *If you can only make **vague** mathematical connections from our course to your topic of interest ("this idea reminded me of quadratic equations"), you will not complete the assignment successfully. However, I encourage you to discuss with me/your TA anyway, so that we can give you advice on making a more specific connection. Then, you can take that feedback into account to complete this assignment on a later day.*
  - *If you can make **specific** mathematical connections from our course to your topic of interest ("this idea reminded me of how we model some data using quadratic equations, so maybe we can approximate the relative maximum using the data"), you will be more likely to complete the assignment successfully. Even if you do not have all of the details, I can help you fill in a few gaps along the way.*
- Lead the discussion. That is, the discussion should be centered on your own perspective and insights, even I ask clarifying questions or mention related ideas.

Each time you complete this assignment, you will earn 40 Engagement Points. Your overall score will be posted at the end of the term. Keep track of how many times you have successfully completed this assignment in your Final Grade Tracker.

Note that I will have limited availability during the first week of December to facilitate these assignments, and all of these assignments must be completed by Friday, December 4, at 5pm. You should not plan to complete more than one "When Will I Ever Use This?" assignment in a week.

Rationale/Reflection:

When the course is too large or dense for me to assign a formal project to students, I give students the option to complete a mini-project. I include the instructions from my College Algebra 1 course from the Fall 2026 semester, where many of my students are not STEM majors and some are scared of math.

Because my primary objective in an intro-level course is for students to demonstrate understanding of the Big Ideas of the course, I do not want to include a required project as a distracting element. So, this assignment is one of several that can be used to earn Engagement Points in the course. If completing a project will help a student learn the course material, I want them to be able to complete a project!

I speak about my Engagement Points system in great detail in a blog post I wrote for the Grading for Growth Blog: Supplementing Standards with Engagement Points.

5.3 Take-Home Assignment: Extra Practice Problem Sets

Math 0120 Worksheet 6

Practice Problems for Big Idea 5: Graph Analysis

Learning Outcomes:

(a) I can determine where a function is increasing/decreasing, as well as the locations of any local minima/maxima, perhaps based on a relevant graph.

(b) I can determine where a function is concave up/down, as well as the locations of any inflection points, perhaps based on a relevant graph.

(c) I can find the x- and y-intercepts, vertical asymptotes, and horizontal asymptotes of a function.

(d) I can sketch the graph of a function given some or all of the above information.

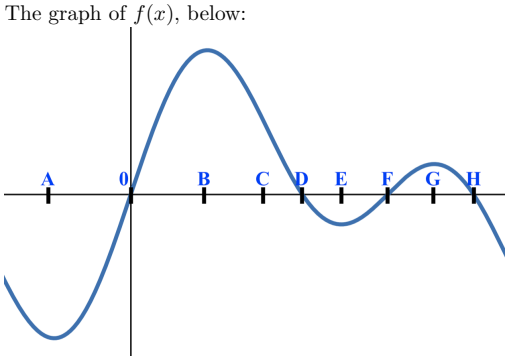
Instructor: Edison Hauptman

All items marked with ✪ are challenge questions, or are otherwise outside of the scope of the Progress Checks.

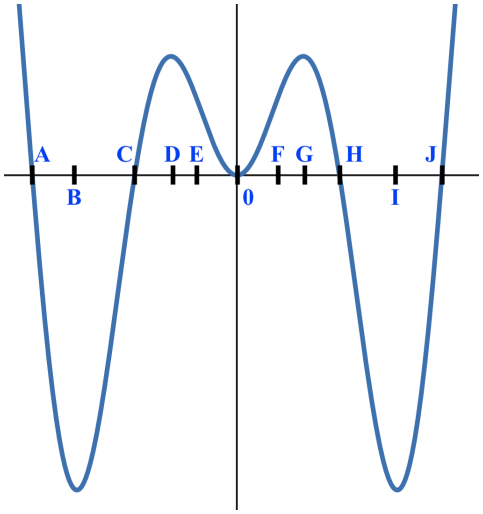
For questions 1-8, use the following graphs to:

- Determine where the function is increasing/decreasing,
- Identify the locations of any minima/maxima,
- Determine where the function is concave up/concave down,
- Identify the locations of any inflection points.

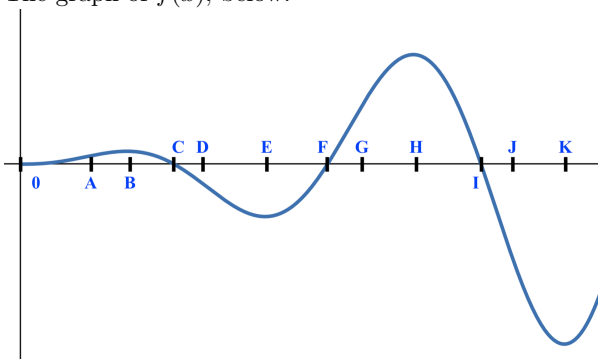
1. The graph of $f(x)$, below:



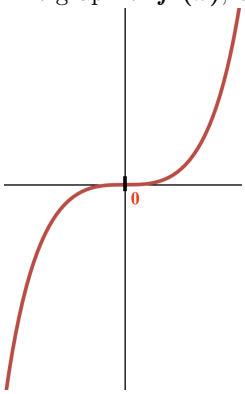
2. The graph of $f(x)$, below:



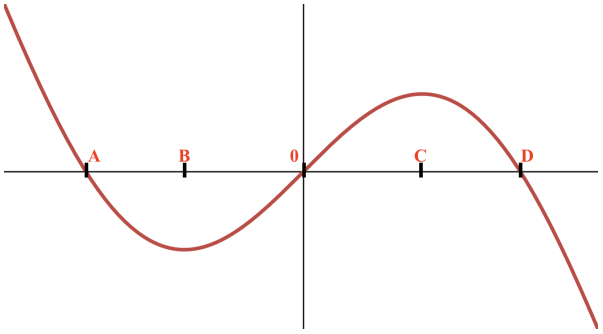
3. The graph of $f(x)$, below:



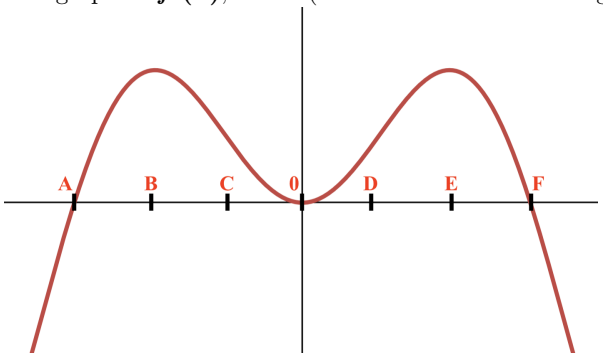
4. The graph of $\boldsymbol{f'(x)}$, below (note that this is **not** the graph of $f(x)$):



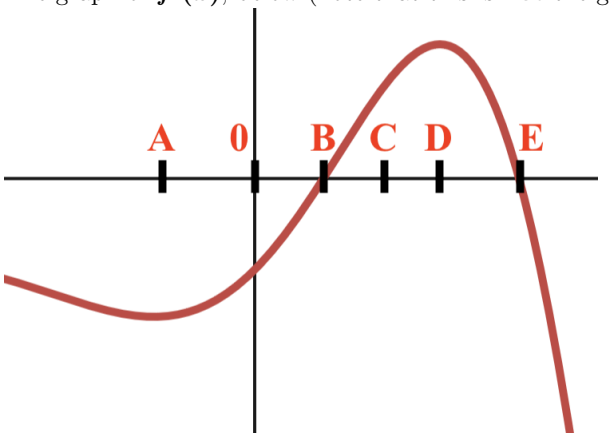
5. The graph of $\boldsymbol{f'(x)}$, below (note that this is **not** the graph of $f(x)$):



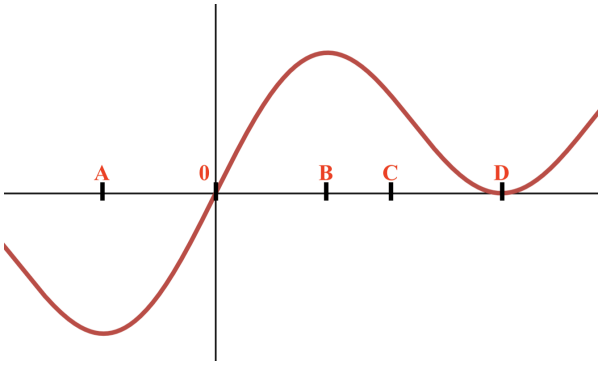
6. The graph of $\boldsymbol{f'(x)}$, below (note that this is **not** the graph of $f(x)$):



7. The graph of $\boldsymbol{f'(x)}$, below (note that this is **not** the graph of $f(x)$):



8. The graph of $\boldsymbol{f'(x)}$, below (note that this is **not** the graph of $f(x)$):



For questions 9-30:

- Determine where the function is increasing/decreasing,
- Identify the locations of any minima/maxima,
- Determine where the function is concave up/concave down,
- Identify the locations of any inflection points.

Note that the first and second derivatives would be given to you in a reasonable form on a Progress Check. You may use a derivative calculator or some other Computer Algebra System (do **not** use AI) to calculate and simplify the first and second derivatives of these functions. Or, you may calculate and simplify the derivatives yourself for some additional practice. For items marked with ♣, I recommended that you use a derivative calculator to calculate the first two derivatives.

9. $f(x) = x^3 - x$
10. $f(x) = x^2 - 6x + 5$
11. $f(x) = x^3 + 3x$
12. $f(x) = x^4 - 4x^3$
13. $f(x) = 2x^3 - 6x^2 + 6x - 18$
14. $f(x) = e^{-2x}$
15. $f(x) = \ln(3x)$
16. $f(x) = \dfrac{4}{x-5}$
17. ♣ $f(x) = \dfrac{2}{x^2 - 2x + 1}$
18. ♣ $f(x) = \dfrac{-3}{x^2 + 8x + 15}$

19. $f(x) = \dfrac{1}{1+x^2}$
20. $f(x) = \dfrac{2+x}{3-x}$
21. $f(x) = \sqrt{x+1} - \sqrt{x-1}$
22. $f(x) = xe^x$
23. $f(x) = xe^{-x/2}$
24. $f(x) = \dfrac{e^x}{x}$
25. $f(x) = e^{3x} - e^x$
26. ♣ $f(x) = \dfrac{x^2}{x-1}$
27. ♣ $f(x) = \dfrac{1}{\sqrt{x^2+4}}$
28. ♣ $f(x) = \dfrac{1}{x^2-1}$
29. ♣ $f(x) = \dfrac{\sqrt{x^2+1}}{x}$
30. ♣ $f(x) = \dfrac{\sqrt{x}}{x+1}$

For questions 31-52, find all x and y-intercepts, along with all vertical and horizontal asymptotes, for the following functions:

31. $f(x) = x^3 - x$
32. $f(x) = x^2 - 6x + 5$
33. $f(x) = x^3 + 3x$
34. $f(x) = x^4 - 4x^3$
35. ✪ $f(x) = 2x^3 - 6x^2 + 6x - 18$
36. $f(x) = e^{-2x}$
37. $f(x) = \ln(3x)$
38. $f(x) = \dfrac{4}{x-5}$
39. $f(x) = \dfrac{2}{x^2 - 2x + 1}$
40. $f(x) = \dfrac{-3}{x^2 + 8x + 15}$

41. $f(x) = \dfrac{1}{1+x^2}$

42. $f(x) = \dfrac{2+x}{3-x}$

43. $f(x) = \sqrt{x+1} - \sqrt{x-1}$

44. ✪ $f(x) = xe^x$

45. ✪ $f(x) = xe^{-x/2}$

46. ✪ $f(x) = \dfrac{e^x}{x}$

47. ✪ $f(x) = e^{3x} - e^x$

48. $f(x) = \dfrac{x^2}{x-1}$

49. $f(x) = \dfrac{1}{\sqrt{x^2+4}}$

50. $f(x) = \dfrac{1}{x^2-1}$

51. ✪ $f(x) = \dfrac{\sqrt{x^2+1}}{x}$

52. $f(x) = \dfrac{\sqrt{x}}{x+1}$

For questions 53-65, draw sketches of functions with the given properties. Label all relevant values.

53.
- $f'(x) > 0$ on $(-\infty, \infty)$,
- $f''(x) > 0$ on $(-\infty, 0)$ and $f''(x) < 0$ on $(0, \infty)$,
- $f(x)$ does not have a vertical asymptote.

54.
- $f'(x) > 0$ on $(-\infty, -2) \cup (5, \infty)$ and $f'(x) < 0$ on $(-2, 5)$,
- $f''(x) < 0$ on $(-\infty, 1) \cup (7, \infty)$ and $f''(x) > 0$ on $(1, 7)$.

55.
- $f(x)$ is defined on $[-1, \infty)$,
- $f(x)$ has intercepts at $(-1, 0)$, $(0, -1)$, and $(3, 0)$,
- $f(x)$ has a local minimum at $x = 1$,
- $f(x)$ is concave up on $(-1, \infty)$.

56.
- $f'(x) > 0$ on $(-\infty, 0) \cup (0, \infty)$,
- $f''(x) > 0$ on $(-\infty, 0)$ and $f''(x) < 0$ on $(0, \infty)$,
- $f(x)$ has a vertical asymptote at $x = 0$,

- As $x \to -\infty$, $f(x)$ has a horizontal asymptote at $y = 1$, and as $x \to \infty$, there is no horizontal asymptote.

57.
- As x goes to $-\infty$, $f(x)$ has a horizontal asymptote at $y = 0$, and as x goes to $+\infty$, $f(x)$ has a horizontal asymptote at $y = 0$,
- The only intercept for $f(x)$ is at $(0, 0)$,
- $f(x)$ is decreasing on $(-\infty, -1) \cup (1, \infty)$, and $f(x)$ is increasing on $(-1, 1)$,
- $f(x)$ has inflection points at $x = -2$ and $x = 3$, and nowhere else.

58.
- $f(x)$ is increasing on $(1, 6) \cup (6, \infty)$ and $f(x)$ is decreasing on $(-\infty, 1)$,
- There is a local minimum at $x = 1$,
- $f(x)$ is concave up on $(-\infty, 3) \cup (6, \infty)$ and possibly elsewhere,
- There is an inflection point at $x = 6$,
- As $x \to -\infty$, $f(x)$ has a horizontal asymptote at $y = 0$.

59.
- $f'(x) > 0$ on $(2, 4)$ and possibly elsewhere,
- The only local minimum is at $x = 2$ and the only local maximum is at $x = 4$,
- $f''(x) < 0$ on $(-\infty, 0)$ and $f''(x) > 0$ on $(6, \infty)$,
- There are inflection points at $x = 0$ and $x = 6$,
- As $x \to -\infty$, $f(x)$ has a horizontal asymptote at $y = 5$, and as $x \to \infty$, $f(x)$ has a horizontal asymptote at $y = -5$.

60.
- $f(x) \geq 0$ for all values of x.
- $f(x)$ is decreasing on $(-\infty, 0)$ and possibly elsewhere,
- $f(x)$ is concave down on $(-\infty, -1)$ and $f(x)$ is concave up on $(-1, \infty)$,
- There is a local minimum at the point $(0, 0)$,
- As $x \to -\infty$, there is no horizontal asymptote.

61.
- $f(x)$ is non-negative for all values of x.
- $f(x)$ has intercepts at $(0, 1)$, $(1, 0)$, and $(4, 0)$.
- $f(x)$ is concave up on $(-1, 2) \cup (3, \infty)$ and $f(x)$ is concave down on $(-\infty, -1) \cup (2, 3)$.

62.
- $f(x)$ has vertical asymptotes at $x = -4$, $x = 1$, and $x = 3$,
- As x goes to $-\infty$, $f(x)$ has a horizontal asymptote at $y = 0$, and as x goes to $+\infty$, $f(x)$ has a horizontal asymptote at $y = 1$,

- $f'(x) > 0$ on $(0,1) \cup (1,2)$ and $f'(x) < 0$ on $(-\infty,-4) \cup (-4,0) \cup (2,3) \cup (3,\infty)$,
- $f(x)$ has no x-intercepts.

63.
- $f'(x) = 0$ at $x = -3$, $x = 3$, and $x = 6$, and nowhere else,
- $f''(x) > 0$ on $(-\infty,2) \cup (4,6)$ and $f''(x) < 0$ on $(2,4) \cup (6,\infty)$,
- There are no horizontal or vertical asymptotes.

64.
- $f(x)$ is increasing on $(-\infty,-3) \cup (-3,0) \cup (2,5) \cup (5,\infty)$ and $f(x)$ is decreasing on $(0,2)$,
- $f(x)$ is concave down on $(-\infty,-3)$ and possibly elsewhere,
- $f(x)$ has inflection points at $x = -3$, $x = -1$, $x = 1$, $x = 5$, and $x = 6$, and nowhere else.

65.
- $f(x)$ has a vertical asymptote at $x = 2$,
- As x goes to $-\infty$, $f(x)$ has a horizontal asymptote at $y = 3$,
- $f(x)$ has intercepts at the points $(0,0)$, $(3,0)$, $(6,0)$, and nowhere else,
- $f(x)$ has a local minimum at $x = 3$ and a local maximum at $x = 5$,
- $f''(x) < 0$ on $(-\infty,2) \cup (4,\infty)$ and $f''(x) > 0$ on $(2,4)$.

66. ✪ Using only functions we have learned in this class, give the algebraic definition of some function satisfying the conditions given by the above curve-sketching questions, 53-65. Consider each as one question.

67. For additional practice, draw the functions from questions 9-30 and 31-52, using the information you've obtained from those questions. Consider each as one question. Note that a lot of the information you've obtained will be self-reinforcing; if you find yourself working with contradictory information, you have probably made a mistake in an earlier question.

Rationale/Reflection:

In addition to questions offered by the textbook, I provided a list of extra practice problems for each Big Idea in my Spring 2026 Business Calculus course.

In a coordinated, standards-based grading course, a bank of questions related to each learning outcome helps facilitate practice. I was not able to dedicate the time to make a formal bank of questions, nor did I look into banks of questions where I would have to check each question. So, I made my own questions.

I include practice problems from this Big Idea (and a corresponding Progress Check) because I have previously seen graph analysis questions presented in four parts. Given one function, identify intervals of increasing/decreasing; identify intervals of concave up/down; find zeros and asymptotes; and graph the function using all the previous information. However, because much of this information reinforces other parts, small mistakes (including trivial sign errors) can result in the graphing question having no comprehensible solution.

Composing information together to write a complete solution is an important skill to develop, but in Business Calculus, I emphasize this skill in the more applied sections of the course, such as Consumers'/Producers' surplus and Optimization. In the Graph Analysis Big Idea, I wanted to evaluate students' understanding of the shape of graphs separately from their ability to carry out computations and algebraic manipulations. Hence, the practice problems (and the corresponding Progress Check 6.1) include pure graphing questions that do not rely on any prior computation.

When I gave a similar Big Idea in Calculus 1 in the previous term, I grouped all of the material together. In this course, I was able to give much better feedback on the graphing question. I could also ask questions about a wider range of functions without needing to make an approachable all-in-one question.

6 Sample Assessments

6.1 Sample Progress Check: Business Calculus

NAME: _____ **EMAIL ID:** _____

Math 0120 Spring 2026, Section 1180 (Version A)

Progress Check 5.3: Graph Analysis

(a) I can determine where a function is increasing/decreasing, as well as the locations of any local minima/maxima, perhaps based on a relevant graph.

(b) I can determine where a function is concave up/down, as well as the locations of any inflection points, perhaps based on a relevant graph.

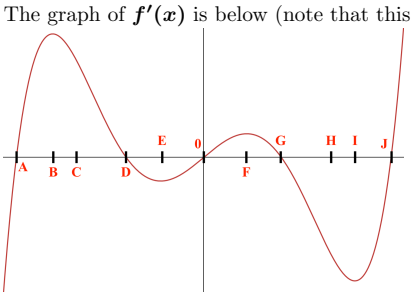
(c) I can find the x- and y-intercepts, vertical asymptotes, and horizontal asymptotes of a function.

(d) I can sketch the graph of a function given some or all of the above information.

You must show your work or otherwise justify your answers to receive credit.

1

The graph of $\boldsymbol{f'(x)}$ is below (note that this is **not** the graph of $f(x)$):



Specify all intervals where $f(x)$ is increasing/decreasing, and give the x-values of local maxima/minima, if they exist.

2

Specify all intervals where $f(x)$ is concave up/concave down, and give the x-values of inflection points, if they exist.

3

For the given function $g(x)$, find the x-intercepts, y-intercept, and any vertical/horizontal asymptotes, if they exist.

$$g(x) = \frac{x^2}{x-2}$$

4

Consider a function $h(x)$ with the following properties:

- $h(x)$ has vertical asymptotes at $x = 0$ and $x = 3$,
- As $x \to -\infty$, $h(x)$ has a horizontal asymptote at $y = 2$.
- $h(x)$ is concave down everywhere it is defined,
- $h(x)$ has a local max at $x = 5$, and possibly elsewhere.

Based on this information, draw $h(x)$ below. It may help you to create sign charts first.

6.2 Sample Progress Checks: College Algebra 1

NAME: _____ **EMAIL ID:** _____

Math 0010 Fall 2026, Section 1070

Progress Check 3.1: Linear Functions and Inequalities

(a) I can draw the graph of a line.

(b) Given two pieces of information about a line (either point and slope, or two points), I can determine its equation.

(c) Given the graph of a line, I can determine its equation.

(d) I can determine whether two lines are parallel or perpendicular.

(e) I can compute the equation of a line which is parallel/perpendicular to another line.

(f) I can solve for the unknown value in a linear equation or inequality.

You must show your work or otherwise justify your answers to receive credit.

1

Determine the equation of the line that passes through the points $(3, -1)$ and $(4, 4)$.

2

Solve the following inequality for x:

$$-4x - 5 \leq 8 + x$$

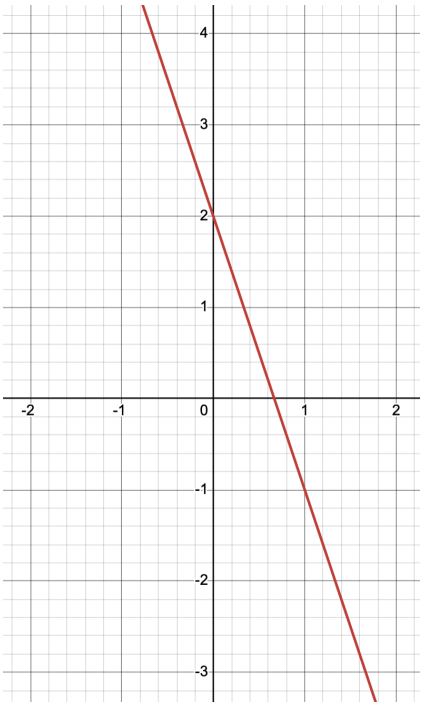
3

Are the following lines parallel, perpendicular, or neither? Justify your answer.

$$3x - 2y = 6, \qquad y + \frac{2}{3}x = 9$$

4

Determine the equation of the line pictured in the following graph:



NAME: _____ **EMAIL ID:** _____

Math 0010 Fall 2026, Section 1070

Progress Check 3.2: Linear Functions and Inequalities

(a) I can draw the graph of a line.

(b) Given two pieces of information about a line (either point and slope, or two points), I can determine its equation.

(c) Given the graph of a line, I can determine its equation.

(d) I can determine whether two lines are parallel or perpendicular.

(e) I can compute the equation of a line which is parallel/perpendicular to another line.

(f) I can solve for the unknown value in a linear equation or inequality.

You must show your work or otherwise justify your answers to receive credit.

1

Determine the equation of the line that passes through the points $(-2, 5)$ and $(-4, -3)$.

2

Solve the following inequality for x:

$$-3x + 4 > -5 + 6x$$

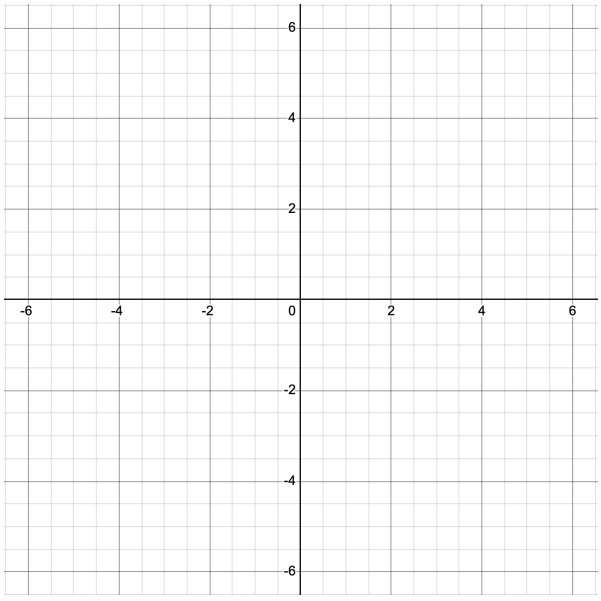
3

Determine the equation of the line that passes through the point $(-4, -2)$ and is perpendicular to the line $y = -2x + 7$.

4

Use the grid lines below to draw the line $4x + 2y = 8$.

Hint: Start by writing this line in slope-intercept form!



A Student Evaluations of Teaching

Highlighted Questions:

1. The instructor created an atmosphere that kept me engaged in course content.
2. The instructor treated students with respect.
3. The instructor provided feedback that was helpful to me.
4. Would you recommend this lecture instructor to a friend taking this course?
5. The instructor creates an inclusive learning environment for all students.
6. Grading standards and policies were clearly communicated.
7. The specific assessment and grading structure for this course facilitated my learning.

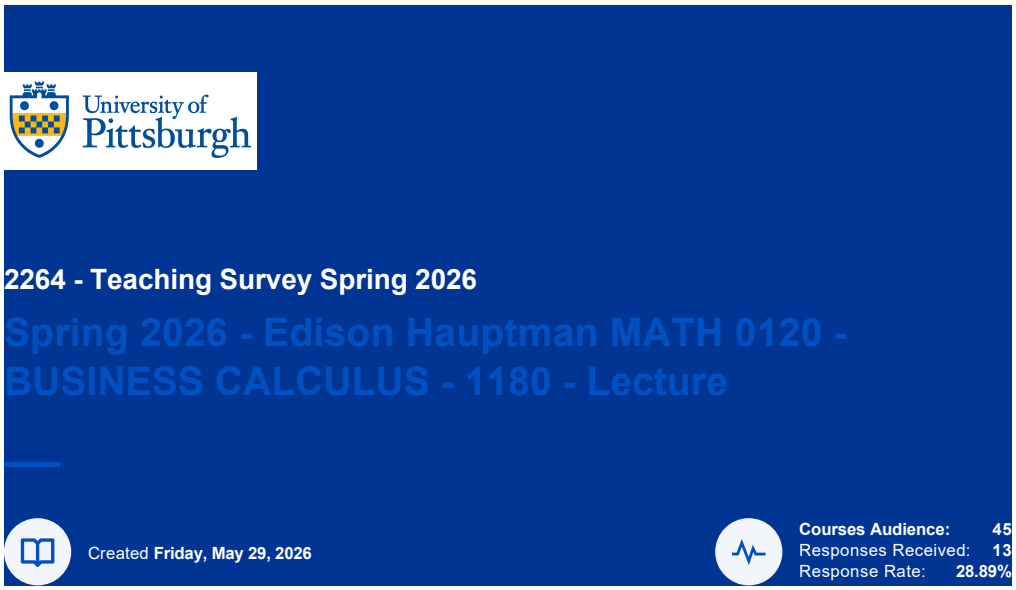
Course	Business Calculus, S26 (45 students)			Calculus 1, F25 (Morning) (71 students)			Calculus 1, F25 (Evening) (68 students)		
Questions	Neg.	Neu.	Pos.	Neg.	Neu.	Pos.	Neg.	Neu.	Pos.
1	1	0	12	7	9	35	3	5	49
2	0	0	13	2	1	47	1	1	55
3	1	0	10	7	8	36	4	5	47
4	1	0	12	8	4	36	2	4	50
5	Mean: 4.77/5			Mean: 4.52/5			Mean: 4.52/5		
6	0	0	13	2	12	33	5	9	41
7	2	1	10	11	6	30	6	4	43

Course	Calculus 2, Su24 (11 students)			Calculus 2, Su23 (18 students)			Calculus 2, Su22 (24 students)		
Questions	Neg.	Neu.	Pos.	Neg.	Neu.	Pos.	Neg.	Neu.	Pos.
1	0	0	3	0	0	6	1	1	6
2	0	0	3	0	0	6	0	0	8
3	0	0	3	0	0	5	1	0	7
4	0	0	3	0	0	5	3	N/A	5
5	0	0	3	0	0	6	1	0	7
6	N/A			N/A			N/A		
7	N/A			N/A			N/A		

I started asking questions 6 and 7 in my evaluations starting in Fall 2025, when I began implementing a standards-based grading structure in a large-enrollment context.

A complete set of my teaching evaluations as an instructor of record, along with student comments, is available on the next pages.

A.1 Full Student Evaluation Reports



Report Comments

Teaching Surveys

Included in this report:

- Summary of responses to scaled questions
- Response breakdowns
- Student comments
- Results to instructor added custom questions (if applicable)

Understanding and using student feedback:

- We have resources to help you interpret and use results including our faculty worksheet with guided prompts and space to record summaries of feedback, actions, and outcomes.
- Members of our Pedagogy, Practice, & Assessment team are available for consultations and can help with:
  - Interpreting OMET results and developing a course of action if necessary.
  - Exploring various methods of assessment to improve teaching.
- In the future:
  - Discuss, teach, and model giving meaningful feedback with your students and give them multiple opportunities to practice giving feedback.
    - Gather important information about students at the beginning of the term by giving a pre-course survey.
    - Check in with students half way through the term by giving a midterm course survey.
- The Teaching Center offers multiple resources to support teaching and learning.

Office of Measurement and Evaluation of Teaching (OMET)
Contact us

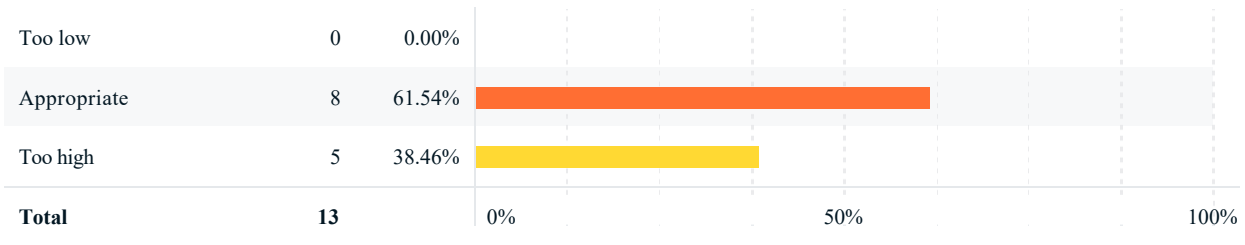
Dietrich School of Arts and Sciences Questions

Summary table

	Invited Count	Response Count	Response Rate	Mean	Median	Mode	SD
The instructor created an atmosphere that kept me engaged in course content.	45	13	28.89%	4.38	5.00	5	1.12
The instructor was prepared for class.	45	13	28.89%	4.62	5.00	5	0.51
The instructor treated students with respect.	45	13	28.89%	4.77	5.00	5	0.44
The instructor was available to me (in-person, electronically, or both).	45	11	24.44%	4.64	5.00	5	0.67
The instructor evaluated my work fairly.	45	11	24.44%	4.00	4.00	5	1.26
The instructor provided feedback that was helpful to me.	45	11	24.44%	4.27	5.00	5	1.19
I learned a lot from this course. If there is no basis to judge or not applicable, answer N/A.	45	13	28.89%	4.15	5.00	5	1.07
Overall of All Questions	315	85	26.98%	4.40	-	-	0.95

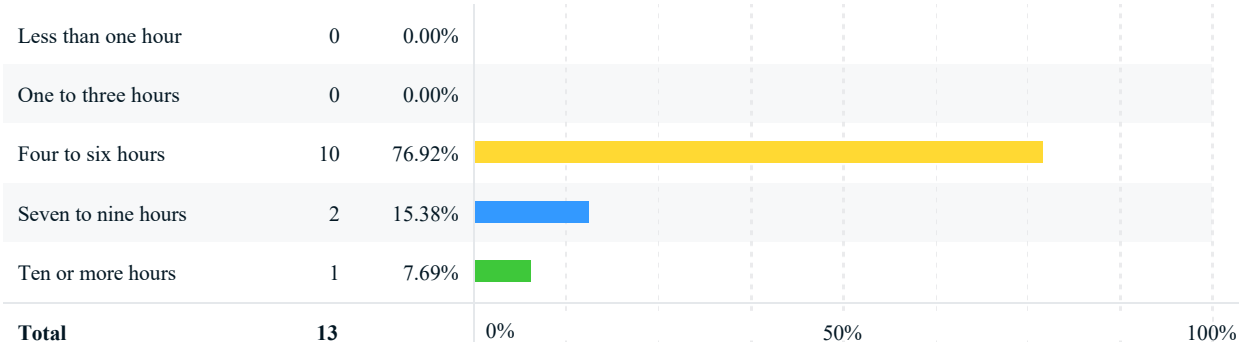
The standards the instructor set for me were:

The standards the instructor set for me were:



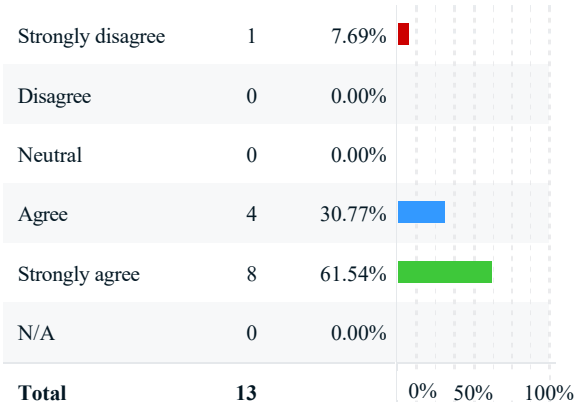
How many hours per week did you usually spend working on this course outside of classroom time?

How many hours per week did you usually spend working on this course outside of classroom time?

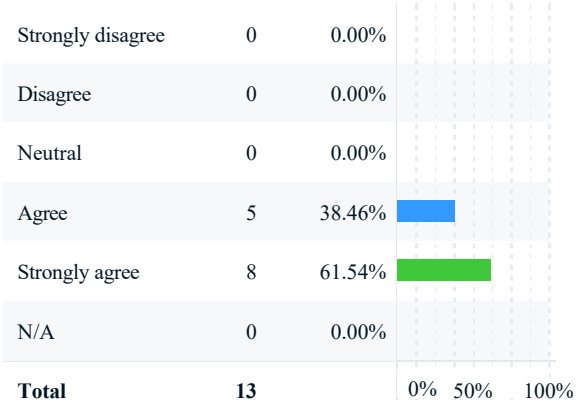


Response breakdown

1. The instructor created an atmosphere that kept me engaged in course content.

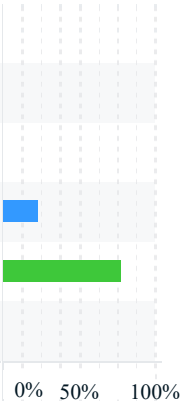


2. The instructor was prepared for class.



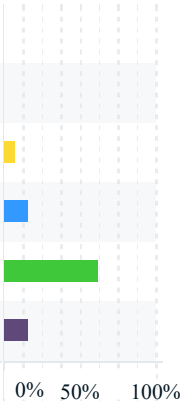
3. The instructor treated students with respect.

Strongly disagree	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	3	23.08%
Strongly agree	10	76.92%
N/A	0	0.00%
Total	**13**	



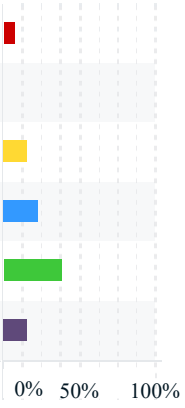
4. The instructor was available to me (in-person, electronically, or both).

Strongly disagree	0	0.00%
Disagree	0	0.00%
Neutral	1	7.69%
Agree	2	15.38%
Strongly agree	8	61.54%
N/A	2	15.38%
Total	**13**	



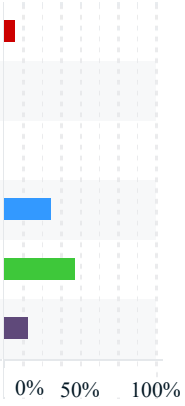
5. The instructor evaluated my work fairly.

Strongly disagree	1	7.69%
Disagree	0	0.00%
Neutral	2	15.38%
Agree	3	23.08%
Strongly agree	5	38.46%
N/A	2	15.38%
Total	**13**	



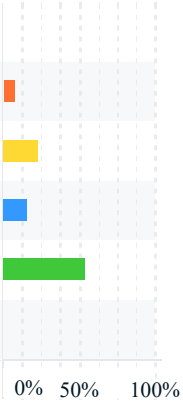
6. The instructor provided feedback that was helpful to me.

Strongly disagree	1	7.69%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	4	30.77%
Strongly agree	6	46.15%
N/A	2	15.38%
Total	**13**	



7. I learned a lot from this course. If there is no basis to judge or not applicable, answer N/A.

Strongly disagree	0	0.00%
Disagree	1	7.69%
Neutral	3	23.08%
Agree	2	15.38%
Strongly agree	7	53.85%
N/A	0	0.00%
Total	**13**	



DSAS Comments

What did you like best about how the course was taught?

Comments
Very great in class examples that genuinely taught the material
The math was well explained, and the grading system is very good.
The style of lecture and how the course was set up.
The professor was extremely engaging and I can tell he's very passionate about calculus and his job. If he's going into teaching after graduating, he'd be great at it. He provided great examples in class that helped with the Progress Checks. He also drew a lot of graphs and showed some from desmos, so i really understood some concepts, like the saddle point and the demand curve, better. He also was extremely great with our class's questions and never made us feel like we were dumb. He was even excited to answer questions and see things in a new perspective.
I liked how complex topics were broken down into easier concepts. I also liked that the structure was forgiving of mistakes and made it easier to learn topics.
I like the set up of the class, and I liked lectures.
No tests
I loved how enthusiastic he was about math, it definitely made it much easier to learn about this difficult material
I liked how the final grade wasn't just dependent on a couple of exams, rather it was broken up into multiple smaller quizzes.
I had to teach myself. My recitation teacher SO mean and his test felt impossible. He went really fast.
caring teac her with appropriate material for all levels
I loved the enthusiasm and actual manner of teaching. It was clear, concise, well–organized, and left room for me to problem–solve on my own. I wish there could have been more group work and collaboration between students, but I understand the restriction of time and the fast–paced nature of the course hindering this.

If you were teaching this course, what would you do differently?

Comments
N/A
I would go a little slower and not skip certain algebra steps during calculations.
I wouldn't do anything differently except maybe provide an answer key for the extra set practiced problems.
This is out of his control, but I think taking lectures slower would be better. Like I think this class would greatly benefit if it were a 2 hour lecture twice a week so we could really dwell on the examples.
I would make the arithmetic on some of the problems a little easier. I understood the concepts, but I felt that the math was a little too complex. That being said, the chances for retakes make up for that issue.
Maybe show harder examples during class.
Nothing
nothing
I would have made it so the second attempts of an idea were the following Tuesday instead of the following Thursday. This way, students would only have to focus on one quiz that Thursday instead of potentially two.
Don’t give impossible questions when we can do the same with with less complex problems bc we only have 15 minutes to do a test that determines our whole grade every week
nothing
I disagree with the entire grading system. I understand why it's set up the way it is – to encourage mastery – but I think it leaves too little room for error, especially for students who aren't strong test takers. Even despite the leniency of the retakes and the availability of practice problems, I found myself failing consistently. The weekly test–taking schedule is rigorous and I never felt like I was adequately prepared for the Progress Checks nor for the 15 minutes we had to take them. Despite my continued efforts in this course outside of the tests, I didn't succeed nearly as much as I'd hoped to.

Math TA/TF Additional Items

Please answer the following questions about your instructor:

1. Did you experience difficulty in comprehending your lecture instructor's spoken language in class?

No difficulty a…	12	92.31%	
Small amount o…	1	7.69%	
Moderate amo…	0	0.00%	
Severe difficulty	0	0.00%	
Total	**13**		0% 50% 100%

2. Did your lecture instructor experience difficulty in comprehending the questions that were asked by students in class?

No difficulty a…	11	84.62%	
Small amount o…	2	15.38%	
Moderate amo…	0	0.00%	
Severe difficulty	0	0.00%	
Total	**13**		0% 50% 100%

The lecture instructor's writing on the chalkboard was legible.

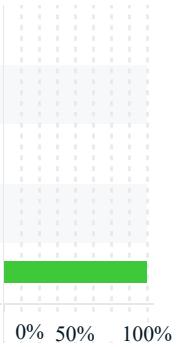
The lecture instructor's writing on the chalkboard was legible.

Seldom	0	0.00%	
Sometimes	0	0.00%	
About half the …	0	0.00%	
Usually	6	46.15%	
Always	7	53.85%	
Total	**13**		0% 50% 100%

The lecture instructor's attitude toward the subject was enthusiastic.

The lecture instructor's attitude toward the subject was enthusiastic.

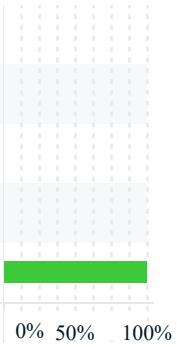
Hardly at all	0	0.00%
To a small degree	0	0.00%
To a moderate …	0	0.00%
To a considera…	0	0.00%
To a very high …	13	100.00%
Total	**13**	



The lecture instructor arrived for class on time.

The lecture instructor arrived for class on time.

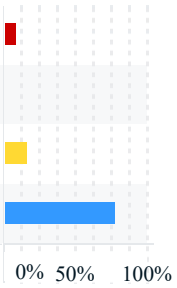
Rarely (less th…	0	0.00%
Seldom (20-40…	0	0.00%
About half the …	0	0.00%
Usually (70-90…	0	0.00%
Over 90% of t…	13	100.00%
Total	**13**	



Please answer the following questions about your instructor:

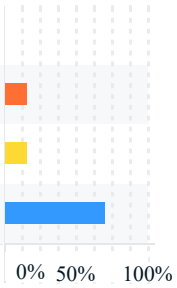
1. The lecture instructor provided the opportunity for questions.

Very seldom	1	7.69%
About half the …	0	0.00%
Frequently	2	15.38%
Almost always	10	76.92%
Total	**13**	



2. Helpful answers were given to questions raised in class.

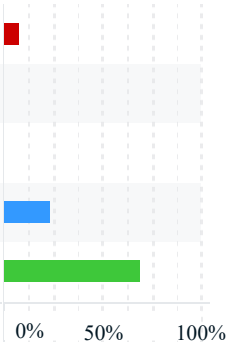
Very seldom	0	0.00%
About half the …	2	15.38%
Frequently	2	15.38%
Almost always	9	69.23%
Total	**13**	



Would you recommend this lecture instructor to a friend taking this course?

Would you recommend this lecture instructor to a friend taking this course?

Not at all	1	7.69%
Unlikely	0	0.00%
Don't know	0	0.00%
Maybe	3	23.08%
Definitely	9	69.23%
Total	**13**	



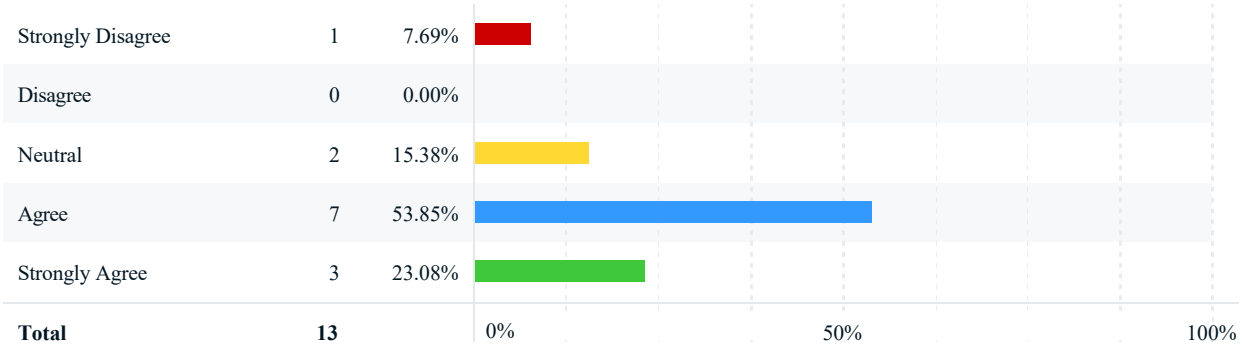
Diversity and Inclusion

Question	Response Count	Response Rate	Mean	Mode	Median	SD
The instructor creates an inclusive learning environment for all students.	13	28.89%	4.77	5	5.00	0.44

Personalized Questions

Course assessments were directly related to what was covered/practiced in the course.

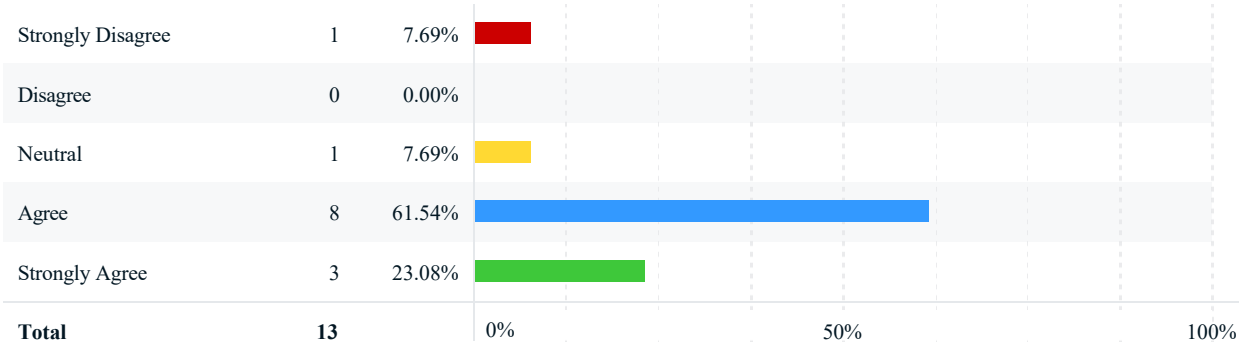
Course assessments were directly related to what was covered/practiced in the course.



Statistics	Value
Mean	3.85
Standard Deviation	1.07

Course assessments provided the opportunity for me to demonstrate an understanding of the course material.

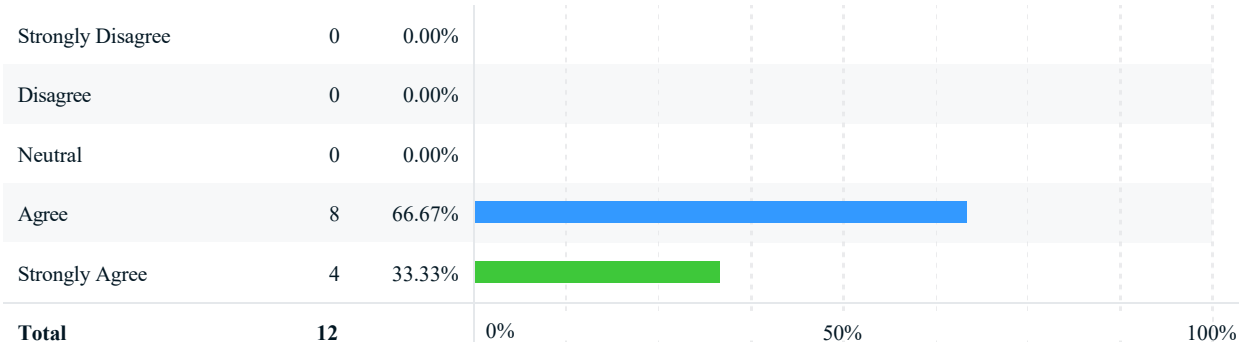
Course assessments provided the opportunity for me to demonstrate an understanding of the course material.



Statistics	Value
Mean	3.92
Standard Deviation	1.04

This course's online materials in Canvas were well organized.

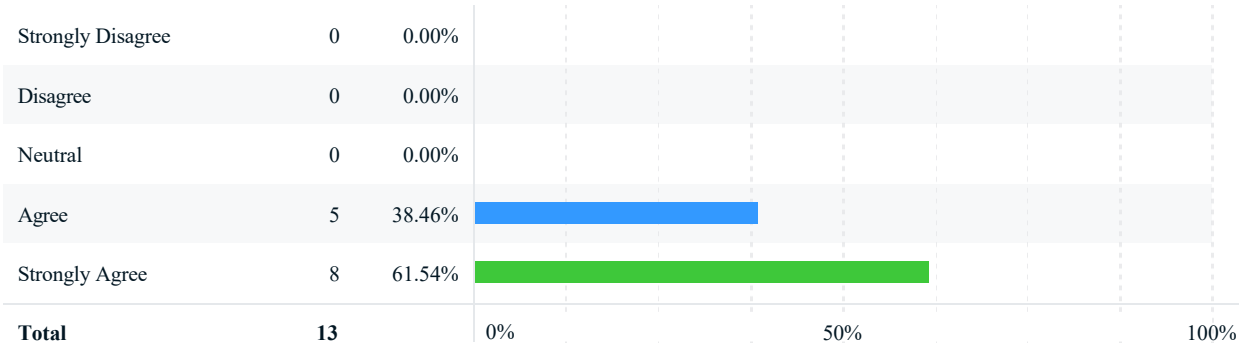
This course's online materials in Canvas were well organized.



Statistics	Value
Mean	4.33
Standard Deviation	0.49

Grading standards and policies were clearly communicated.

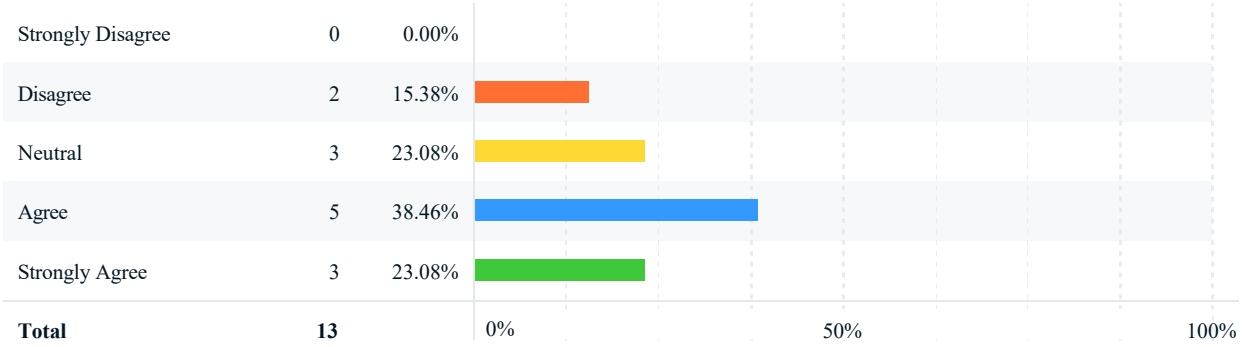
Grading standards and policies were clearly communicated.



Statistics	Value
Mean	4.62
Standard Deviation	0.51

Group work contributed to my understanding of the course material.

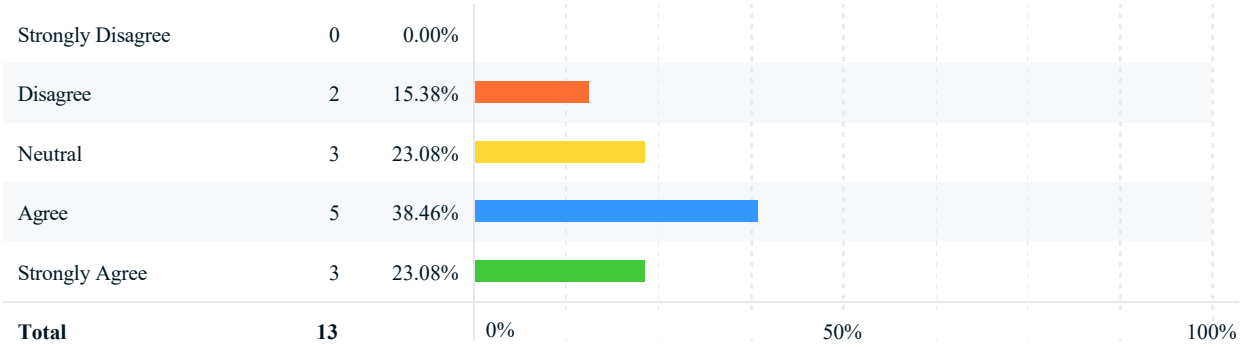
Group work contributed to my understanding of the course material.



Statistics	Value
Mean	3.69
Standard Deviation	1.03

Group work helped to develop my skills in problem-solving.

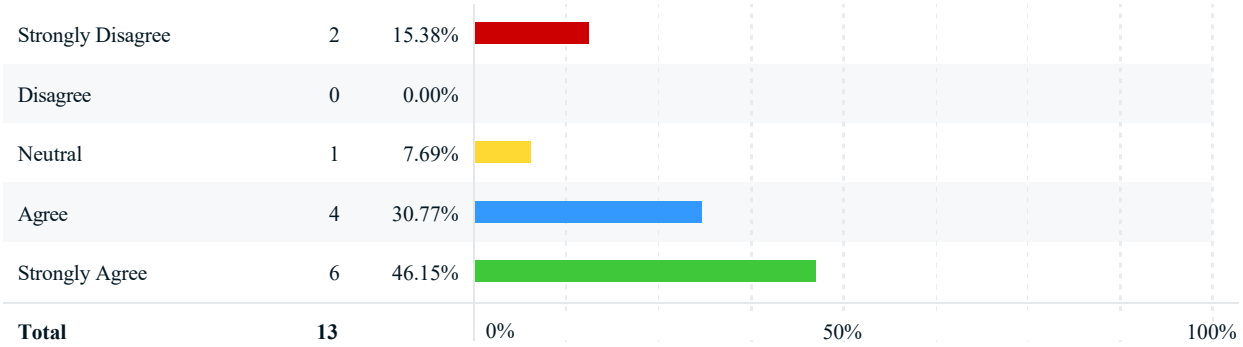
Group work helped to develop my skills in problem-solving.



Statistics	Value
Mean	3.69
Standard Deviation	1.03

The specific assessment and grading structure for this course facilitated my learning.

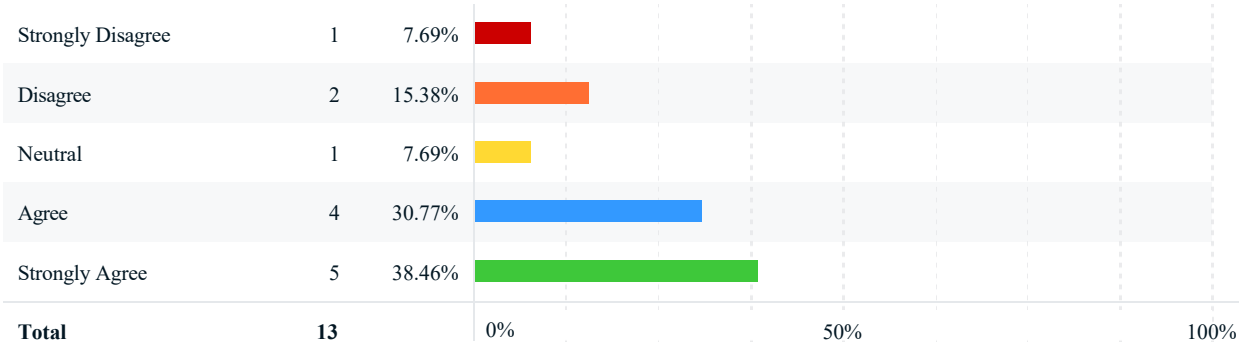
The specific assessment and grading structure for this course facilitated my learning.



Statistics	Value
Mean	3.92
Standard Deviation	1.44

The specific assessment and grading structure for this course was preferable to having 3-4 big exams.

The specific assessment and grading structure for this course was preferable to having 3-4 big exams.



Statistics	Value
Mean	3.77
Standard Deviation	1.36

Please share your thoughts on the assessment and grading structure for this course. If the general idea was good, what should I keep, and what should I change? If the general idea was bad, please say so!

Students
I love this structure. It was fun!
I feel that the grading system was good and honestly kind of fun too! The progress checks are very high stakes, but the retakes are helpful.
The structure of the course if great, the quizzes are much easier to grasp topic by topic than having to take a bigger test with more on it.
The structure was really confusing at the start of the year, but as we progressed, it made more sense. I think the grading is extremely fair, like how we have three attempts at essentially every big idea. I like that if you do amazingly on the super final, then that overwrites your grade, no matter how poor you did. I also like that if you bomb it, the progress checks prevent you from failing the course.
I really liked the assessment and grading structure.
Quizzes were good, but too difficult compared to how we are taught in class.
It was great
Personally, I am not the best under pressure, so 15 minute quizzes just don't work well with how I operate. But it is a college level course so to say that it is unfair with the way things are run would be ignorant, especially due to the 3 chances we are given to learn from our mistakes
I liked the current structure, I would just make it so the second attempt would be on a Tuesday instead of the following Thursday
A test every week was SO stressful it was horrible
very good pls keep i don't understand why people say this structure is hard it was super nice
I would prefer that the Engagement Points and the work involved with getting had more of a weight for grading. I ended the semester with over 200, but because I didn't pass many of the Progress Checks, my grade ultimately suffered in spite of my best efforts. I enjoyed learning calculus and did learn a lot, but my grade didn't reflect it.

If you used the Extra Practice Problem sets or some other outside resource, were they useful? If applicable, compare to the course textbook.

Students
As a way to get Engagement Points yes, but lectures/recitations were the best.
The extra practice problems were helpful, and I mainly used them for study.
Yes, they were useful as they somewhat mirrored problems that were on the quizzes. The only thing I would make sure to have for these extra practice problems is the answer key as it does get difficult if you don't know what you're doing.
Yes, but I really wish they had an answer key so I wouldn't have to use AI to see if I was doing things right. The homework isn't really useful when you don't really get feedback. The practice problems were great in comparison to the progress checks though, and the textbook problems didn't really help much but they did have an answer key.
The Extra Practice Problem sets were very useful. They covered every topic and math scenario needed and were completely sufficient on their own for studying. I found them more useful than the textbook because the problems were more diverse.
I liked the textbook the most. It helped knowing the answer and if I was right or wrong.
Yes they were useful
very useful, It would be very beneficial though if there was an answer key to the problems, so i can check my work after
They were useful and that was my main study tool
More useful then textbook
yes very
The extra practice problem sets were the most useful part of this course for me. I didn't really bother with the textbook since they were available, which was nice.

Is there anything else I should know?

Students
Nope, loved the class
Love the course!
n/a
I think it would be really helpful to have some recorded lectures uploaded. Like when I'd go after class to see examples, none of them were as complex as the progress checks so I really couldn't teach myself (btw this isn't a failure on you, but I never understand math unless I actually take time to understand what's going on, rather than writing it down. that's also why i was absolutely terrible at the tophat questions, but i liked how you only docked off 0.1 point. like compared to my econ class where 0.6 of a point was knocked off, it was really helpful and i wasn't really penalized for getting things wrong). I also think you would benefit from writing on like a blank sheet on the computer and presenting that to the class, instead of the whiteboard, because when you can't see the board, the writing is kind of useless. there were some points of the year when I just couldn't see the board.
The course was very enjoyable!
No.
No
Thanks for the great semester!!
MAKE EASIER TESTS
nope
Even if I'm unhappy about my grade, thank you helping me!

University of Pittsburgh

2261 - Teaching Survey Fall 2025

Fall 2025 - Edison Hauptman MATH 0220 - ANALYTC GEOMETRY & CALCULUS 1 - 1500 - Lecture

Created **Wednesday, December 17, 2025**

Courses Audience: 71
Responses Received: 51
Response Rate: **71.83%**

Report Comments

Teaching Surveys

Included in this report:

- Summary of responses to scaled questions
- Response breakdowns
- Student comments
- Results to instructor added custom questions (if applicable)

Understanding and using student feedback:

- We have resources to help you interpret and use results including our faculty worksheet with guided prompts and space to record summaries of feedback, actions, and outcomes.
- Members of our Pedagogy, Practice, & Assessment team are available for consultations and can help with:
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- In the future:
  - Discuss, teach, and model giving meaningful feedback with your students and give them multiple opportunities to practice giving feedback.
    - Gather important information about students at the beginning of the term by giving a pre-course survey.
    - Check in with students half way through the term by giving a midterm course survey.
- The Teaching Center offers multiple resources to support teaching and learning.

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Contact us

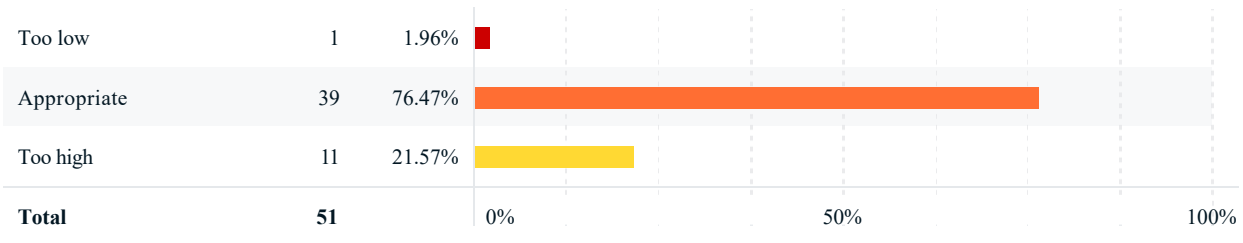
Dietrich School of Arts and Sciences Questions

Summary table

	Invited Count	Response Count	Response Rate	Mean	Median	Mode	SD
The instructor created an atmosphere that kept me engaged in course content.	71	51	71.83%	3.84	4.00	5	1.17
The instructor was prepared for class.	71	50	70.42%	4.30	4.50	5	0.95
The instructor treated students with respect.	71	50	70.42%	4.52	5.00	5	0.81
The instructor was available to me (in-person, electronically, or both).	71	49	69.01%	4.43	5.00	5	0.94
The instructor evaluated my work fairly.	71	51	71.83%	4.08	4.00	4,5	1.00
The instructor provided feedback that was helpful to me.	71	51	71.83%	3.88	4.00	4,5	1.13
I learned a lot from this course. If there is no basis to judge or not applicable, answer N/A.	71	48	67.61%	3.88	4.00	4	0.84
Overall of All Questions	497	350	70.42%	4.13	-	-	0.99

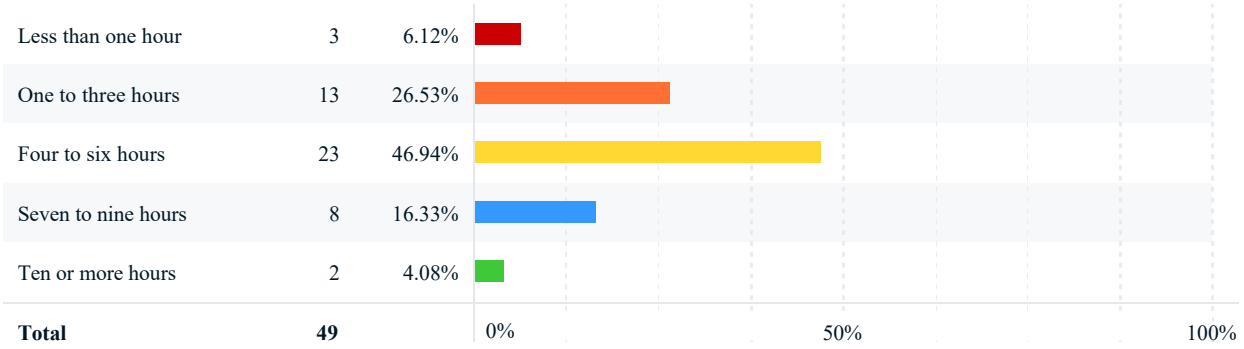
The standards the instructor set for me were:

The standards the instructor set for me were:



How many hours per week did you usually spend working on this course outside of classroom time?

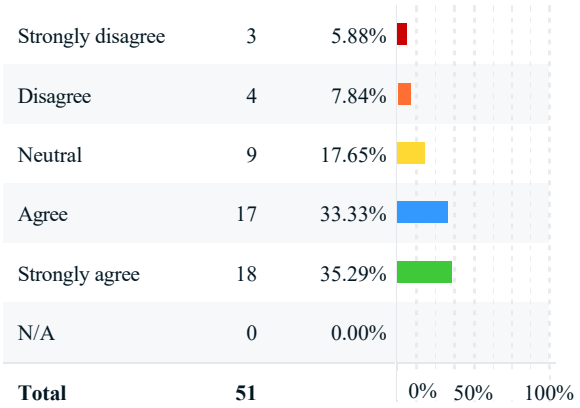
How many hours per week did you usually spend working on this course outside of classroom time?



Less than one hour	3	6.12%
One to three hours	13	26.53%
Four to six hours	23	46.94%
Seven to nine hours	8	16.33%
Ten or more hours	2	4.08%
Total	**49**	

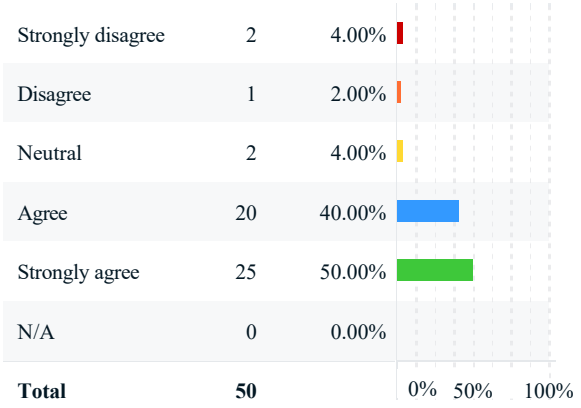
Response breakdown

1. The instructor created an atmosphere that kept me engaged in course content.



Strongly disagree	3	5.88%
Disagree	4	7.84%
Neutral	9	17.65%
Agree	17	33.33%
Strongly agree	18	35.29%
N/A	0	0.00%
Total	**51**	

2. The instructor was prepared for class.



Strongly disagree	2	4.00%
Disagree	1	2.00%
Neutral	2	4.00%
Agree	20	40.00%
Strongly agree	25	50.00%
N/A	0	0.00%
Total	**50**	

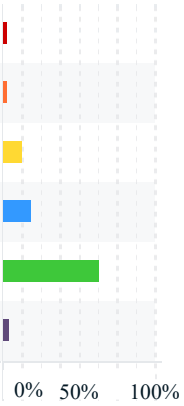
3. The instructor treated students with respect.

Strongly disagree	1	2.00%
Disagree	1	2.00%
Neutral	1	2.00%
Agree	15	30.00%
Strongly agree	32	64.00%
N/A	0	0.00%
Total	**50**	



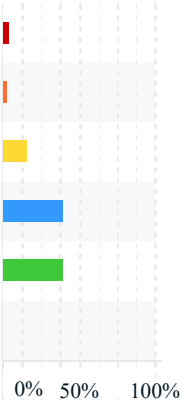
4. The instructor was available to me (in-person, electronically, or both).

Strongly disagree	1	1.96%
Disagree	1	1.96%
Neutral	6	11.76%
Agree	9	17.65%
Strongly agree	32	62.75%
N/A	2	3.92%
Total	**51**	



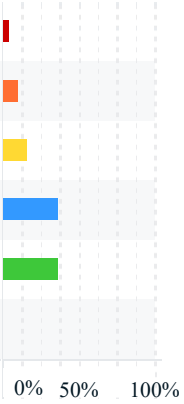
5. The instructor evaluated my work fairly.

Strongly disagree	2	3.92%
Disagree	1	1.96%
Neutral	8	15.69%
Agree	20	39.22%
Strongly agree	20	39.22%
N/A	0	0.00%
Total	**51**	

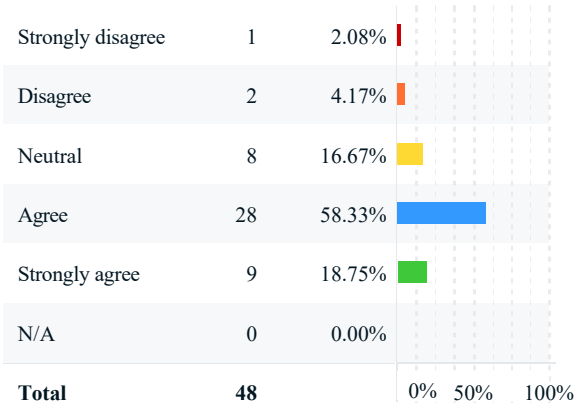


6. The instructor provided feedback that was helpful to me.

Strongly disagree	2	3.92%
Disagree	5	9.80%
Neutral	8	15.69%
Agree	18	35.29%
Strongly agree	18	35.29%
N/A	0	0.00%
Total	**51**	



7. I learned a lot from this course. If there is no basis to judge or not applicable, answer N/A.



DSAS Comments

What did you like best about how the course was taught?

Comments
the grading system and the way we were tested was so helpful because it tested you once a week on one topic instead of every few weeks on a bunch of topics so everything was still fresh on your mind. grading system makes it so you have many opportunities to save your grade
The professor did an amazing job at lecturing about difficult calculus topics. He didn't just show us how to solve certain problems, but he also taught us the foundations of different concepts and why/how they work. Overall amazing job at teaching this course!
I liked weekly assessments over large cumulative exams, allowed me to spend more time on each individual topic and make sure I understood them really well.
I liked how you could go at your own pace.
Nothing.
I liked how professor explaining the examples. He showed us examples with explanation, and then we asked question.
I liked the essentially correct idea, it truly felt like a calculus class.
The grading system
The recitations helped since we went over the progress checks in them.
The structure is great. Easy to follow examples when introducing a new topic are great.
I like the way he show us the examples.

Comments
I liked how there were progress checks every week instead of a bunch of exams.
Derivatives
I like how the Big Ideas were structured so if we made a mistake, we can retest the week after.
no midterm and mandatory final
the ability for retakes in theory
I liked the flexibility of the course structure.
The week by week tests
I liked how there were top hat questions that kept students engaged throughout the lecture. I also liked how the professor provided multiple attempts to a progress check rather than giving full exams.
Instructor clearly enjoys mathematics, good about teaching proofs, empathetic about student health struggles.
The formating kept me engaged
I liked the lectures
I liked that you could tell HE liked what he was doing...
Very Engaging
I liked the practice problems.
I liked that it was engaging, and there were many opportunities to talk through problems in class instead of having it just be a lecture.
he really knew how to teach complex ideas in a simple way for all of us to understand better
I liked the layout of the class, and how fast paced it is.
There was many resources available for practice and help
The weekly progress check with many attempts. A blessing and a curse…
Lectures were full of great information. His lectures was perfect for a class like this one.
nothing
I love how the structure of topics was taught. I really thought that the homework and Lon Capa were Helpful.
I liked all of the opportunities there were to earn more points.
No exams
The preparation given and real–life application and connection to previous math topics
I liked how there were multiple chances to improve your grade.
I liked the top hat to reinforce our learning.

Comments
No tests
I liked the system of Big Ideas and Learning outcomes,
It was clear what you wanted us to take away from the course.
More open for the understanding of the student

If you were teaching this course, what would you do differently?

Comments
nothing
The only change I would make is maybe adding another buffer week in the middle of the semester to allow some people to get caught up on quizzes and stop the snowball effect that can build with having to take two quizzes every week. But I do love the structure of the course and how the grading system works.
I believe it would've been easier to give students more of an "ease–in" period to the grading system, which was highly confusing at the start and is still slightly confusing as of now.
Make sure all the TAs had the week's assessment before Monday recitation. Sometimes I would come in and Olivia wouldn't have the new Progress check so she couldn't answer all of our questions.
I would get rid of the practice problem sets and have anyone practice from any website they want to, just thought the practice problem sets were busy work rather than actually practicing.
Actually take into account that students may not have a calculus background.
I wouldn't do really differently. I think that this is already great.
Give more quiz level questions during the lecture to try in our free time.
Add class notes on canvas
Explain concepts more in depth instead of briefly mentioning them and moving on.
The jokes aren't funny but I'm glad you still made an effort despite it being an 8am class. Good vibes
Maybe more example question.
Probably put resources online when it comes to problems you did in class. Sometimes if you zone out for a second you miss something and have no idea what happened in the last 30 seconds.
I wish the big Ideas didnt exist
Nothing. This class was low–stress but I could actually learn the topics in–depth because how Edison lectured
i would not include learning outcomes in final grading tracker, just big ideas and engangement pts, easier questions on progress checks, return progress check way quicker because that was the biggest factor in what i should be studying, but i had to wait to get it back and then i only have 2 busy days to study
Explain reasons for doing things, not have the grading set up the way it is

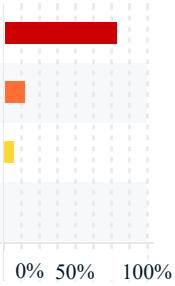
Comments
I would make provide more suggestions at the beginning of the semester of how to handle the workload because it can be very overwhelming trying to do all the assignments while properly studying for the weekly progress checks. I would also make it more lenient on the the progress checks, instead of having to get 3/4 for the progress check to count I would make it 2/4.
Have learning outcomes be cumulative while big ideas still need to get the 3/4.
I would provide more examples to the topics that we are learning about
Be more specific on what is being tested on progress checks, more classroom resources posted for knowledge checks.
Probably nothing tbh
Change the grading
The weekly quiz structure was horrible, especially the retake portion. While it is appreciated, having to take two quizzes makes it so that you're now studying DOUBLE the material each week. We only get 5 days (not even) to know TWO topics well enough to pass his trick questions. It was impossible for me, personally.
N/A
I would dumb the concepts down more.
Maybe have more top hat questions.
i wouldnt make the learning outcomes count for nothing if it is less than 3/4. i often get 2/4 just because i am more focused on 1 big idea quiz mastery than the other.
More group excursions
I would record lecture/ post lecture notes!
Quizzes instead of exams. I understand that it helps him make sure we are learning everything that we need, but getting two questions wrong and failing never feels good. On exams, partial credit can be awarded, and it makes the students life less stressful.
i would answer students' questions thoughtfully and respectfully
I'm not sure, everything was very overwhelming though.
Maybe go more into definitions and concepts rather than just doing practice problems
I would provide more in class examples and possible questions
I would make it so there's way to earn learning outcomes outside of quizzes
I would structure the notes a bit more, starting with the topic and then doing examples rather than the other way around.
Grading differently
Probably the LonCapa homework system. Furthermore, group studying in recitation for the Final for the upcoming weeks.
No weekly quizzes, they were hard to keep up with.
Nothing

Math TA/TF Additional Items

Please answer the following questions about your instructor:

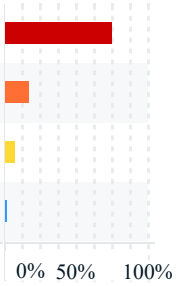
1. Did you experience difficulty in comprehending your lecture instructor's spoken language in class?

No difficulty a…	38	79.17%
Small amount o…	7	14.58%
Moderate amo…	3	6.25%
Severe difficulty	0	0.00%
Total	**48**	



2. Did your lecture instructor experience difficulty in comprehending the questions that were asked by students in class?

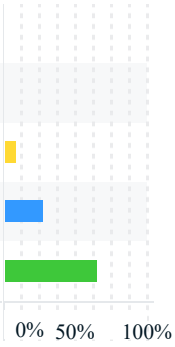
No difficulty a…	35	74.47%
Small amount o…	8	17.02%
Moderate amo…	3	6.38%
Severe difficulty	1	2.13%
Total	**47**	



The lecture instructor's writing on the chalkboard was legible.

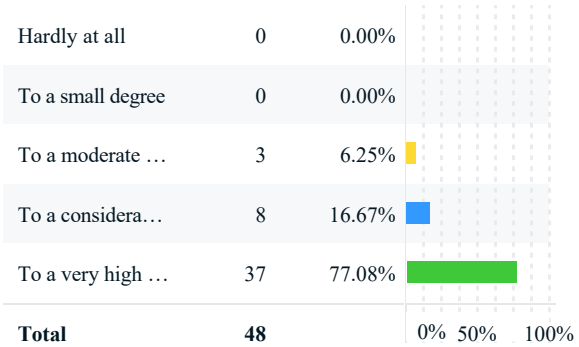
The lecture instructor's writing on the chalkboard was legible.

Seldom	0	0.00%
Sometimes	0	0.00%
About half the …	4	8.33%
Usually	13	27.08%
Always	31	64.58%
Total	**48**	



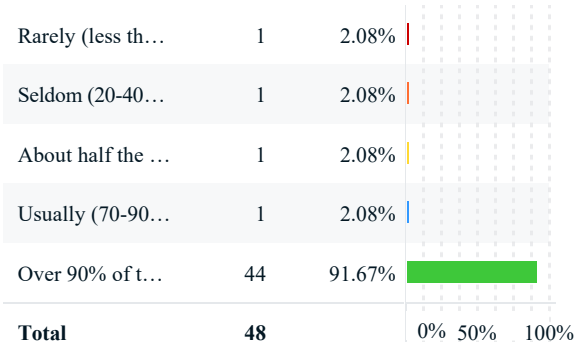
The lecture instructor's attitude toward the subject was enthusiastic.

The lecture instructor's attitude toward the subject was enthusiastic.



The lecture instructor arrived for class on time.

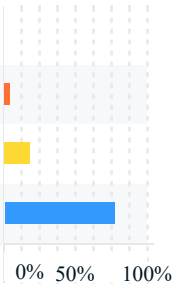
The lecture instructor arrived for class on time.



Please answer the following questions about your instructor:

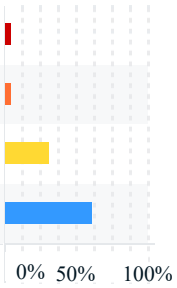
1. The lecture instructor provided the opportunity for questions.

Very seldom	0	0.00%
About half the …	2	4.17%
Frequently	9	18.75%
Almost always	37	77.08%
Total	**48**	



2. Helpful answers were given to questions raised in class.

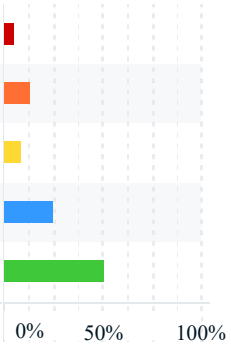
Very seldom	2	4.17%
About half the …	2	4.17%
Frequently	15	31.25%
Almost always	29	60.42%
Total	**48**	



Would you recommend this lecture instructor to a friend taking this course?

Would you recommend this lecture instructor to a friend taking this course?

Not at all	2	4.17%
Unlikely	6	12.50%
Don't know	4	8.33%
Maybe	12	25.00%
Definitely	24	50.00%
Total	**48**	



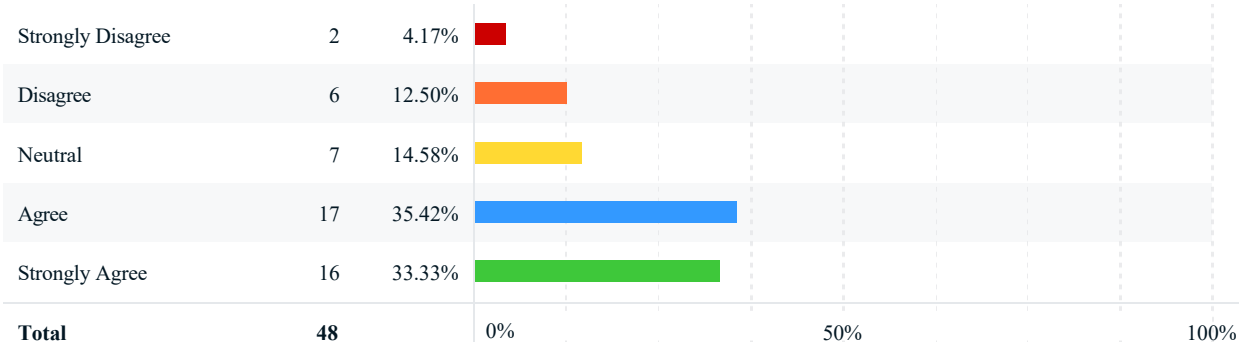
Diversity and Inclusion

Question	Response Count	Response Rate	Mean	Mode	Median	SD
The instructor creates an inclusive learning environment for all students.	48	67.61%	4.52	5	5.00	0.74

Personalized Questions

Course assessments were directly related to what was covered/practiced in the course.

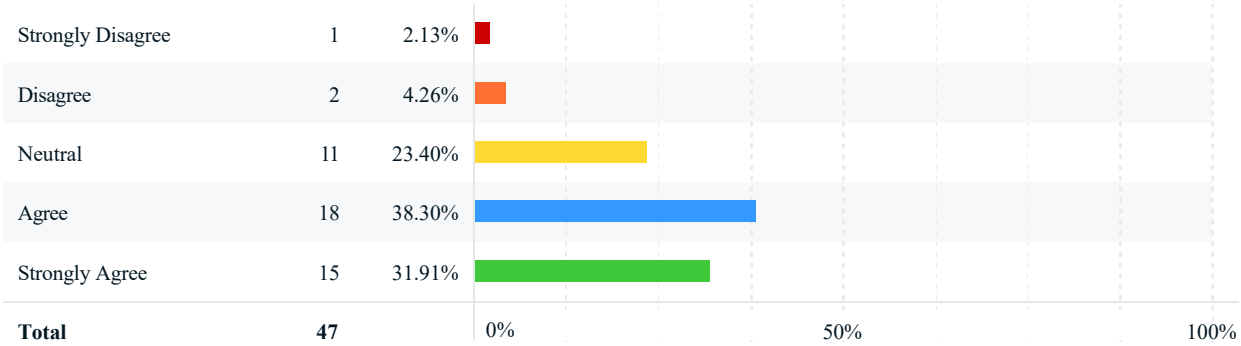
Course assessments were directly related to what was covered/practiced in the course.



Statistics	Value
Mean	3.81
Standard Deviation	1.16

Course assessments provided the opportunity for me to demonstrate an understanding of the course material.

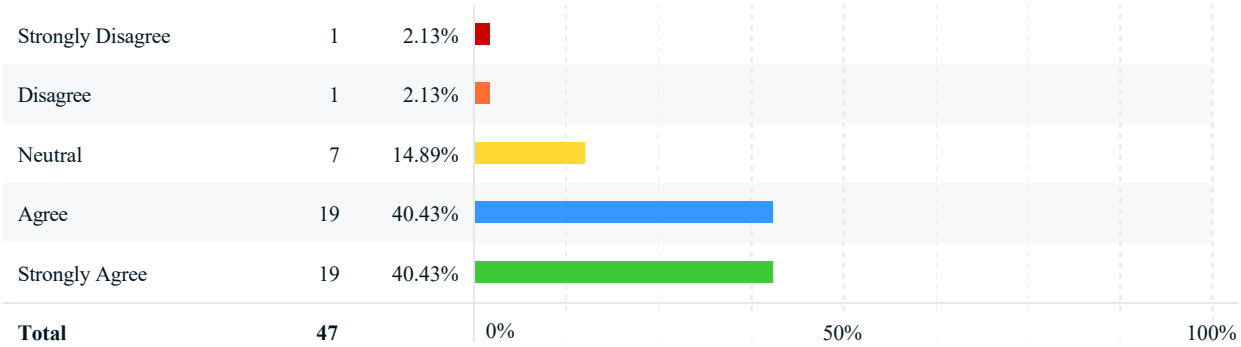
Course assessments provided the opportunity for me to demonstrate an understanding of the course material.



Statistics	Value
Mean	3.94
Standard Deviation	0.96

This course's online materials in Canvas were well organized.

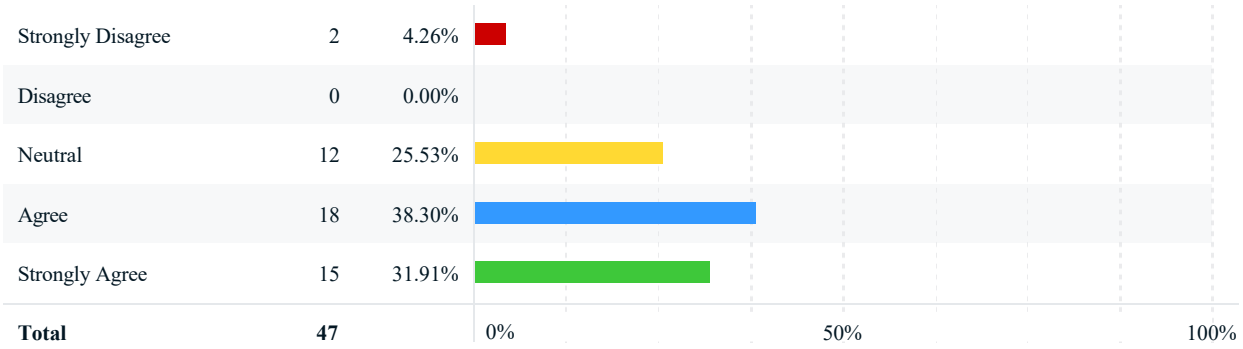
This course's online materials in Canvas were well organized.



Statistics	Value
Mean	4.15
Standard Deviation	0.91

Grading standards and policies were clearly communicated.

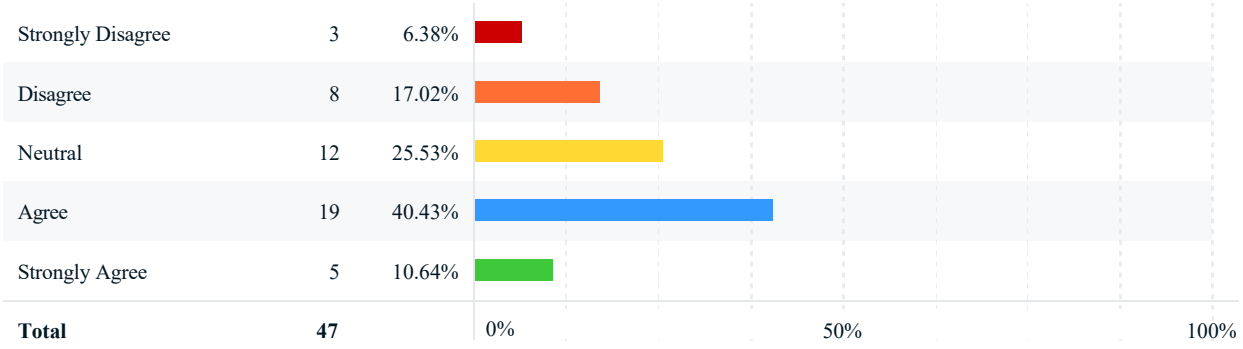
Grading standards and policies were clearly communicated.



Statistics	Value
Mean	3.94
Standard Deviation	0.99

Group work contributed to my understanding of the course material.

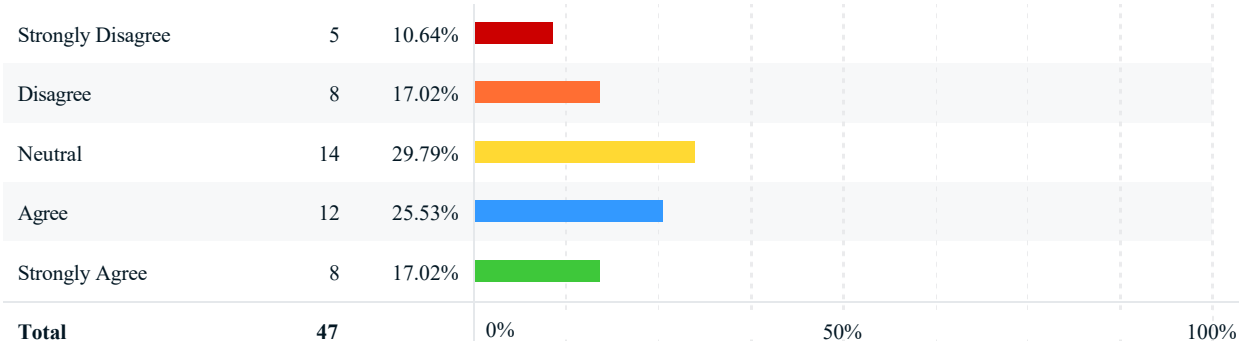
Group work contributed to my understanding of the course material.



Statistics	Value
Mean	3.32
Standard Deviation	1.09

Group work helped to develop my skills in problem-solving.

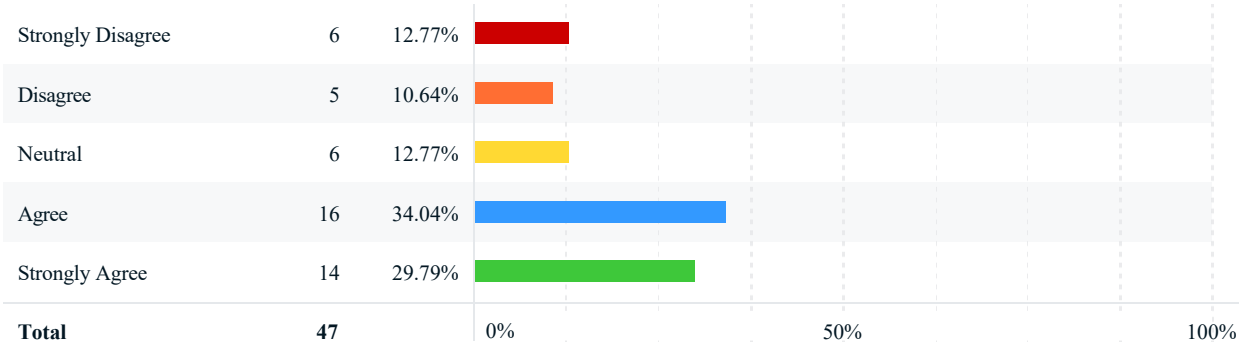
Group work helped to develop my skills in problem-solving.



Statistics	Value
Mean	3.21
Standard Deviation	1.23

The specific assessment and grading structure for this course facilitated my learning.

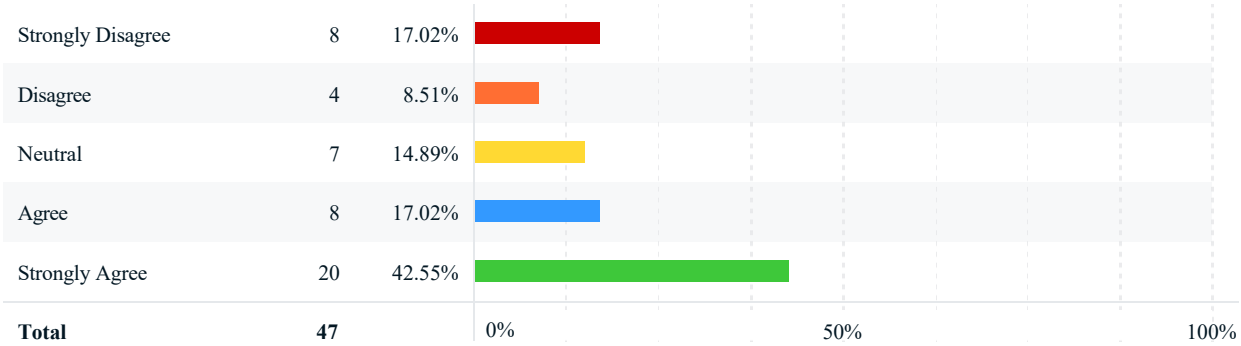
The specific assessment and grading structure for this course facilitated my learning.



Statistics	Value
Mean	3.57
Standard Deviation	1.36

The specific assessment and grading structure for this course was preferable to having 3-4 big exams.

The specific assessment and grading structure for this course was preferable to having 3-4 big exams.



Statistics	Value
Mean	3.60
Standard Deviation	1.53

Please share your thoughts on the assessment and grading structure for this course. If the general idea was good, what should I keep, and what should I change? If the general idea was bad, please say so!

Students
made everything about taking calc less stressful because there were many ways you could get more points in this class. I wouldn't change anything
Was definitely a great grading structure.
While I understand the grading system, I would argue that certain expectations of content knowledge were extremely high when we went in to take certain quizzes. I would argue some of the content was even more challenging or confusing than we would expect going in. I would've preferred more time between assessments, more trial and error with the professor, and so forth to get a more steady idea. As someone who typically struggles with math, and someone who has not taken Calc 1 before, I believe this made the overall experience very hard.
I really liked this grading system, the learning outcomes and big ideas on the final are still a little confusing to me, but overall I loved my individual assessments being worth more than a cumulative exam!
I thought the assessment and grading structure were good and should be kept the way they are.
It was fine, I just hated it because I was terrible at Calc.
I really liked that we didn't have big exams like other courses. Weekly small exam was keep putting me on the track.
I like the idea, I wish there was like one or two more opportunities for engagement points. But other than that I really like the ability to try again, since for me I feel like getting feedback really helps me do better on the next one. Or sometimes I just need a little bit longer to fully grasp the concept.
I thought the grading system was good
I feel the grading idea was bad. The concept of the big ideas doesn't give students a fair judgement on their knowledge because the questions were challenging. The fact that getting 2 right and getting the whole big idea not counted for doesn't seem right.
I thought the progress checks were a wonderful idea. Group excursions need work (recitation in general... well I think that its just because of my recitation TA–but that's not a critique your lectures.)
It was great the way it was laid out for the most part.
Everything was good. The course is very manageable if you actually study and keep on top of work.
It seems like the idea has potential but in practice it's frustrating getting 2/4s on the assessments which are then just 0's. I honestly would've preferred larger style exams where partial credit could be earned for work.
I think the general idea was good, it however is very overwhelming the first few weeks because of all the options that are presented to you to get participation points. So recommendations of what exactly is needed to do to get certain grades (not just how many points you need by x time in the semester to be on track for x grade) would be great. The weekly progress checks are good, especially being able to retake it the following week was great for learning from previous mistakes and improving.
I loved the structure of the tests
I think you should keep the multiple attempts at progress checks, although, I found some material on the progress checks to be very difficult to complete compared to our lecture and textbook examples. In general, it is a good idea

Students
For me personally I feel maybe 3–4 exams and a final is better (with lowest dropped) which spans multiple units up to 15 questions would be better.. I feel the concept of multiple retakes has led me to have bad time management and I feel 4 questions is sometimes not enough to truly assess a students understanding.. I sometimes feel the big idea questions can be a bit vague of what concepts are precisely being tested on some occasions.. However with having health issues early on that are now being resolved and all past papers avaliable perhaps 3rd attempt will be better. One last thing is maybe Ion cappa should be less engagement points.
I really liked the structure because while it was more work to study every week for tests it was less overwhelming than studying for a large exam
The idea of it was good in theory, however, the explantation of the grading system, and the weekly exam schedule was horrid. I think you should keep the retakes, but throw away the weekly quizzes, as a student taking the normal amount of credits i had to neglect my other classes just to barely get by in this one, and i do believe this is because of the structure and not me as a student.
N/A
I really liked the assessment structure please keep it.
I liked everything except that when learning derivatives, we were required to use certain algebra rules instead of trying to take harder derivatives. I'm much better at derivatives than algebra so that was hard for me.
it is extremely hard to "master" a big idea sometimes because of the stress of having to finish 2 quizzes in a time span. if there was just 1 quiz a week and retakes could be given in recitation, that would make things a lot more doable and less stressful.
I like the smaller progress check format as it allows for easier preparation instead of rushed and in less detail that usually comes with bigger exams.
the idea was very good the group assignments felt like a wast of time, no one o spoke to found any benefit to them other then getting the points for it.
I honestly would prefer exams. If you were to keep the structure of having one quiz a week, maybe make it just a bit longer so partial credit can be given.
i think i actually prefer having 3–4 exams weirdly enough. it was just a lot of pressure weekly
I would change the grading system. It is tough for students to score a good grade in the class. The big ideas and learning outcomes system is not great. It is very disappointing when you score a 3/4 and get a big idea, but just get 2 learning outcomes because you didn't get both the questions correct for a single learning outcome. I would rather have flexible quizzes and keep it simple, or a midterm would be better. I also don't like the fact that the lowest score you get from the three (big ideas, learning outcomes, engagement points) is your final grade. I would teach more practice problems in class and spend a lot of time with a lot of different examples. The content that was on the progress checks was a little advanced and never really taught at that level in class.
It is very well structured if you stay on top of things like you should be! If not then its very easy to fall back and it makes it very hard to recover, but its like that in any course.
I really enjoyed the style. Maybe the Tuesday before lecture though, go over what is on the Big idea on Wednesday because it did not always correlate to what we were learning the day before.
I wish there was more opportunity for engagement points and more leniency to how many big ideas we have to get
I think the general idea is good but it could be frustrating to miss big ideas and learning outcomes because the structure of the questions were unfamiliar from attempt to attempt
I liked the idea overall, just some weeks the double quizzes were hard to study for.

Students
I think it’s okay
I liked the grading system alot, and the lenient grading system.
I found it hard to understand the grading structure for this course, but once I understood, it was a little too late. I appreciate the multiple attempts at the quizzes, but I found it hard to keep up with. The homework and LON CAPA didn't really help me with the progress checks. Until around the end of the course did I figure out what I could expect from the progress checks.
I think the grading is a little harsh with trying to get 3/4 even though you udnerstand the learning outcome.

If you used Paul's Online Math Notes or some other outside resource, was it useful? If applicable, compare to the course textbook.

Students
i tried pauls once or twice but preferred the textbook
Yes, I used Paul's Online Math Notes, but I also used ChatGPT a lot to provide me with practice problems of different varieties that where similar to in class examples, and that was a huge help for me in preparing for quizzes.
Sometimes. I felt as though Paul's Online Math Notes, while good, tended to lack certain areas of complication compared to some of the questions (or became too hard).
I really liked Paul's OMN, there was some things not explicitly sectioned off like integration by parts, but overall it was useful as my main source of practice problems.
It was useful to the course material, I think it resembled the progress checks very similarly.
No.
I used Paul's Online Math Notes, and it was helpful.
It was, I used Paul's Online Math Notes and it was almost perfect. The only problem was sometimes I couldn't find enough questions for the problem sets.
I used the textbook
Yes, Paul's online math notes was useful. I feel it was better than the textbook.
Yes Paul's is helpful I used it a couple times.
I am using Paul's Online Math Notes, and it's pretty helpful.
Pauls online math notes was great. I didn’t use the textbook.
I used Paul's Online Math Notes. Sometimes the textbook went too far with the difficulty of the problem or it was too tedious with its parts.
going back and forth with chatgpt asking it to explain and create problems and my TA helped me most.
Paul's Online Math Notes were pretty helpful but I found most of my help from Khan Academy and Organic Chemistry Tutor on youtube.

Students
I mainly used the course textbook.
Yes pauls online math notes was helpful
NA
Organic chemistry tutor was very helpful. The course textbook layout for graph questions is horrid online, other questions from textbook are fine.
I only used the course text book
I liked pauls up until the integral units only because you had nuances in what was going to be on quizzes that they didnt cover. I think the textbook and pauls though are pretty equal as i had this porblem with the textbook as well.
N/A
I used Paul's online math notes and the organic chemistry tutor's calc videos on yt very often they were useful
I only used the textbook
yes it was very helpful
Yes, I used Paul's Online Math Notes for all of my practice problems and found them helpful
n/a, used textbook.
Yes, Paul's online math notes were great and made it so I did not need to purchase a textbook.
it was useful. i used pauls online math notes
Yes, the notes and textbook were helpful
I mostly used the textbook but I found both to be helpful!
I used Paul's Online Math notes, I though it was useful for practice problem sets
yes
Paul's Online Math Notes was useful but finding exactly what we were covering could be difficult
Yes it was useful but there were some topics that were not covered on Paul's Online Math Notes.
Only the online notes
I liked the course textbook.
I didn't find either the course textbook or Paul's Online Math Notes helpful. I didn't know what to focus on, and would do random questions in the question pool that wouldn't appear in the class lecture or the progress check, so I wasted some time figuring out how to complete those questions. It was also hard to finish each section of the homework because sometimes it would not be a topic in class that we had covered yet, but then I would not have enough time to complete the assignment, or I would have to rush.
Only the textbook

Is there anything else I should know?

Students
no
You did a great job! Thank you!
I appreciate all the support and respect you show your students so, so much. But I do feel as though the grading system just did not click with me as a student.
Nope!
The professor should go over terms and definitions before jumping into a new topic.
N/A
N/A
No
Keep up the positive energy! I love seeing a real passion behind a teacher.
So I feel like we went over limits in the beginning of the semester quite fast and then never used them again. They supposedly help you with derivatives but they did not help me at all if anything they hurt me. I didn't get the first two big ideas of limits, but then I got every single big idea after that including all learning objectives. This doesn't make a lot of sense to me because I understood everything to do with derivatives but somehow I can't get limits. Therefore my opinion is that limits really had no purpose or benefit in my learning experience whatsoever, but I do understand that you probably need to teach them.
No.
N/A
N/a
NA
The ta selections are good.
Thanks !
I think a better job has to be done with timing, pace, and practice. As you have say, its all about practice, so more time to do so would be beneficial.
N/A
No
the practice problem sets should be worth more engagement points because they take a long time to do
N/A
you are new at this, you did good, but about halfway through i figured our how to learn it at home without spending extra time on material, this worked better for me only attending necessary test days, most recitations, and most group work days.
I absolutely love your attitude for math. Your lectures were great because of your enthusiasm, and you did a wonderful job of conveying what you were trying to teach. The only issue I had with this course was the quizzes, but if that were to change, I think it would be a perfect class! Great job this semester Edison.
nope, thank you

Students
Not that I can think of
No
It would be helpful to know what to study for the progress checks.
No

University of Pittsburgh

2261 - Teaching Survey Fall 2025

Fall 2025 - Edison Hauptman MATH 0220 - ANALYTC GEOMETRY & CALCULUS 1 - 1025 - Lecture

Created **Wednesday, December 17, 2025**

Courses Audience: 68
Responses Received: 57
Response Rate: **83.82%**

Report Comments

Teaching Surveys

Included in this report:

- Summary of responses to scaled questions
- Response breakdowns
- Student comments
- Results to instructor added custom questions (if applicable)

Understanding and using student feedback:

- We have resources to help you interpret and use results including our faculty worksheet with guided prompts and space to record summaries of feedback, actions, and outcomes.
- Members of our Pedagogy, Practice, & Assessment team are available for consultations and can help with:
  - Interpreting OMET results and developing a course of action if necessary.
  - Exploring various methods of assessment to improve teaching.
- In the future:
  - Discuss, teach, and model giving meaningful feedback with your students and give them multiple opportunities to practice giving feedback.
    - Gather important information about students at the beginning of the term by giving a pre-course survey.
    - Check in with students half way through the term by giving a midterm course survey.
- The Teaching Center offers multiple resources to support teaching and learning.

Office of Measurement and Evaluation of Teaching (OMET)
Contact us

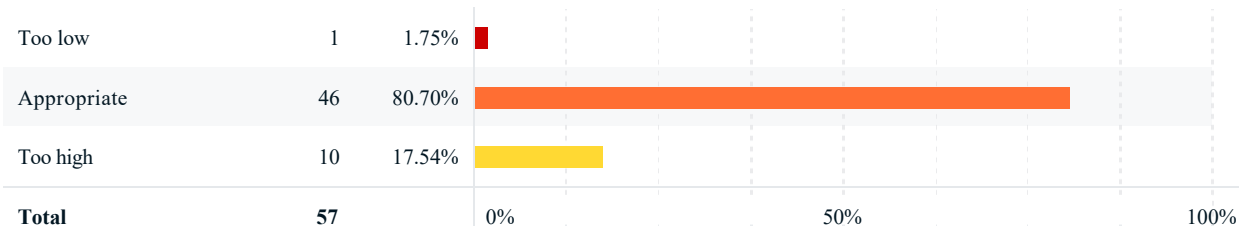
Dietrich School of Arts and Sciences Questions

Summary table

	Invited Count	Response Count	Response Rate	Mean	Median	Mode	SD
The instructor created an atmosphere that kept me engaged in course content.	68	57	83.82%	4.18	4.00	4	0.87
The instructor was prepared for class.	68	57	83.82%	4.37	5.00	5	0.77
The instructor treated students with respect.	68	57	83.82%	4.58	5.00	5	0.71
The instructor was available to me (in-person, electronically, or both).	68	54	79.41%	4.46	5.00	5	0.86
The instructor evaluated my work fairly.	68	56	82.35%	4.39	5.00	5	1.00
The instructor provided feedback that was helpful to me.	68	56	82.35%	4.21	4.50	5	1.06
I learned a lot from this course. If there is no basis to judge or not applicable, answer N/A.	68	56	82.35%	3.93	4.00	4	0.99
Overall of All Questions	476	393	82.56%	4.30	-	-	0.90

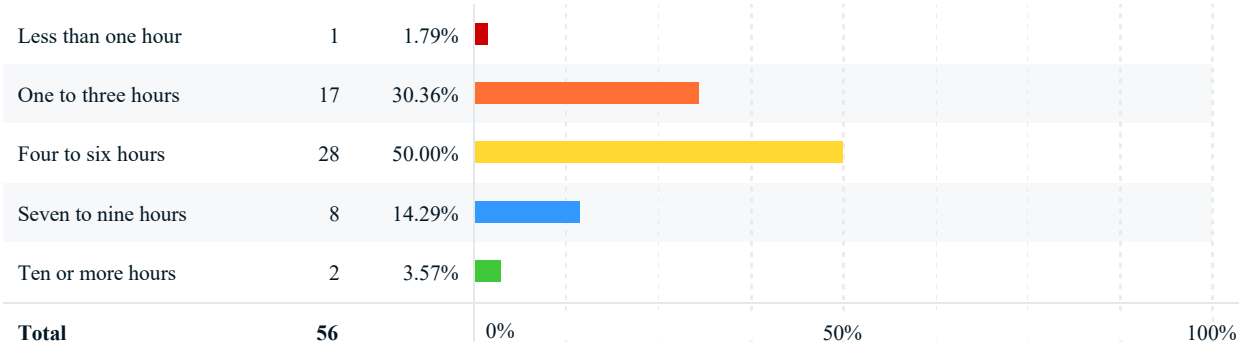
The standards the instructor set for me were:

The standards the instructor set for me were:



How many hours per week did you usually spend working on this course outside of classroom time?

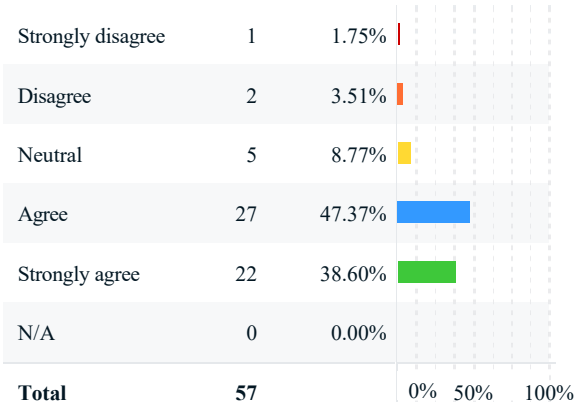
How many hours per week did you usually spend working on this course outside of classroom time?



Less than one hour	1	1.79%
One to three hours	17	30.36%
Four to six hours	28	50.00%
Seven to nine hours	8	14.29%
Ten or more hours	2	3.57%
Total	**56**	

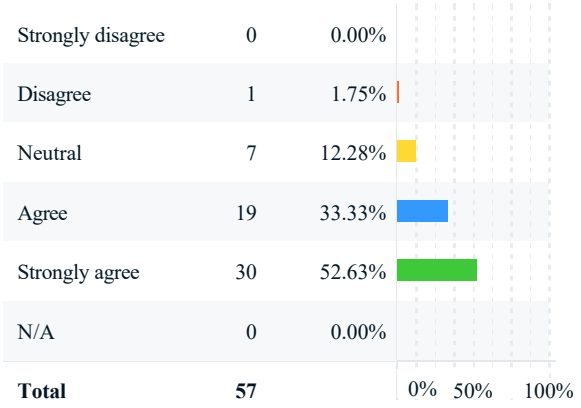
Response breakdown

1. The instructor created an atmosphere that kept me engaged in course content.



Strongly disagree	1	1.75%
Disagree	2	3.51%
Neutral	5	8.77%
Agree	27	47.37%
Strongly agree	22	38.60%
N/A	0	0.00%
Total	**57**	

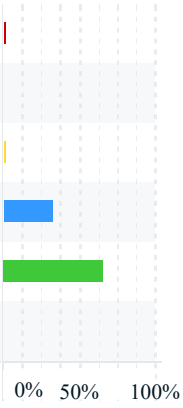
2. The instructor was prepared for class.



Strongly disagree	0	0.00%
Disagree	1	1.75%
Neutral	7	12.28%
Agree	19	33.33%
Strongly agree	30	52.63%
N/A	0	0.00%
Total	**57**	

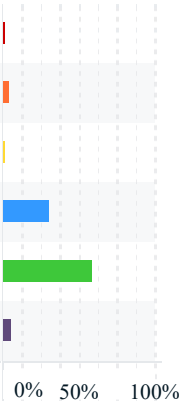
3. The instructor treated students with respect.

Strongly disagree	1	1.75%
Disagree	0	0.00%
Neutral	1	1.75%
Agree	18	31.58%
Strongly agree	37	64.91%
N/A	0	0.00%
Total	**57**	



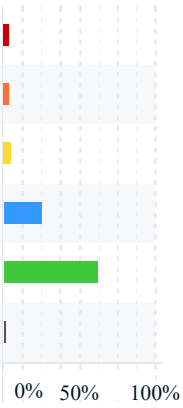
4. The instructor was available to me (in-person, electronically, or both).

Strongly disagree	1	1.75%
Disagree	2	3.51%
Neutral	1	1.75%
Agree	17	29.82%
Strongly agree	33	57.89%
N/A	3	5.26%
Total	**57**	



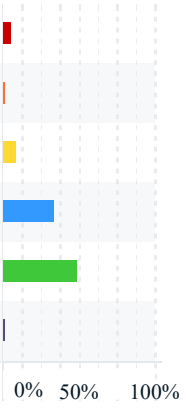
5. The instructor evaluated my work fairly.

Strongly disagree	2	3.51%
Disagree	2	3.51%
Neutral	3	5.26%
Agree	14	24.56%
Strongly agree	35	61.40%
N/A	1	1.75%
Total	**57**	

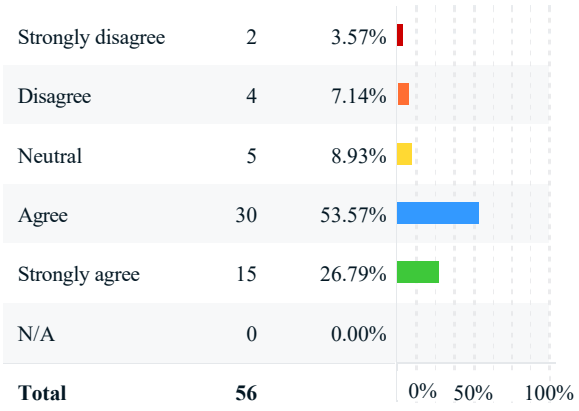


6. The instructor provided feedback that was helpful to me.

Strongly disagree	3	5.26%
Disagree	1	1.75%
Neutral	5	8.77%
Agree	19	33.33%
Strongly agree	28	49.12%
N/A	1	1.75%
Total	**57**	



7. I learned a lot from this course. If there is no basis to judge or not applicable, answer N/A.



DSAS Comments

What did you like best about how the course was taught?

Comments
All of the point opportunities.
the quizzes method
Nothing, the professor was absolutely horrible at teaching concepts in a way for non math students to understand
I like how the canvas was organized!
He is very engaging and everything about the way the course is set up is perfect 10/10. No exams, but weekly checks to stay on topic and chances to retake and earn more points.
I liked how he established the grading system day one which i really liked it made learning calc very very easy
engagement
The grading structure is better than the typical calc class
There was so much flexibility in the course/curriculum. This allowed me to remain focused, and not stressed out.
I like how the quizzes where set up. This format helped me to master topics more easily.
The engagement point style of the grading allowed me to construct my own path to the grade I wanted
Was interactive
I liked how the professor is very understanding and if you need help on something, he will help you, so you don't fall behind.

Comments
I enjoyed the theory he provided for each calculus concept. Providing greater in–depth analysis surrounding topics while also clarifying simple misunderstandings.
I liked the progress checks a lot. It gave me time to understand the concepts instead of cramming and forgetting the information.
progress checks
I liked the progress checks every week instead of a few big exams.
I liked the structure with multiple chances at each big idea.
It was very interactive and dove into concepts deeply, making them easier to understand.
I really valued the way the material was taught in lecture, the examples were concrete and his lecture kept me engaged.
i thought it was taught at a good pace
I liked going through the examples
All lectures were super detailed and easy to understand
I liked how the material was presented during the lectures.
Honestly nothing, I felt Edison was not respectful of me when I tried to ask questions before or after lecture. He moved to fast for me to understand what he was saying, and explained all concepts in a way that made it more challenging to understand.
The instructor went at a pace that was comfortable for the class and went as fast as we were ready.
I liked how this course was designed in a way to track your progress and didn't depend on 1 or 2 huge exams. It made the class a lot more approachable and far less stressful.
I enjoyed learning the topics he provided and sowed what we have to know and stuff not to worry too much about. I liked how there are many opportunities to get engagement points and stuff like that. There were some hard topics but he did try his best to break it down slowly and show how each topic functions.
i like how there was no midterm
I liked how it was structured and organized.
I liked the evaluation method
I really liked the grading structure since progress checks ensure you're learning the main ideas of Calc 1, while extra learning outcomes and group excursions add to the engagements points. There's more incentive to learn with this structure, rather than a "Will I pass the exam" mindset and then forgetting the information later.
very interactive with the students
I liked having multiple quiz attemots
I liked how he explained things
n/a
The way how to solve indefinite integrals.
I really liked the flexible structure of the course, especially how there were no midterms but rather weekly quizzes.

Comments
I love and hate how this class was structured. It was so different from any class I have taken before but with a wonderful teacher I think this class was great. I really like lectures and even though it was a late class I was engaged for most of it. But if the teacher was bad then I could definitely see this class as being awful.
I liked the idea of weekly progress checks, but they were difficult
Teaching Style
I really like the Big Idea concept. I made it more manageable to learn every part of calculus.
He was very concerned about student progress and ensuring that students had multiple opportunities to demonstrate mastery of the material. Was very available to students as well and provided multiple opportunities for practice to be graded.
The instructor taught each concept with many examples and different ways to solve the problems.
I like that when ideas were given, background was explained so we understood conceptually. We were taught the “why?” behind equations, formulas, and theories which was very helpful.
I had no exams.
I really liked how each week you could focus on a topic because it keeps you up to date with what you need to study. Otherwise, I can push work off to the very end and then try to cram stuff in, but that usually means I wouldn't have been prepared. I really appreciate how you decided to take a different path to calculus, and I think it helped me learn better and become more motivated. It was by far the most engaging math class that I have taken at Pitt so far.
I like how every student was treated fairly.
i really like the grading system and i think it’s super fair to let us retake the quizzes because it really is about how we improve in the course and not just what we can regurgitate out on some certain day.
It was well structured and bridges into each other smoothy
I appreciated that Edison was open to answering many questions during the lecture.
I like how during recitations, the TA would go over example problems based on the upcoming quizzes

If you were teaching this course, what would you do differently?

Comments
Integrate more of the homework into the actual lectures (going over problems or referencing past problems).
Everything
I wish the the professor and TA's had a bit more communication.
nothing
i wish that we could still earn learning out comes even with a 2/4
show more examples
Less learning outcomes required to achieve a better grade

Comments
Nothing
Nothing.
Be more transparent about how the engagement points worked from the beginning as they can be quite confusing if you have no experience with that grading style
Nothing
If I were to teach this course, I would most likely not make the quizzes so difficult. I felt like the quizzes were very difficult outside of the scope of our understanding.
I would push progress checks to the second recitation because Monday progress chekcs were hard if some material was not covered yet.
study tips/guidance
I would make worksheets or upload problems to canvas so students could look ahead.
N/A
N/A
I think the format for this class was very different and also very great at keeping constant engagement throughout the semester. I think the one issue that I had with it is that some questions on the progress checks were very difficult (the 2nd optimization problem) and so your whole perception on that third progress check becomes skewed because you don't know what to expect. It creates a thinking that you may just have to skip over that topic in order to put time and do good in the others.
i would do more extra credit
Less time on the proofs because it became confusing
Explain in a different way
The grading system is honestly corrupt. I hate how if you get 2 out of 4 on a quiz, it basically counts for nothing. I would take 50% over 0% any day. At least a 50% shows that I understood something, but the way it's set up now, it feels like all that work just disappears. This system is dragging my overall grade down for no good reason. You can say "grades don't matter," but they absolutely do. When everything is calculated in the end, every single percentage matters. I'd rather have even 25% counted toward my final grade than a straight zero just because I only got one question right. It makes no sense that partial understanding earns the exact same score as not trying at all. How is that supposed to motivate anyone? The whole system feels like it punishes effort instead of rewarding progress. If the goal is to reflect what we actually know, then the grading should show that and not erase it. Yes, there are multiple attempts, but if I still fail to reach 3 out of 4 questions I get 0% on that unit. Like I would take partial credit anytime. The grading system has to change because it doesn't motivate and there's no reward for partial understanding. Please change the grading system!!!!!!!!!!!!!!!!!!!!
Everything honestly.
I would try to assign more concrete homework problems with answers already worked out for them to check over.
Honestly, nothing. The course was set up in a very unique and fun way that kept the class engaging.
If I did something differently I would change up the teaching like download a file online took take some notes and from there I would project to write down what to write on the notes and show example problems and I would mix it with writing on the board too. If someone or many people are struggling I would be glad to help them right away and show easier ways to solve a problem.

Comments
instead of the retakes being harder make them close to the same difficulty because the kids have to study for another big idea and the retake which was always harder. the increase in difficulty made it hard for people who passed the first attempt to do it let alone people who were already struggling on the topic. So i would make the retake around the same difficulty just so the student can prove they know how to do it, not punish them for not doing well on the first attempt
I would do nothing differently.
I wouldn't change anything
I wouldn't change anything.
nothing
The quizzes were much to hard compared to the lecture content. Lectures rarely,y had multiple worked out examples to study from let alone examples that matched the rigor of the quizzes. I also think the all or nothing aspect of the quizzes leaves little room for error– perhaps on a quiz retake you would only need to get the questions wrong that you missed the first time to earn the credit. It was very difficult to study the amount I needed to for this class on top of doing what I had to do for the other requirements and my other courses. Created a stressful environment weekly
Nothing at all
n/a
I will be faster in explaining the basic concepts and show more example problems in class.
Nothing really. I just didn't like how late the class was but again, that's not really in the instructor's control
I would just tweak the final grading tracker slightly and maybe shift all the big idea requirements down one for each grade.
For the progress checks, I grew to like them. I would have liked to see them more spaced out, like a culminating exam every two weeks or so. That way i wouldn't have to worry every sunday
I will do the same
I would make sure that the practice problems aligned with what we were doing in class. Sometimes the practice problems were not related to what was on the Big Idea for the week so it felt like I was doing extra work for no reason.
I would offer a mock final exam and give an estimate of how much time each problem should to take while taking the exam or quizzes.
Sometimes we would get lost in problems during lecture which made concepts hard to grasp. Better going through them before lecture would be beneficial just to work out any kinks in the problem.
Try to effectively solve problems and show each step.
I wouldn't do much differently. I think you can refine the system of learning outcomes, etc, depending on the data/results you get from the end of the year.
I would not have a Bid Ideas, I would just have mid terms. Personally I do better in math when particle credit is given and this course did not have this.
I felt like the quiz questions were too hard sometimes or very complex compared to what other teachers were testing their students on. It felt too much at times just to get a 3/4 knowing that one of the 4 questions would be an extremely difficult one.
Change the grading system. Each practice problem sets should have an option do practice more problems and thus gain more engagement points. Additionally intentionally practicing as seperate harder set of questions would award more point.

Comments
Make study problems for each module we go over. I didn't really like having to use the book or even the LON–CAPA to do practice questions.
The homework's that were due the Monday right before the quiz weren't usually related to the topic of the quiz, so I would change it to where they were always related.

Math TA/TF Additional Items

Please answer the following questions about your instructor:

1. Did you experience difficulty in comprehending your lecture instructor's spoken language in class?

No difficulty a…	47	83.93%	
Small amount o…	5	8.93%	
Moderate amo…	4	7.14%	
Severe difficulty	0	0.00%	
Total	**56**		0% 50% 100%

2. Did your lecture instructor experience difficulty in comprehending the questions that were asked by students in class?

No difficulty a…	42	75.00%	
Small amount o…	13	23.21%	
Moderate amo…	1	1.79%	
Severe difficulty	0	0.00%	
Total	**56**		0% 50% 100%

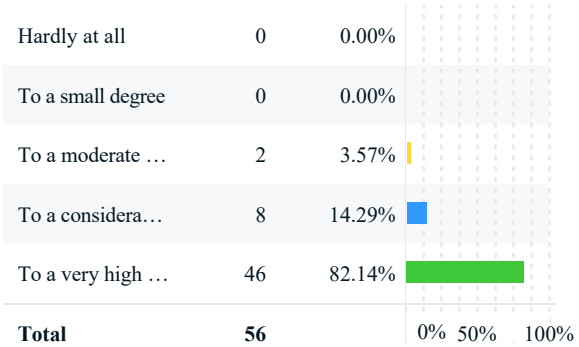
The lecture instructor's writing on the chalkboard was legible.

The lecture instructor's writing on the chalkboard was legible.

Seldom	1	1.79%	
Sometimes	0	0.00%	
About half the …	2	3.57%	
Usually	22	39.29%	
Always	31	55.36%	
Total	**56**		0% 50% 100%

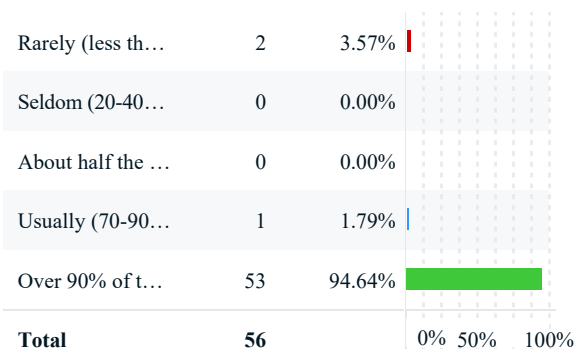
The lecture instructor's attitude toward the subject was enthusiastic.

The lecture instructor's attitude toward the subject was enthusiastic.



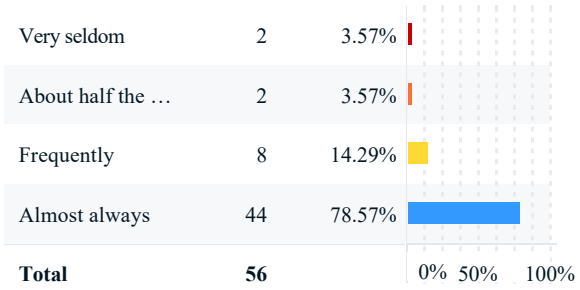
The lecture instructor arrived for class on time.

The lecture instructor arrived for class on time.

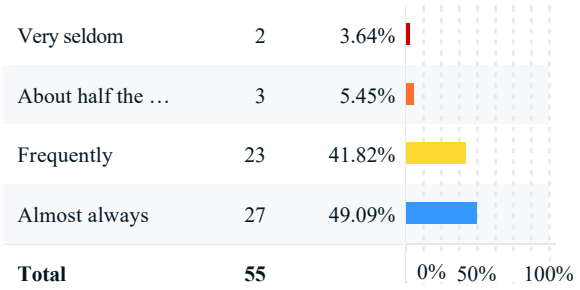


Please answer the following questions about your instructor:

1. The lecture instructor provided the opportunity for questions.

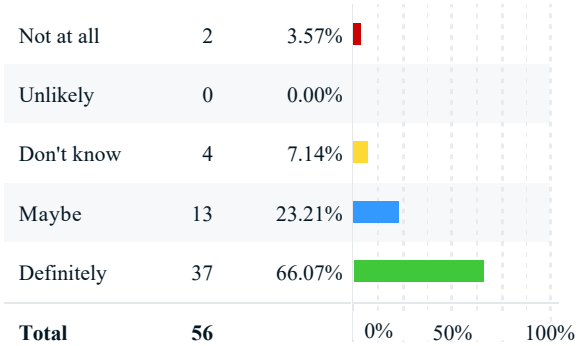


2. Helpful answers were given to questions raised in class.



Would you recommend this lecture instructor to a friend taking this course?

Would you recommend this lecture instructor to a friend taking this course?



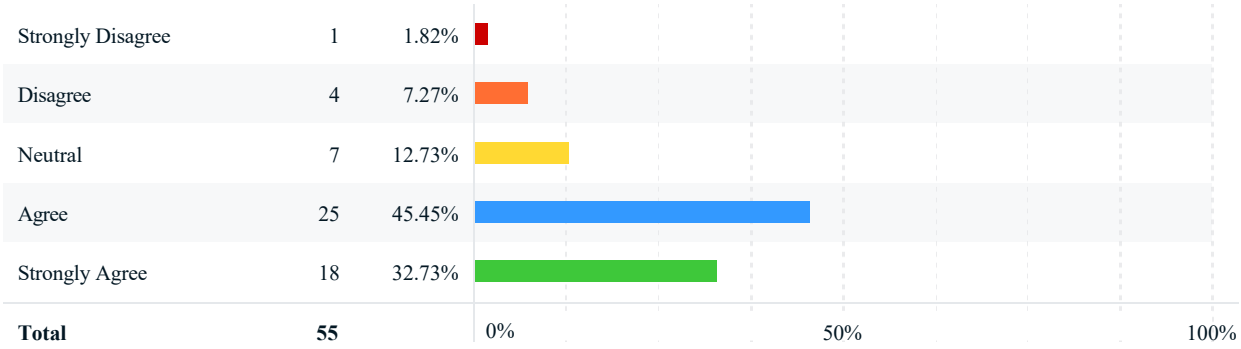
Diversity and Inclusion

Question	Response Count	Response Rate	Mean	Mode	Median	SD
The instructor creates an inclusive learning environment for all students.	56	82.35%	4.52	5	5.00	0.74

Personalized Questions

Course assessments were directly related to what was covered/practiced in the course.

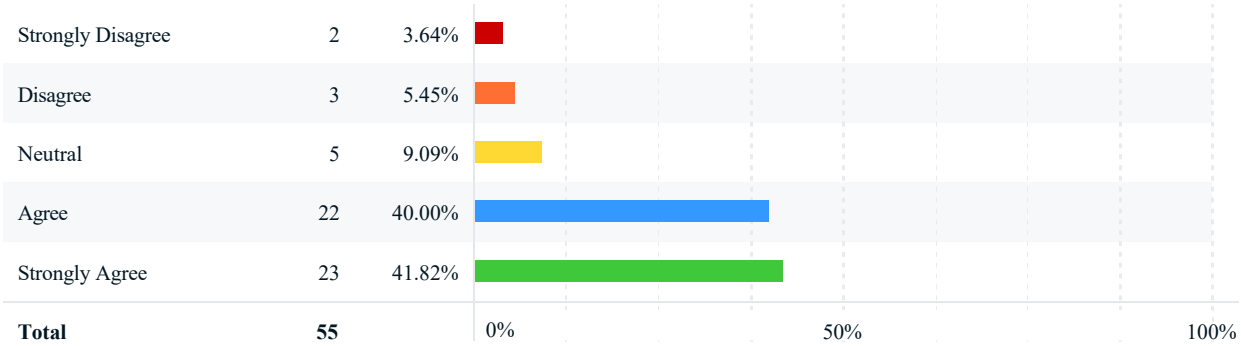
Course assessments were directly related to what was covered/practiced in the course.



Statistics	Value
Mean	4.00
Standard Deviation	0.96

Course assessments provided the opportunity for me to demonstrate an understanding of the course material.

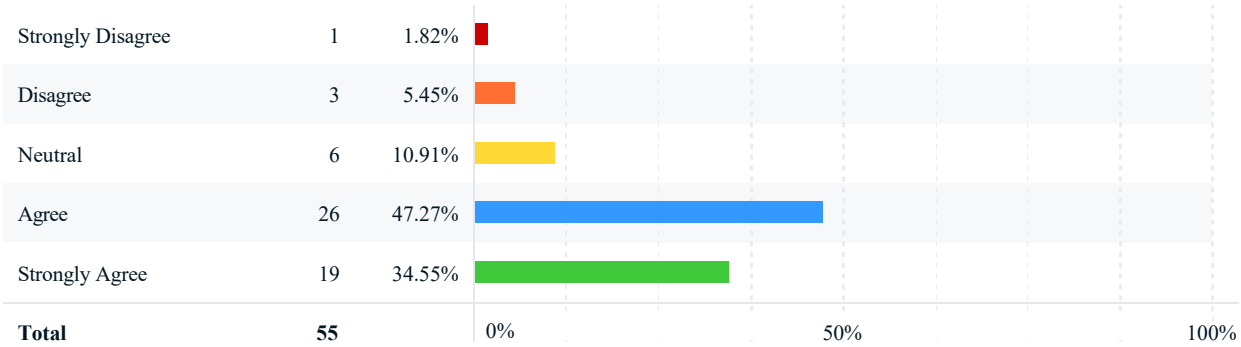
Course assessments provided the opportunity for me to demonstrate an understanding of the course material.



Statistics	Value
Mean	4.11
Standard Deviation	1.03

This course's online materials in Canvas were well organized.

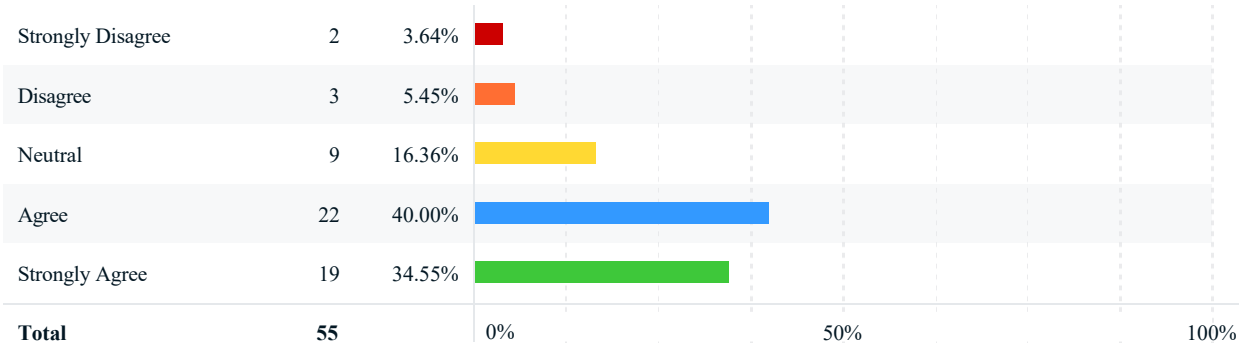
This course's online materials in Canvas were well organized.



Statistics	Value
Mean	4.07
Standard Deviation	0.92

Grading standards and policies were clearly communicated.

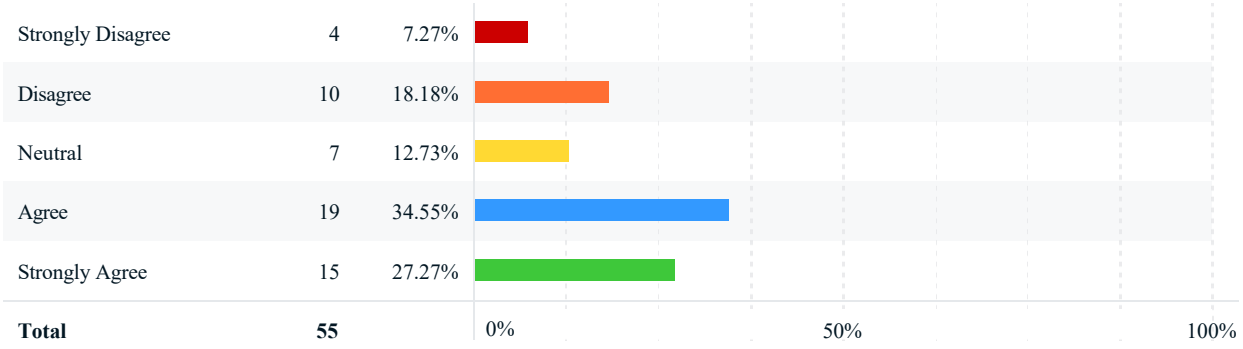
Grading standards and policies were clearly communicated.



Statistics	Value
Mean	3.96
Standard Deviation	1.04

Group work contributed to my understanding of the course material.

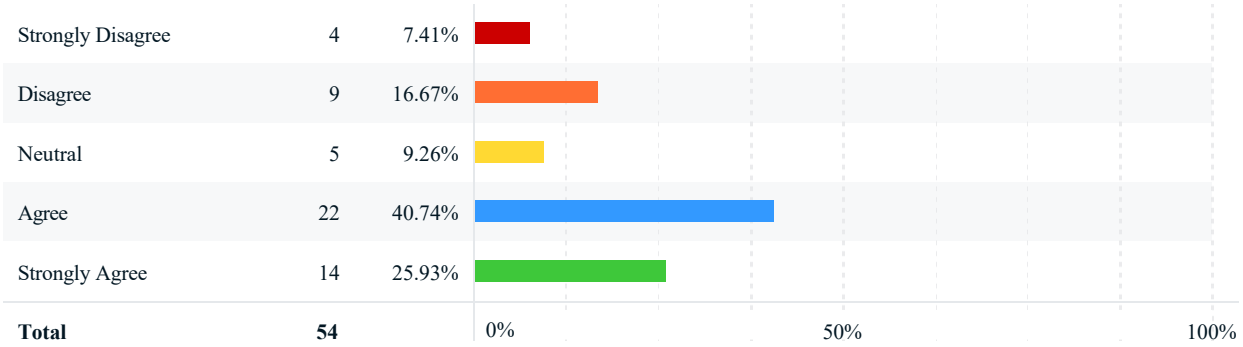
Group work contributed to my understanding of the course material.



Statistics	Value
Mean	3.56
Standard Deviation	1.27

Group work helped to develop my skills in problem-solving.

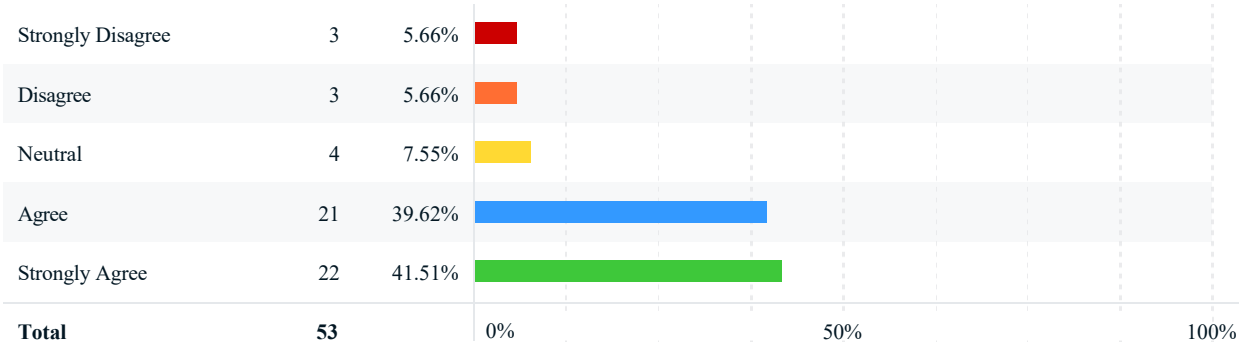
Group work helped to develop my skills in problem-solving.



Statistics	Value
Mean	3.61
Standard Deviation	1.25

The specific assessment and grading structure for this course facilitated my learning.

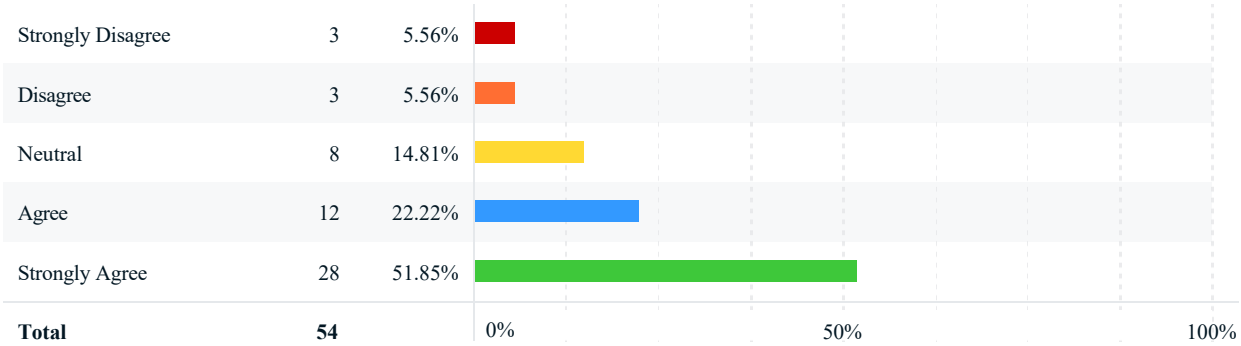
The specific assessment and grading structure for this course facilitated my learning.



Statistics	Value
Mean	4.06
Standard Deviation	1.12

The specific assessment and grading structure for this course was preferable to having 3-4 big exams.

The specific assessment and grading structure for this course was preferable to having 3-4 big exams.



Statistics	Value
Mean	4.09
Standard Deviation	1.19

Please share your thoughts on the assessment and grading structure for this course. If the general idea was good, what should I keep, and what should I change? If the general idea was bad, please say so!

Students
Too many learning outcomes required to pass since many of the big ideas only had two learning outcomes
The general idea was great, and provided a lot of benefits compared to a standard 4–exam curriculum. I do think that the grading sheet that we were all given was a little confusing/overwhelming, but other than that, it was great.
I loved the assessment and grading structure of this course. I would not change anything about it.
Keep the learning checks and the big ideas
Honestly, I loved this course so much. I loved how we don't have quizzes or midterms, but not only that, but I do agree that our progress checks are harder. Compared to what my friends have on their quizzes, the ones on the progress checks are way harder. Even though its way harder, I feel like it does exercise our brains more, making me think more, which does prepare me for the finals. I feel like you can continue the way you teach.
I loved the way the class was structured because it gave me room to make errors and correct myself based on them. The group excursions were fun, but I didn't really learn much from them.
great idea, gives us many chances to prove our knowledge of course material
The general idea is good, you should keep the weekly checks and the amount of retakes.
I liked the of it, made it more attainable to really understand each topic.
I thought the grading structure was nice, but it would also be beneficial to have more questions on the progress checks.
I thought the general idea was pretty amazing, I think that having this structure allows you to not become defeated easily if you do bad on one exam. That keeps people who are not traditionally from a math background motivated to keep going in this course. I really don't think it was that difficult to comprehend (just different). I think you should keep the progress checks, I would maybe consider changing the learning outcomes to one for each individual question. 4 learning outcomes per progress check, just because with some of them, if you get the first one wrong then there's no way you get the learning outcome.
i think that it is all very balanced
I thought it was fine it was just a little overwhelming sometimes
I didn't really like the grading system. I would've much rather had a midterm than the big idea progress checks.
The grading system is honestly corrupt. I hate how if you get 2 out of 4 on a quiz, it basically counts for nothing. I would take 50% over 0% any day. At least a 50% shows that I understood something, but the way it’s set up now, it feels like all that work just disappears. This system is dragging my overall grade down for no good reason. You can say “grades don’t matter,” but they absolutely do. When everything is calculated in the end, every single percentage matters. I’d rather have even 25% counted toward my final grade than a straight zero just because I only got one question right. It makes no sense that partial understanding earns the exact same score as not trying at all. How is that supposed to motivate anyone? The whole system feels like it punishes effort instead of rewarding progress. If the goal is to reflect what we actually know, then the grading should show that and not erase it. Yes, there are multiple attempts, but if I still fail to reach 3 out of 4 questions I get 0% on that unit. Like I would take partial credit anytime. The grading system has to change because it doesn't motivate and there's no reward for partial understanding. Please change the grading system!!!!!!!!!!!!!!!!!!!!
General idea was good, execution wasn't great. Get rid of group excursions they dod not add to anything, change entire structure of recitations they were not heplful, make the progress checks more approachable.

Students
I do think that the grade tracker is a good idea but it can get complex at times. I will say I do prefer a traditional grading scale.
I loved the idea. Please keep it as it is. I am positive many other students will feel similarly.
Maybe change the grading a bit, instead of 3/4 to get a big idea maybe go down to 2/4 or a way for many of the class to get a big idea each week. Homework wise I think that is okay so far. The format of the weekly big ideas were good it's that many people tend to forget what to on a problem because of maybe it is set up and are not used to seeing as to how it would be on a homework or notes. Maybe break each topic down so that students know secret ways to solve a problem. Other than that I did enjoy the class and how you taught the class, you did your absolute best in teaching the class and sharing your work.
i like the idea, i just don’t like the jump in difficulty for each retake. This is usually not the students only class and having to study for both of them usually can’t fit in their schedule
I think overall everything is good and well organized, nothing I would change.
I liked it because I had to understand the concept and not some crazy complicated problem
I believe that the grading structure is very good, though some changes could be made. Only four questions is not always enough to demonstrate what you understand, and those four questions are similar on progress check attempts 2 and 3 as well. While that means there is no good excuse to why you can't get the big idea on attempts 2 or 3 if not 1, some students may simply struggle with comprehending or mastering a particular idea that's guaranteed on one of the questions. Maybe having 6 questions per progress checks would be better.
i loved it
The idea was good but the execution made my life very stressful especially right now with a lot of pressure in the last two weeks of classes. Two to three big exams would help me to focus my studying whereas this structure meant I had to learn and master content in sometimes under a week since you had to get all the questions right if you wanted to earn the credit. The grading scale leaves little room for error which is stressful.
I like the general Idea
I like it now that im passing but for the first 4–6 weeks I was struggling. having the difficulty raised for the second attempt feels backwards. If a student didnt understand the material the first time increasing the difficulty for the next one is frustrating. I was playing catch up after idea 2 and it cost me probably 2 other big ideas in the process. Having the first quiz be easier is nice for grades but it really makes me question my abilities when I try the second for practice and get a 1/4. I would make the difficulty of both equal so if someone falls behind or tries to focus on a retake they dont shoot themselves in the foot by ignoring the current idea.
The website "Paul's math notes" is very good.
I really liked the overall structure of the course. I personally prefer lectures or independent work over group excursions, so those activities were not as helpful for me. That said, I recognize that different formats benefit different learners, and the group components may have been valuable for others even if they did not make a difference for me.
I think the grading structure is actually very good for the most part. It really allows each student to work on what they struggle with and progress. My only gripe with the grading would be how harsh it to get a good grade. Like I said previously changing the big idea requirements down by 1 for every grade would make it all more achievable to get a desired grade.
N/A
You are good.
I like the general idea of not having big exams which take pressure off of having to memorize everything and encourages you to really learn the topic. Sometimes it's a bit frustrating when a big idea is only 2 learning outcomes so you could only get 1 learning outcome if you get 3/4. Maybe trying to make each test more even. Other than that everything was good.

Students
I think the grading structure was fine; however, I believe students should be able to earn the learning objectives without fully mastering a big idea. I think this would be beneficial to student grades, as they would have more opportunities to directly practice what they got wrong and improve it.
The sctructure is honestly the best I experienced in college so far.
The Big Ideas quizzes should be kept as it helped me understand the material better. I don't think anything has to be changed.
For big ideas, having only four questions to prove my understanding was very difficult for me. Even when I felt I had a good grasp on the material at hand, stupid mistakes would make it hard for me to do well with so few questions with point opportunities. I like the big idea structure over exams, but think the structure would benefit with a few more questions to balance it out. Maybe two questions per learning outcome?
I like the idea of a no exam class since it lessened the stress but having a quiz every monday was a bit rough because I would only have 5 days to master material.
I thought that the general idea was good, I talked about it in the other answer parts, but the structure kind of forces you to stay engaged compared to leaving everything at the end, which then can cause me to not study enough. I felt that even though it was a different format I put more work into passing the class because it felt more achievable as I knew where I stood at all times.
I would recommend that partial credit is given but more questions on the test to help balance it out.
The assesment and grading structure was good. But I dislike the amount of engagement points earned for each activity. i think that is pretty flawed. Some activities deserve more points for the amount of effort put into earning them. Lastly i believe the progress checks should have more questions. 4 questions per big idea is pretty limiting and does tn allow one to showcase all thats learned. Instead having more small learning outcomes per progress check would help out a lot.
Having a quiz every week was really stressful for me, and honestly it was tough for me to find time to study because I had other exams and quizzes every week.
It was definitely confusing, but after understanding the grading system, it was pretty forgiving and fair.
I like it, a lot of opportunity to make up points
Terrible and unclear
I really liked this structured, please don't change it!
I loved everything, even though I started slacking off near the end, I appreciate the chances for retakes and the homework points and everything how you formatted it. I wish all professors were like this.
Keep the structure however include more examples in class

If you used Paul's Online Math Notes or some other outside resource, was it useful? If applicable, compare to the course textbook.

Students
Yes it was useful, it was very much like the textbook.
I feel that these resources were useful.

Students
I used the course textbook which I had from another course prior. The textbook was very helpful, but it would be more helpful if the class all looked at these problems and the teacher discussed these problems.
I didn't used Paul's Online Math Notes, because I have already taken calc in high school, so almost everything was a review. I can mostly get/ understand all the problems in the calculus book, which I feel like its a great book. It starts out easy, but then when you get to the end, they're will be harder problems.
I used the textbook and nothing else.
yes, it was similar to textbook
It was useful but confusing sometimes because the titles in the problem set assignment was different from the titles in Pauls Online Notes so I didn't always know which problems to do. Maybe provide numbers to do in Pauls Online Notes like you do for the textbook.
I feel like I would have rather used the textbook but Paul's wasn't bad.
Yes. It supplemented the textbook.
I just used pauls online math notes, greatest thing ever and I will be using if for my future Calc classes. I don't think anybody would need anything else to do good in this course.
yes it was helpful
Paul’s was useful but at some point, it would not have some topics or have enough questions for that topic to hit the required amount for the problem set that week
yes! It had a lot of very useful notes and practice problems.
Khan academy is my best friend, love love love.
Didn't use it
Paul's Online Math Notes was very useful and a lot clearer as to what topic's questions you are doing as opposed to the textbook.
I mainly used the textbook which was certainly helpful for the progress checks.
Yes I used Paul's math notes and other outside sources that helped me understand topics over the course.
It was definitely useful in order to complete assignments required and helped me learn.
Yes there is good practice
I used Paul's Online Math Notes frequently for both studying and homework. The course textbook is good if you want more rigorous and in–depth questions to really test your understanding, but POMN can help you initially understand the calculus idea better. For instance, if I'm struggling with related rates, then reading POMN sections and watching a video and checking class notes, then doing some easier problems on POMN will set me up better rather than just jumping into the exercises provided in the book.
yes
Pauls was pretty useful
N/A
It is useful.

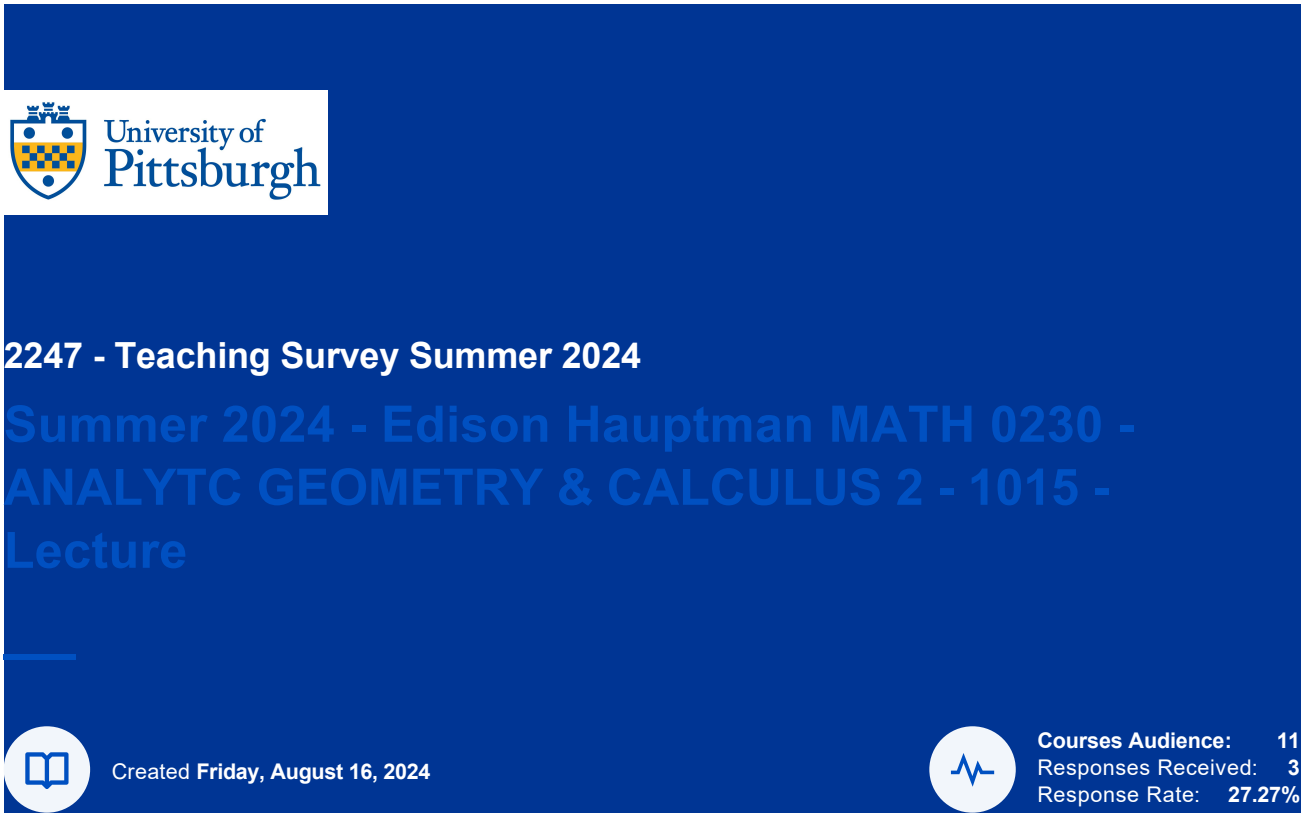
Students
Yes, it was really helpful. When I didn't fully understand a concept, the notes section on Paul's Online Math Notes gave clear explanations and walked me through difficult topics and example problems. In many cases, it was easier to follow than the textbook and helped me reinforce what we were learning in class.
I used Paul's online math notes for all of the homework assignments and overall I felt like it did a decent job of helping me. It was just unfamiliar and hard to find all the questions that we needed to answer compared to the textbook problems. So maybe in the practice problem sets on canvas put where they could find those specific problems on Paul's online math notes.
N/A
it is useful.
Yes. I like using paul's online math notes because it when threw the steps of the problems and explained it very well. The textbook often times does not show the answer so it is harder to know if you're doing it right.
I think the problems for Paul's were way more helpful than the textbook.
I did not use any outside sources.
I used both and found both very useful!
The textbook problems were good practice and so was pauls online notes.
I used Paul's Online Math Notes, and when I did that, I liked how I could solve the problem and then get step–by–step answers. It helped me know how I was wrong and what to fix in the future.
Pauls Online math notes were not useful at all.
Compared to the textbook, Paul's online math notes offered concise, to the point explanations with examples.
I watched a lot of organic chemsitry tutor videos and those helped
Yes, it was helpful. Paul's Online Math Notes helped out a lot.
yes
Yes because I learned nothing from the professor so all I could do was use outside sources
It was useful to to clarify stuff that my notes after class didn't make sense.
i didnt use it as much.
Yes. More comprehensible than the textbook

Is there anything else I should know?

Students
Keep teaching.
A little less enthusiasm might make the class a little more enjoyable.
No, have a good winter break

Students
N/A
It was a very fun semester!
n/a
No
Great style of teaching, kept everyone engaged.
I think you are a great professor who shows a lot of enthusiasm and commitment to teaching. Please keep in environments that allow you to do so.
no
Not especially
No
NA
N/A
Nope :)
No comment
Nope
Nope.
no
N/A
Thanks for a great semester, this was both challenging and rewarding.
Please list the title of what we should do of the titles on the internet.
No! Thank you for such a great semester!!
I think for not being a professor you did an incredible job even better then some professors I had this semester.
N/A
no
Nothing I can think of.
No. I enjoyed the class.
N/A – overall your enthusiasm regarding the material kept me interested and engaged even when the concepts were hard! Thank you for that!
I will give my very honest feedback. A good way to get people to learn is to stay focused on the problems and how to solve them. Don't use tons of different colored markers and over–explain all the nitty gritty details of the theory.

Students
I think you're a great teacher, and I hope you can keep on finding great ways to teach calc with any school.
Edison was a great teacher but his grading style was not for me.
N/A
Very engaging energy
Don’t be a professor
N/A
nope
no



Report Comments



Included in this report:

- Summary of responses to scaled questions
- Response breakdowns
- Student comments
- Results to instructor added custom questions (if applicable)

Understanding and using student feedback:

- We have resources to help you interpret and use results including our faculty worksheet with guided prompts and space to record summaries of feedback, actions, and outcomes.
- Members of our Pedagogy, Practice, & Assessment team are available for consultations and can help with:
  - Interpreting OMET results and developing a course of action if necessary.
  - Exploring various methods of assessment to improve teaching.
- In the future:
  - Discuss, teach, and model giving meaningful feedback with your students and give them multiple opportunities to practice giving feedback.
    - Gather important information about students at the beginning of the term by giving a pre-course survey.
    - Check in with students half way through the term by giving a midterm course survey.
- The Teaching Center offers multiple resources to support teaching and learning.

Office of Measurement and Evaluation of Teaching (OMET)
Contact us

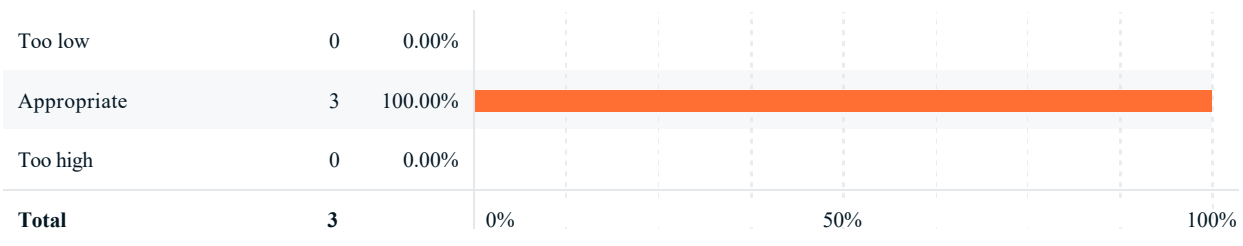
Dietrich School of Arts and Sciences Questions

Summary table

	Invited Count	Response Count	Response Rate	Mean	Median	Mode	SD
The instructor created an atmosphere that kept me engaged in course content.	11	3	27.27%	5.00	5.00	5	0.00
The instructor was prepared for class.	11	3	27.27%	5.00	5.00	5	0.00
The instructor treated students with respect.	11	3	27.27%	5.00	5.00	5	0.00
The instructor was available to me (in-person, electronically, or both).	11	3	27.27%	5.00	5.00	5	0.00
The instructor evaluated my work fairly.	11	3	27.27%	5.00	5.00	5	0.00
The instructor provided feedback that was helpful to me.	11	3	27.27%	5.00	5.00	5	0.00
I learned a lot from this course. If there is no basis to judge or not applicable, answer N/A.	11	3	27.27%	4.67	5.00	5	0.58
Overall of All Questions	77	21	27.27%	4.95	-	-	0.22

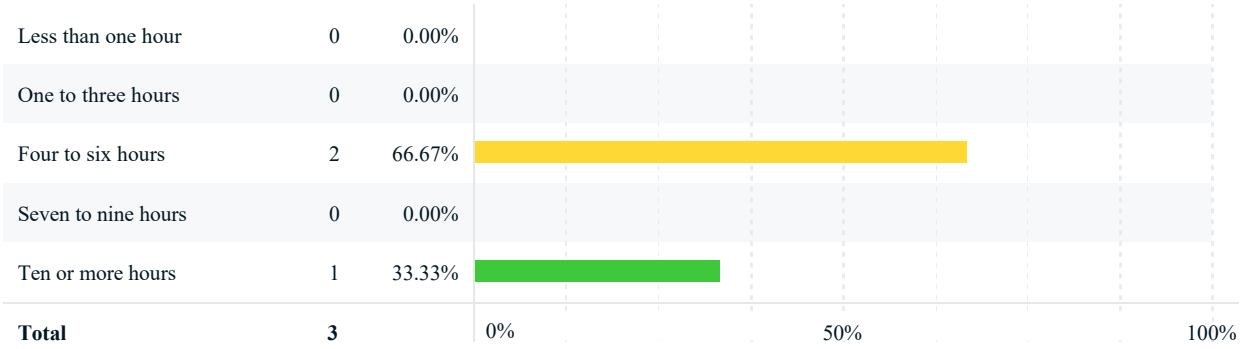
The standards the instructor set for me were:

The standards the instructor set for me were:



How many hours per week did you usually spend working on this course outside of classroom time?

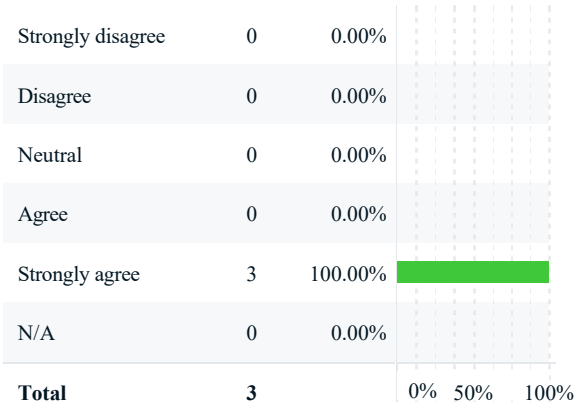
How many hours per week did you usually spend working on this course outside of classroom time?



Less than one hour	0	0.00%
One to three hours	0	0.00%
Four to six hours	2	66.67%
Seven to nine hours	0	0.00%
Ten or more hours	1	33.33%
Total	**3**	

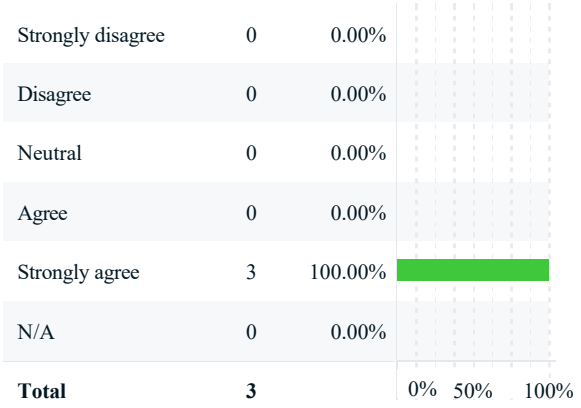
Response breakdown

1. The instructor created an atmosphere that kept me engaged in course content.



Strongly disagree	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	0	0.00%
Strongly agree	3	100.00%
N/A	0	0.00%
Total	**3**	

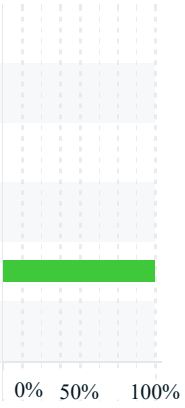
2. The instructor was prepared for class.



Strongly disagree	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	0	0.00%
Strongly agree	3	100.00%
N/A	0	0.00%
Total	**3**	

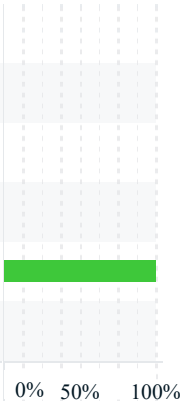
3. The instructor treated students with respect.

Strongly disagree	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	0	0.00%
Strongly agree	3	100.00%
N/A	0	0.00%
Total	**3**	



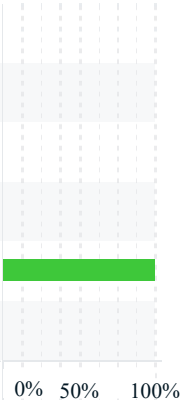
4. The instructor was available to me (in-person, electronically, or both).

Strongly disagree	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	0	0.00%
Strongly agree	3	100.00%
N/A	0	0.00%
Total	**3**	



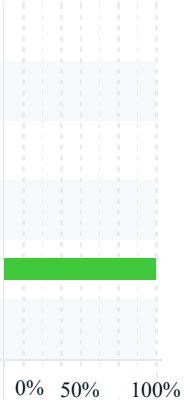
5. The instructor evaluated my work fairly.

Strongly disagree	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	0	0.00%
Strongly agree	3	100.00%
N/A	0	0.00%
Total	**3**	



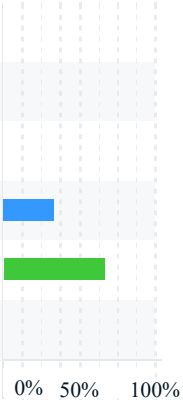
6. The instructor provided feedback that was helpful to me.

Strongly disagree	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	0	0.00%
Strongly agree	3	100.00%
N/A	0	0.00%
Total	**3**	



7. I learned a lot from this course. If there is no basis to judge or not applicable, answer N/A.

Strongly disagree	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	1	33.33%
Strongly agree	2	66.67%
N/A	0	0.00%
Total	**3**	



DSAS Comments

What did you like best about how the course was taught?

Comments
The ability to do projects and other assignments for more credit, collaborative work with other students, professor was always prepared and explained things extremely clearly, always allowed room for questions.
I liked the approach of using words like “remember when we solved this in the previous problem” or “our goal is to make it easier/simpler integral” instead of just sharing what we have to do. Sharing what we had to do, makes it harder to recognize problems and understand what the problem is asking us to solve.
The fact that Edison payed attention to students faces and knew when to explain a certain subject in more detail.

If you were teaching this course, what would you do differently?

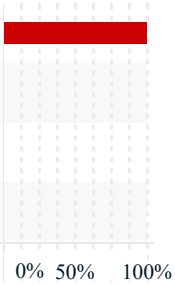
Comments
Honestly, nothing different . I would retake it if it was the same professor and grading system

Math TA/TF Additional Items

Please answer the following questions about your instructor:

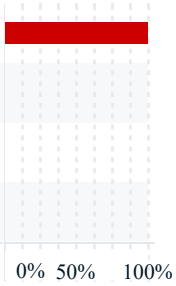
1. Did you expereince difficulty in comprehending your lecture instructor's spoken language in class?

No difficulty a…	3	100.00%
Small amount o…	0	0.00%
Moderate amo…	0	0.00%
Severe difficulty	0	0.00%
Total	**3**	



2. Did your lecture instructor experience difficulty in comprehending the questions that were asked by students in class?

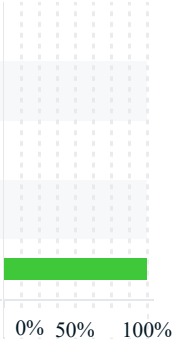
No difficulty a…	3	100.00%
Small amount o…	0	0.00%
Moderate amo…	0	0.00%
Severe difficulty	0	0.00%
Total	**3**	



The lecture instructor's writing on the chalkboard was legible.

The lecture instructor's writing on the chalkboard was legible.

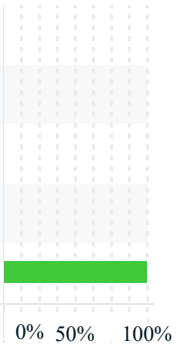
Seldom	0	0.00%
Sometimes	0	0.00%
About half the …	0	0.00%
Usually	0	0.00%
Always	3	100.00%
Total	**3**	



The lecture instructor's attitude toward the subject was enthusiastic.

The lecture instructor's attitude toward the subject was enthusiastic.

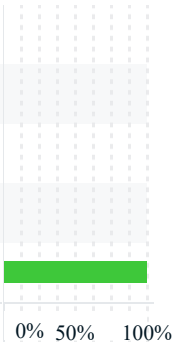
Hardly at all	0	0.00%
To a small degree	0	0.00%
To a moderate …	0	0.00%
To a considera…	0	0.00%
To a very high …	3	100.00%
Total	**3**	



The lecture instructor arrived for class on time.

The lecture instructor arrived for class on time.

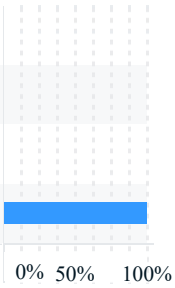
Rarely (less th…	0	0.00%
Seldom (20-40…	0	0.00%
About half the …	0	0.00%
Usually (70-90…	0	0.00%
Over 90% of t…	3	100.00%
Total	**3**	



Please answer the following questions about your instructor:

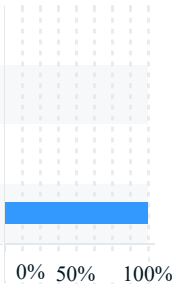
1. The lecture instructor provided the opportunity for questions.

Very seldom	0	0.00%
About half the …	0	0.00%
Frequently	0	0.00%
Almost always	3	100.00%
Total	**3**	



2. Helpful answers were given to questions raised in class.

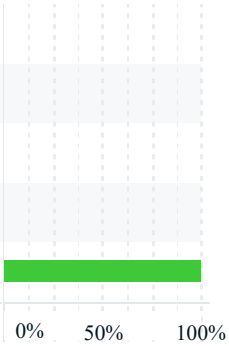
Very seldom	0	0.00%
About half the …	0	0.00%
Frequently	0	0.00%
Almost always	3	100.00%
Total	**3**	



Would you recommend this lecture instructor to a friend taking this course?

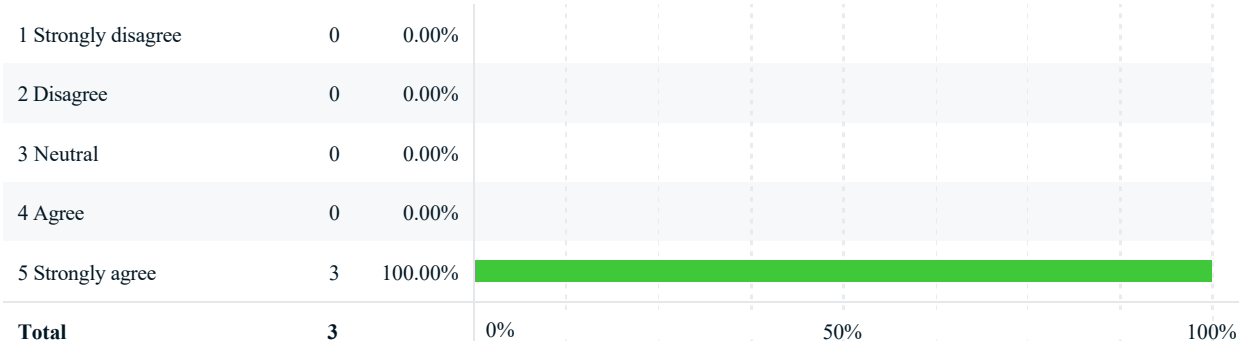
Would you recommend this lecture instructor to a friend taking this course?

Not at all	0	0.00%
Unlikely	0	0.00%
Don't know	0	0.00%
Maybe	0	0.00%
Definitely	3	100.00%
Total	**3**	



Diversity and Inclusion

The instructor creates an inclusive learning environment for all students.



Statistics	Value
Invited Count	11
Response Count	3
Response Ratio	27.27%
Mean	5.00
Median	5.00
Mode	5
Standard Deviation	0.00

University of Pittsburgh

2237 - Teaching Survey Summer 2023

Summer 2023 - Edison Hauptman MATH 0230 - ANALYTC GEOMETRY & CALCULUS 2 - 1015 - Lecture

Created **Friday, August 18, 2023**

Courses Audience: 18
Responses Received: 6
Response Rate: 33.33%

Report Comments

Teaching Survey

Included in this report:

- Summary of responses to scaled questions
- Response breakdowns
- Student comments
- Results to instructor added custom questions (if applicable)

Understanding and using student feedback:

- We have resources that can help with interpreting your teaching survey report.
- Schedule a meeting with a teaching consultant who can help you interpret your results and develop a course of action if necessary.
- In the future:
  - Discuss, teach, and model giving meaningful feedback with your students.
  - Request a midterm survey of your course and give students multiple opportunities to practice giving feedback.

Contact OMET

Dietrich School of Arts and Sciences Questions

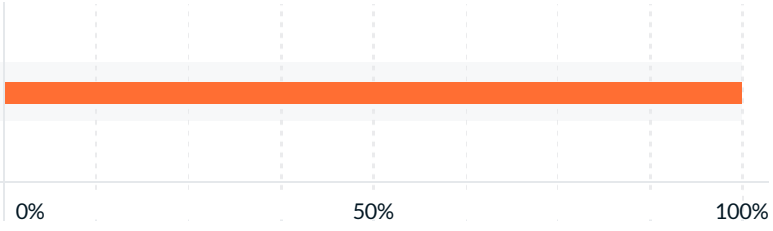
Summary table

	Invited Count	Response Count	Response Rate	Mean	Median	Mode	SD
The instructor created an atmosphere that kept me engaged in course content.	18	6	33.33%	5.00	5.00	5	0.00
The instructor was prepared for class.	18	6	33.33%	5.00	5.00	5	0.00
The instructor treated students with respect.	18	6	33.33%	5.00	5.00	5	0.00
The instructor was available to me (in-person, electronically, or both).	18	5	27.78%	5.00	5.00	5	0.00
The instructor evaluated my work fairly.	18	5	27.78%	5.00	5.00	5	0.00
The instructor provided feedback that was helpful to me.	18	5	27.78%	5.00	5.00	5	0.00
I learned a lot from this course. If there is no basis to judge or not applicable, answer N/A.	18	6	33.33%	5.00	5.00	5	0.00
Overall of All Questions	126	39	30.95%	5.00	-	-	0.00

The standards the instructor set for me were:

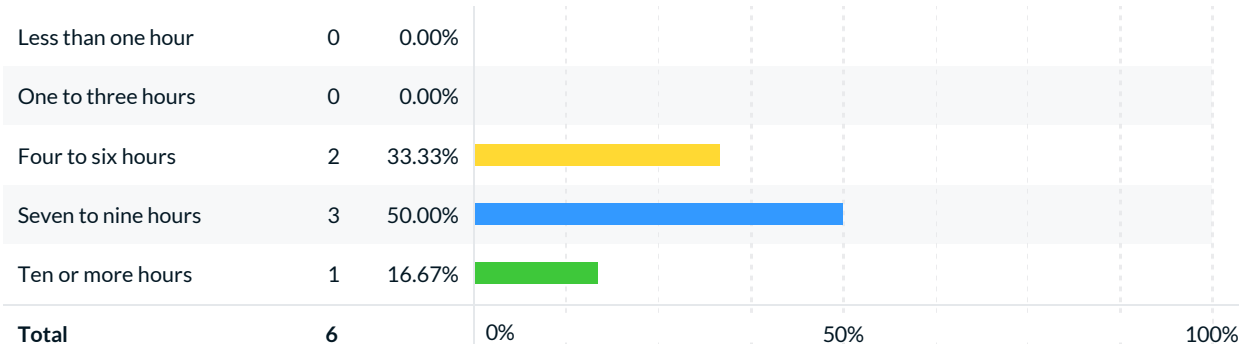
The standards the instructor set for me were:

Too low	0	0.00%
Appropriate	6	100.00%
Too high	0	0.00%
Total	**6**	



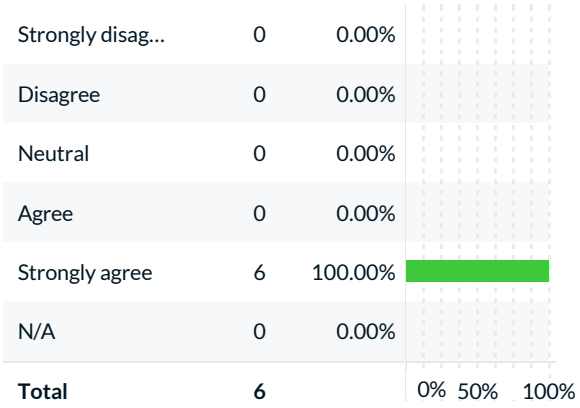
How many hours per week did you usually spend working on this course outside of classroom time?

How many hours per week did you usually spend working on this course outside of classroom time?

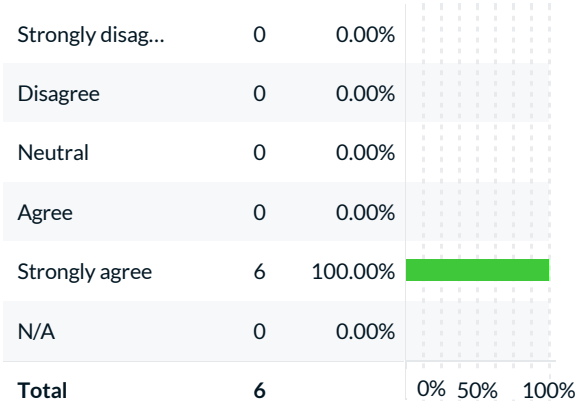


Response breakdown

1. The instructor created an atmosphere that kept me engaged in course content.

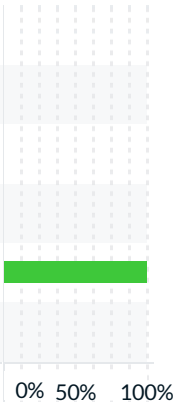


2. The instructor was prepared for class.



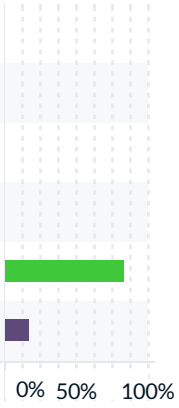
3. The instructor treated students with respect.

Strongly disag…	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	0	0.00%
Strongly agree	6	100.00%
N/A	0	0.00%
Total	**6**	



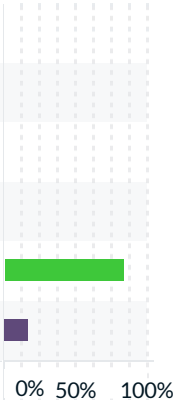
4. The instructor was available to me (in-person, electronically, or both).

Strongly disag…	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	0	0.00%
Strongly agree	5	83.33%
N/A	1	16.67%
Total	**6**	



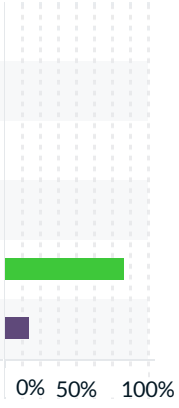
5. The instructor evaluated my work fairly.

Strongly disag…	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	0	0.00%
Strongly agree	5	83.33%
N/A	1	16.67%
Total	**6**	



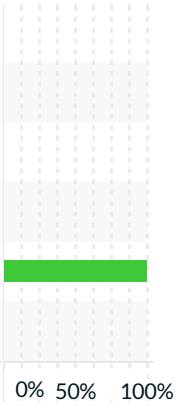
6. The instructor provided feedback that was helpful to me.

Strongly disag…	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	0	0.00%
Strongly agree	5	83.33%
N/A	1	16.67%
Total	**6**	



7. I learned a lot from this course. If there is no basis to judge or not applicable, answer N/A.

Strongly disag…	0	0.00%
Disagree	0	0.00%
Neutral	0	0.00%
Agree	0	0.00%
Strongly agree	6	100.00%
N/A	0	0.00%
Total	**6**	



DSAS Comments

What did you like best about how the course was taught?

Comments
I loved the alternative grading structure and the different choices for assignments. This allowed me to apply the subject of calculus 2 into channels I was both adept in and passionate about. The structure allowed me to succeed as long as I put in enough effort to do so; where as traditional grading structures can punish or are simply neutral to the amount of effort a student puts into the class. You are so enthusiastic about the subject that I actually become engaged through your interest. The passion you have for not only math but the art of teaching is readily apparent and improves the lives of everyone around you. You have wonderful ideas for how to conduct a course and I only wish great things for you in the future of your teaching career. Thank you for making the class I feared the most, truly great.
I liked how the point system took stress off of grades, and with that created more creative opportunities for students to be engaged in the material (ex: teach the rec, projects). The expectations of the course are very clear, and Edison teaches the material in a very engaging way
The engagement and primarily the style of teaching itself made me want to be there and learn the material, analyze the problems, and ask though provoking questions.
Points grading! It incentivized me to do more work and thereby learn more in the process. Further, huge exams werent crazy weighted, allowing me to focus on what I knew was within my realm of capability and made me perform better as the pressure was taken off.
I cannot say enough good things about Edison as an instructor. He engaged the students, met them where they were at, pushed us to learn challenging material, but still made every effort to be compassionate to each student's individual needs. I liked the points system of grading. As someone who is not a strong test taker, I always prefer a take home assignment or project over a timed exam. I liked how I could complete additional assignments to supplement my scores form tests and quizzes. Edison was readily available outside the classroom to answer any questions. I appreciated his sense of humor and I thought he had a warm demeanor which made him very relatable to students from many diverse backgrounds. The course material was challenging and Edison graded accordingly. I thought his grading rubrics were strict but fair. He did not cut corners or give students shortcuts, but instead empowered students with the tools required to approach very difficult problems. I previously took Calculus II at a different university before transferring in. If Edison was my instructor the first time around, I most likely would not have needed to repeat this course. I sincerely think he is not just the best math instructor I've had at Pitt, but the best math teacher I've had ever.

If you were teaching this course, what would you do differently?

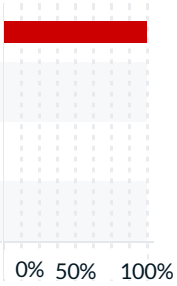
Comments
I would allow the equation sheets to be used during the exams.
Nothing
Not let people talk in class as much
If I was teaching this course, I would not have been able to do anywhere near as good of a job as Edison. That being said, it is very hard to find something I did not like about how the class was taught. The only room I could see for improvement would be a closer watch of the time. I like how each class started with a logic puzzle. It was a fun ice breaker, but this extra time at the beginning of class frequently caused us to go a couple minutes over (no more than 5) at the end of class. If we wrapped up the logic puzzles sooner, we would have been right on time. That is really a minute critique. I cannot stress enough how much I liked this class. If I still had more math classes I needed to take, i would go out of my way to take one taught by Edison.

Math TA/TF Additional Items

Please answer the following questions about your instructor:

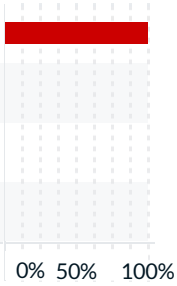
1. Did you expereince difficulty in comprehending your lecture instructor's spoken language in class?

No difficulty at...	5	100.00%
Small amount ...	0	0.00%
Moderate amo...	0	0.00%
Severe difficulty	0	0.00%
Total	**5**	



2. Did your lecture instructor experience difficulty in comprehending the questions that were asked by students in class?

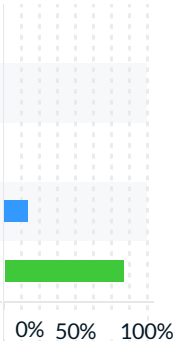
No difficulty at...	5	100.00%
Small amount ...	0	0.00%
Moderate amo...	0	0.00%
Severe difficulty	0	0.00%
Total	**5**	



The lecture instructor's writing on the chalkboard was legible.

The lecture instructor's writing on the chalkboard was legible.

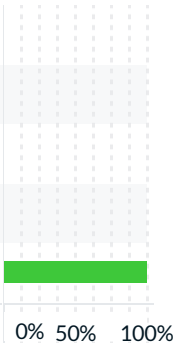
Seldom	0	0.00%
Sometimes	0	0.00%
About half the ...	0	0.00%
Usually	1	16.67%
Always	5	83.33%
Total	**6**	



The lecture instructor's attitude toward the subject was enthusiastic.

The lecture instructor's attitude toward the subject was enthusiastic.

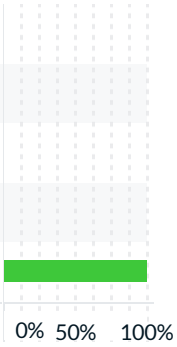
Hardly at all	0	0.00%
To a small degr…	0	0.00%
To a moderate …	0	0.00%
To a considera…	0	0.00%
To a very high …	6	100.00%
Total	**6**	



The lecture instructor arrived for class on time.

The lecture instructor arrived for class on time.

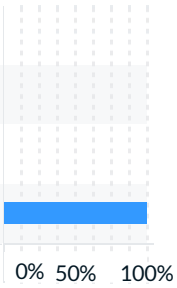
Rarely (less th…	0	0.00%
Seldom (20-40…	0	0.00%
About half the …	0	0.00%
Usually (70-90…	0	0.00%
Over 90% of t…	6	100.00%
Total	**6**	



Please answer the following questions about your instructor:

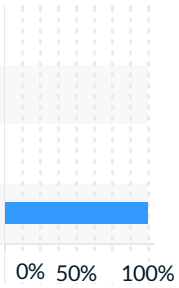
1. The lecture instructor provided the opportunity for questions.

Very seldom	0	0.00%
About half the ...	0	0.00%
Frequently	0	0.00%
Almost always	6	100.00%
Total	**6**	



2. Helpful answers were given to questions raised in class.

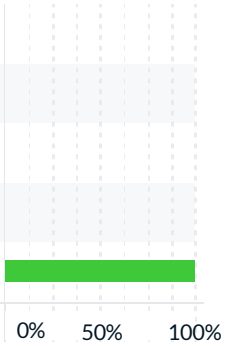
Very seldom	0	0.00%
About half the ...	0	0.00%
Frequently	0	0.00%
Almost always	6	100.00%
Total	**6**	



Would you recommend this lecture instructor to a friend taking this course?

Would you recommend this lecture instructor to a friend taking this course?

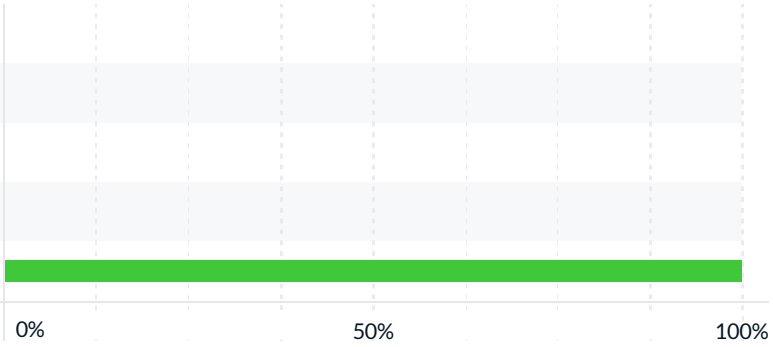
Not at all	0	0.00%
Unlikely	0	0.00%
Don't know	0	0.00%
Maybe	0	0.00%
Definitely	6	100.00%
Total	**6**	



Diversity and Inclusion

The instructor creates an inclusive learning environment for all students.

1 Strongly disagree	0	0.00%
2 Disagree	0	0.00%
3 Neutral	0	0.00%
4 Agree	0	0.00%
5 Strongly agree	6	100.00%
Total	**6**	



Statistics	Value
Invited Count	18
Response Count	6
Response Ratio	33.33%
Mean	5.00
Median	5.00
Mode	5
Standard Deviation	0.00

University of Pittsburgh

Summer 2022 - Edison Hauptman MATH 0230 - ANALYTC GEOMETRY & CALCULUS 2 - 1130 - Lecture

Project Title: **2227 - Teaching Survey Summer 2022**

Courses Audience: **24**
Responses Received: **8**
Response Rate: **33.33%**

Report Comments

Included in this report:

- Responses to numerical questions
- Responses to instructor added questions (if applicable)
- Student comments

Understanding and using student feedback:

- We have resources that can help with interpreting your teaching survey report.
- Meet with a Teaching Consultant who can help you interpret your results and develop a course of action if necessary. Email teaching@pitt.edu to set up a consultation.
- In the future, discuss, teach, and model giving meaningful feedback with your students. Give them multiple opportunities to practice giving feedback. We have several resources that can help guide the discussion and options for gathering student feedback throughout the term.

The Office of Measurement and Evaluation of Teaching (OMET)
Contact us with questions or visit our website to learn more about teaching surveys and collecting student feedback .

Creation Date: **Wednesday, August 24, 2022**

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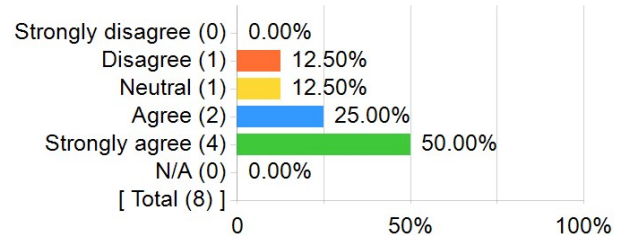
Dietrich School of Arts and Sciences Questions

Summary: 5-point scale - Strongly Disagree to Strongly Agree

Question	Results		
	Response Count	Mean	Standard Deviation
The instructor created an atmosphere that kept me engaged in course content.	8	4.13	1.13
The instructor was prepared for class.	8	4.38	0.52
The instructor treated students with respect.	8	4.63	0.52
The instructor was available to me (in-person, electronically, or both).	8	4.63	0.52
The instructor evaluated my work fairly.	8	4.50	0.53
The instructor provided feedback that was helpful to me.	8	4.13	0.99
I learned a lot from this course. If there is no basis to judge or not applicable, answer N/A.	8	4.25	0.89

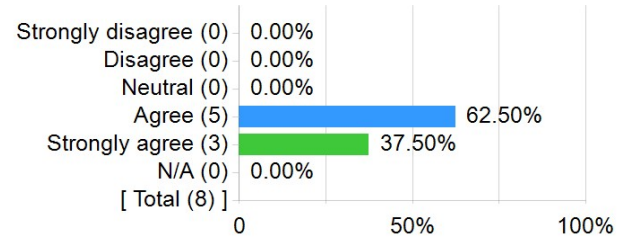
Detailed Responses

1. The instructor created an atmosphere that kept me engaged in course content.



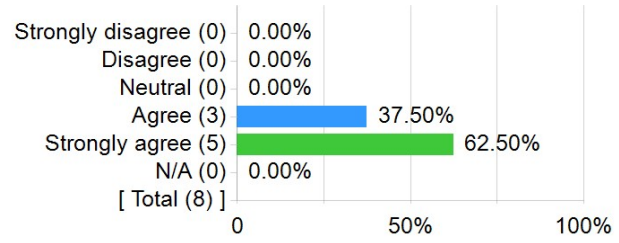
Statistics	Value
Invited Count	24
Response Count	8
Response Ratio	33.33%
Mean	4.13
Median	4.50
Mode	5
Standard Deviation	1.13

2. The instructor was prepared for class.



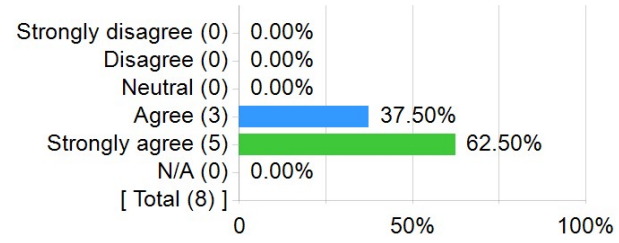
Statistics	Value
Invited Count	24
Response Count	8
Response Ratio	33.33%
Mean	4.38
Median	4.00
Mode	4
Standard Deviation	0.52

3. The instructor treated students with respect.



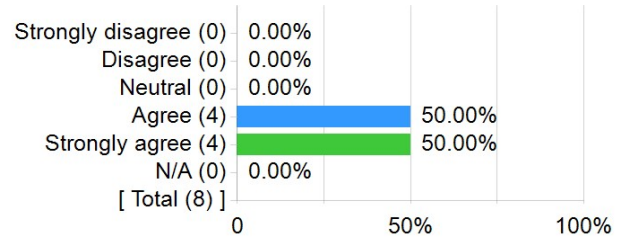
Statistics	Value
Invited Count	24
Response Count	8
Response Ratio	33.33%
Mean	4.63
Median	5.00
Mode	5
Standard Deviation	0.52

4. The instructor was available to me (in-person, electronically, or both).



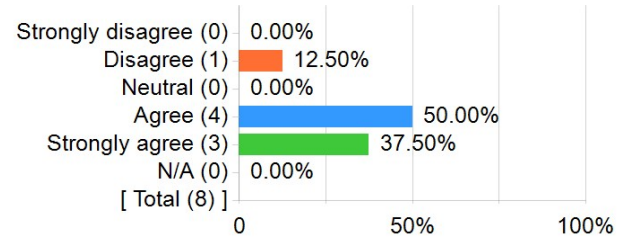
Statistics	Value
Invited Count	24
Response Count	8
Response Ratio	33.33%
Mean	4.63
Median	5.00
Mode	5
Standard Deviation	0.52

5. The instructor evaluated my work fairly.



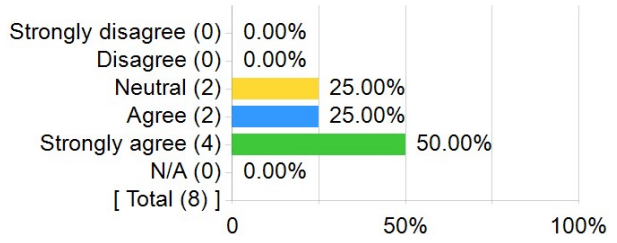
Statistics	Value
Invited Count	24
Response Count	8
Response Ratio	33.33%
Mean	4.50
Median	4.50
Mode	4, 5
Standard Deviation	0.53

6. The instructor provided feedback that was helpful to me.



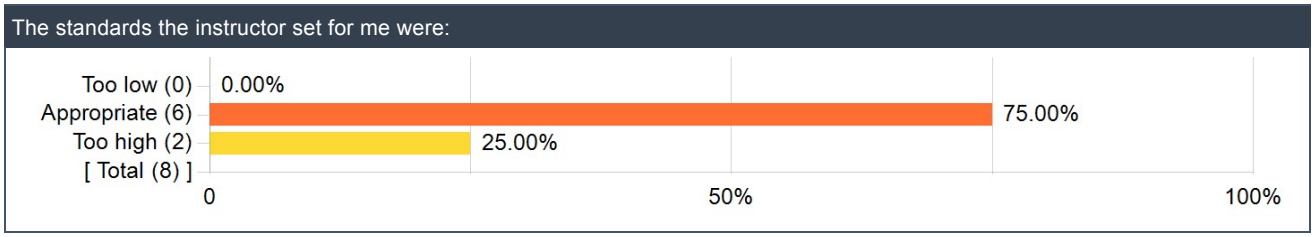
Statistics	Value
Invited Count	24
Response Count	8
Response Ratio	33.33%
Mean	4.13
Median	4.00
Mode	4
Standard Deviation	0.99

7. I learned a lot from this course. If there is no basis to judge or not applicable, answer N/A.

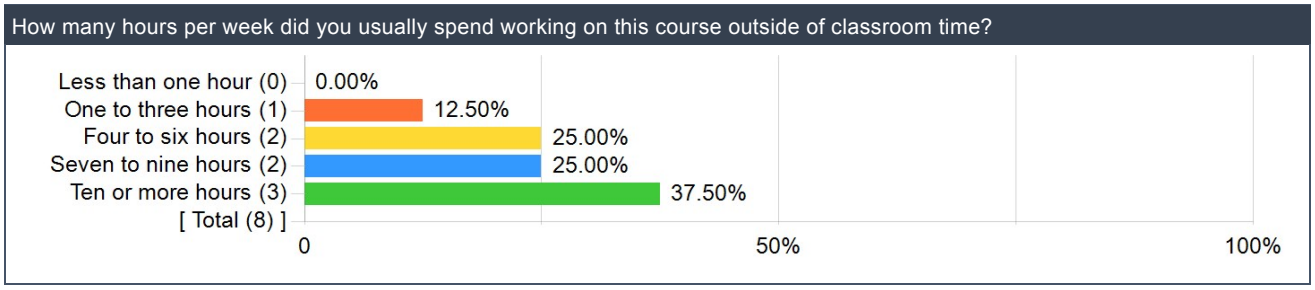


Statistics	Value
Invited Count	24
Response Count	8
Response Ratio	33.33%
Mean	4.25
Median	4.50
Mode	5
Standard Deviation	0.89

The standards the instructor set for me were:



How many hours per week did you usually spend working on this course outside of classroom time?



Comments

What did you like best about how the course was taught?

Comments
The content is taught and tested pretty fairly.
How passionate Edison is!
Edison explained the material really well, and it was easier to understand than learning from the book. However, I do not agree with the thought process of teaching the more difficult problems in class, so that would prepare us for the easier problems when we complete those on our own. I was lost and confused during class, and it did not prepare me well when trying to complete the homework problems on my own. In addition, the problems on the quizzes and the exam were not problems that we touched on during class, so that was also difficult to study for, and it was difficult to see the grade I received afterwards.
The instructor made sure to explain all work he did on the board, so it was extremely easy to follow his process.
The pacing of the course, while clearly faster than a typical 12–week long one, never felt as though it was too much to handle even when the material grew more complex. The general rhythm of going through two topics within a lecture along with a handful of relevant examples for each of them was consistently engaging and informing without being overwhelming. I also deeply appreciated the amount of work in the course, how it was all weighted, and the way in which due dates for homework assignments were handled. Throughout the course, I constantly felt as though I had the opportunities necessary to succeed and overcome mistakes on past assignments. Nothing ever felt unfairly difficult, either. All of the problems that appeared throughout all of the assignments felt as though they were designed to be easily understood for what they were at face value. Very rarely, if at all, did I ever look back at an assignment's answers and wonder how I was supposed to get to it.

If you were teaching this course, what would you do differently?

Comments
Sometimes it seems like Edison is not fully prepared for every example taught in class. It gets confusing when we go through an example and then afterwards he notices a big mistake or an easier way to do the problem. It would be helpful for him to review and complete every single problem shown in class prior to the lecture so that it causes less confusion.
Nothing.
I would use practice problems from the book and use those examples, so we could at least complete them outside of class.
Nothing. I could not have asked for a better teacher for Calculus 2.
The only thing that I would possibly do differently would be to potentially change the timing of the first quiz by pushing it back a day. There's a fair bit of material that is covered in those first two days of the course with what feels like a rather short window of time to practice it all, especially for those that may have also needed to review concepts from Calculus I and Trigonometry. However, I will also admit that this is a single, incredibly minor thing.

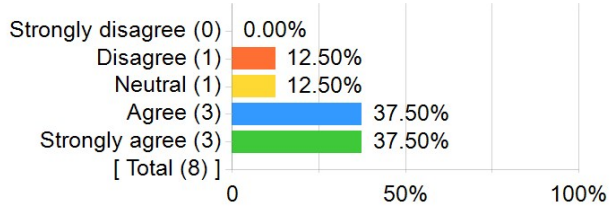
Math Primary/Secondary Instructor Additional Items

Summary: 5-point scale - Strongly Disagree to Strongly Agree

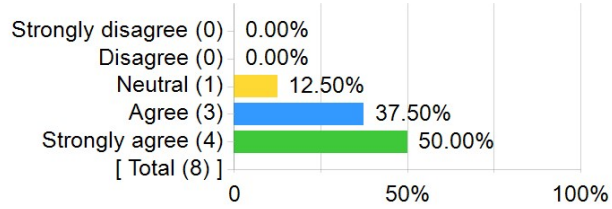
Question	Results		
	Response Count	Mean	Standard Deviation
The instructor presented the course in an organized way.	8	4.00	1.07
The instructor responded to student's questions in an effective manner.	8	4.38	0.74
The instructor used class time effectively to enhance understanding of course topics.	8	4.25	1.04
Based on what I learned in this course, I feel well prepared to use the topics of the course in the future.	8	3.88	1.13

Detailed Responses

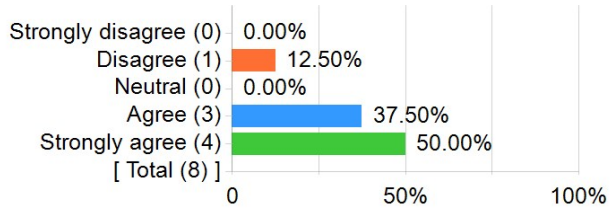
1. The instructor presented the course in an organized way.



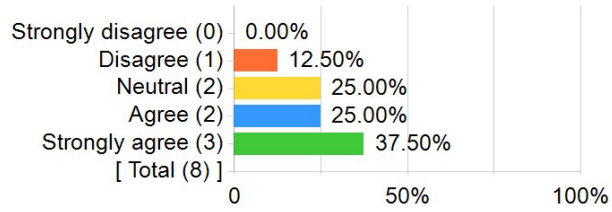
2. The instructor responded to student's questions in an effective manner.



3. The instructor used class time effectively to enhance understanding of course topics.

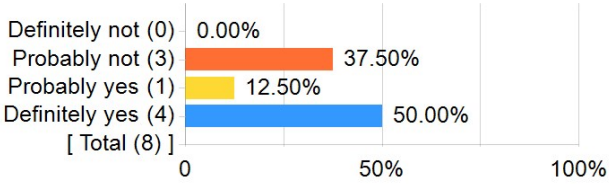


4. Based on what I learned in this course, I feel well prepared to use the topics of the course in the future.

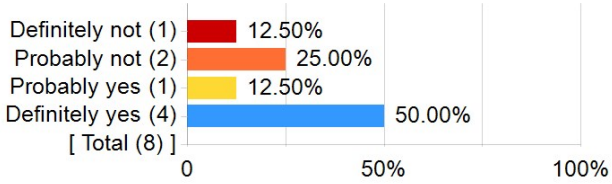


Detailed Responses

1. Would you recommend this instructor to other students?



2. Would you take another course from this instructor?

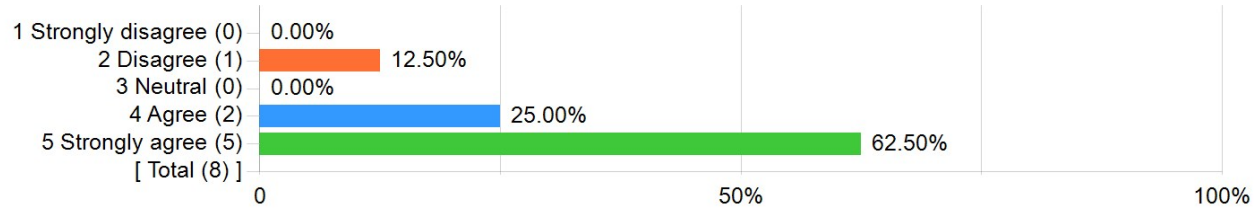


Diversity and Inclusion

Question	Response Count	Mean	Standard Deviation
The instructor creates an inclusive learning environment for all students.	8	4.38	1.06

Details

The instructor creates an inclusive learning environment for all students.



Statistics	Value
Invited Count	24
Response Count	8
Response Ratio	33.33%
Mean	4.38
Median	5.00
Mode	5
Standard Deviation	1.06